SAT Math Question Booklet

1,146 questions classified by topic

Algebra	330
Advanced Math	332
Geometry and Trigonometry	274
Problem-Solving and Data Analysis	210

Every question appears within its domain and skill, Medium before Hard. Each question carries its bank ID, its difficulty and, where recorded, its tool (No calculator needed, Calculator or Desmos). Multiple-choice questions have one correct answer; for student-produced responses (SPR) write the value in the answer box.

This booklet contains questions only: no answers, solutions or explanations.

Mr Malek Thiab

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Algebra (330)

Linear equations in one variable (51)

1. **A-016** · Medium · SPR

What value of x satisfies the equation $\frac{2x+5}{3} = x - 4$?

Answer:

2. **A-022** · Medium · SPR

What value of x satisfies the equation $\frac{3x-4}{2} = x + 5$?

Answer:

3. **A-031** · Medium · SPR

What is the value of x that satisfies $5(x-2) - 3 = 2(x+4)$?

Answer:

4. **A-038** · Medium

What are the solutions to the equation $|2x - 5| = 11$?

A) $x = -3$ or $x = 8$

B) $x = 3$ or $x = -8$

C) $x = -3$ only

D) $x = 8$ only

5. **A-047** · Medium

A taxi in Riyadh charges a base fare of SAR 8 plus SAR 2.50 per kilometer traveled. If a ride costs SAR 33.50, how many kilometers was the ride?

A) 8.2

B) 9.5

C) 10.2

D) 12.0

6. **A-048** · Medium

Fahd's family budgets SAR 450 per month for electricity from the Saudi Electricity Company (SEC). Last month's actual bill was SAR 570, an increase caused entirely by extra usage billed at a marginal rate of SAR 0.6 per kWh. If Fahd's household reduces its usage by k kWh next month, at the same marginal rate, which equation could be solved to find the value of k that would bring next month's bill back down to the SAR 450 budget, assuming all other charges stay the same?

A) $450 - 0.6k = 570$

B) $570 - 0.6k = 450$

C) $570 + 0.6k = 450$

D) $450(0.6) - k = 570$

7. **A-050** · Medium · SPR

A car dealership in Riyadh has a total vehicle inventory T. Of these vehicles, $\frac{1}{3}$ are sedans, $\frac{1}{4}$ are SUVs, $\frac{1}{6}$ are trucks, and the remaining 45 vehicles are motorcycles. What is the total number of vehicles, T, in the dealership's inventory?

Answer:

8. **A-057** · Medium · SPR

A juice company blends a liters of a 15% concentrate juice with 800 liters of a 60% concentrate juice to create a mixture that is 40% concentrate. What is the value of a?

Answer:

9. **A-058** · Medium

A company that roasts and sells coffee beans calculates its monthly profit, in SAR, by subtracting its fixed monthly costs from its monthly sales revenue. The equation $22{,}000 = 3.50x - 6{,}500$ represents this situation for a month where x kilograms of coffee beans are sold. Which statement is the best interpretation of $3.50x$ in this context?

A) The monthly sales revenue, in SAR, from selling x kilograms of coffee beans

B) The monthly sales revenue, in SAR, from each kilogram of coffee beans sold

C) The monthly cost, in SAR, of roasting each kilogram of coffee beans

D) The monthly cost, in SAR, of roasting x kilograms of coffee beans

10. **A-060** · Medium

If $58 = 22 + 3(x - 5)$, what is the value of $3(x - 5)$?

A) 22

B) 36

C) 44

D) 58

11. **A-116** · Medium · SPR

If $x \circledast y = 5y - 3(x + y)$, what is the value of $2 \circledast 6$?

Answer:

12. A-117 · Medium

a, b, and c are positive consecutive even integers, and $c > b > a$. What is the value of b^2 in terms of a?

A) $a^2 + 2$ **B)** $(a+1)^2$

C) $(a+2)^2$ **D)** $a^2 + 4$

13. A-118 · Medium · SPR

m, n, and p are three consecutive positive odd integers such that $m < n < p$. If their sum is 63, what is the value of mp?

Answer:

14. A-119 · Medium · SPR

Yousef is now five times as old as his sister Lama. In 20 years, he will be 16 years older than Lama. What is Yousef's age now?

Answer:

15. A-127 · Medium

If $p = \dfrac{5q-3}{6}$, then $q =$

A) $\dfrac{6p+3}{5}$ **B)** $\dfrac{6p-3}{5}$

C) $\dfrac{5p+3}{6}$ **D)** $\dfrac{p}{5}+3$

16. A-128 · Medium · SPR

If $\dfrac{3m}{4} + 2 = 8$, what is the value of $\dfrac{5}{2m}$?

Answer:

17. A-142 · Medium · SPR

$7(wx+17) = \dfrac{21}{2}x + 182$ In the equation above, w is a constant. If the equation has no solution, what is the value of w?

Answer:

18. A-188 · Medium · SPR

The sum of three consecutive odd integers is 15 less than 4 times the smallest of the three integers. What is the largest of the three integers?

Answer:

19. A-192 · Medium · SPR

The length of a rectangle is 2 less than 3 times its width. The perimeter of the rectangle is 60 centimeters. What is the width, in centimeters?

Answer:

20. A-246 · Medium · SPR

When 3 is subtracted from 5 times a number n, the result is 1 less than twice the sum of n and 8. What is the value of n?

Answer:

21. A-247 · Medium · SPR

The three sides of a triangle have lengths x, $x+3$ and $2x-1$. The perimeter of the triangle is 38. What is the length of the longest side?

Answer:

22. A-248 · Medium · SPR

40% of a number n equals 25% of 64. What is the value of n?

Answer:

23. A-293 · Medium · SPR

In the equation $kx - 7 = 2x + 5$, k is a constant. If $x = 3$ is the solution, what is the value of k?

Answer:

24. A-017 · Hard

If the equation $a(x-3) + 5 = bx - 2a + 11$ is true for all values of x, where a and b are constants, what is the value of a?

A) -6 **B)** -2

C) 2 **D)** 6

25. A-033 · Hard

If $N = \dfrac{Sp}{q-1}$, where N, S, and p are nonzero, what is q in terms of N, S, and p?

A) $q = \dfrac{Sp+N}{N}$ **B)** $q = \dfrac{Sp-N}{N}$

C) $q = \dfrac{Sp}{N} - 1$ **D)** $q = \dfrac{N}{Sp} + 1$

26. A-039 · Hard · SPR

What is the value of x that satisfies $|3x+2| = 5x - 6$?

Answer:

27. A-059 · Hard · SPR

$$\frac{18x+24}{6} - \frac{s}{11} = r(x-9)$$

In the given equation, s and r are constants, and $s > 0$. If the equation has infinitely many solutions, what is the value of s?

Answer:

28. A-061 · Hard · SPR

An amusement park in Riyadh charges 18 SAR per ride ticket. On a Thursday, the park earned 612 SAR in profit from ride-ticket sales after paying daily operating expenses of 216 SAR. How many ride tickets were sold on Thursday? (profit = total revenue − total expenses)

Answer:

29. A-115 · Hard · SPR

The smallest integer in a set of consecutive integers is -11. If the sum of the integers in the set is 12, what is the total number of integers in the set?

Answer:

30. A-122 · Hard

Let m and n be positive integers such that one-third of m is n less than one-half of m. Which of the following is a possible value of m?

A) 16 **B)** 20

C) 24 **D)** 28

31. A-129 · Hard · SPR

If $\frac{3x+5}{4} = \frac{4x-1}{5}$, what is the value of x?

Answer:

32. A-141 · Hard · SPR

$\frac{48x+144}{8} - \frac{c}{15} = d(x-2)$ In the equation above, c and d are constants. If the equation has infinitely many solutions, what is the value of c?

Answer:

33. A-144 · Hard

$7x + 4 = c(x+d)$ In the equation above, c and d are constants. If the equation has no solutions, which of the following must be true? I. $c = 7$ II. $d = \frac{4}{7}$ III. $cd \neq 4$

A) None **B)** I only

C) I and II only **D)** I and III only

34. A-151 · Hard · No calculator needed

$$k(x+3) - 5x = 3x + 2m$$

In the equation above, k and m are constants. If the equation has infinitely many solutions, what is the value of $m - k$?

A) 20 **B)** 4

C) 8 **D)** 12

35. A-160 · Hard · No calculator needed

$$\frac{3x+2}{4} - \frac{x-7}{6} = \frac{5x-1}{12} + 2$$

If x is the solution to the equation above, which of the following intervals contains the value of $x^2 + 4x - 1$?

A) Between 10 and 50 **B)** Greater than 50

C) Between 0 and 5 **D)** Between 5 and 10

36. A-172 · Hard · SPR · No calculator needed

$$k(3x-2) - (x-m) = 8x + 10$$

In the equation above, k and m are constants. The equation is true for every real value of x. What is the value of $k + m$?

Answer:

37. A-173 · Hard · SPR

$$k(x+2) - 3x = 5x + 4m$$

In the equation above, k and m are constants. If the equation has infinitely many solutions, what is the value of $k + m$?

Answer:

38. A-174 · Hard

$$3(2x-a) + 5 = bx - 4$$

In the equation above, a and b are constants. Which of the following pairs of values for a and b gives an equation with no solution?

A) $a = 3,\ b = 6$ **B)** $a = 2,\ b = 6$

C) $a = 5,\ b = 3$ **D)** $a = 6,\ b = -6$

39. A-184 · Hard · SPR

$$k(x-4) + 3x = 7x + m$$

In the equation above, k and m are constants. If the equation has infinitely many solutions, what is the value of m?

Answer:

40. A-185 · Hard · SPR

$$4(ax-3) = 10x + 9$$

In the equation above, a is a constant. If the equation has no solution, what is the value of a? (Enter as a decimal or fraction.)

Answer:

41. A-186 · Hard · SPR

$$3(4x-5)+2=7(4x-5)-18$$

What is the value of $4x-5$?

Answer:

42. A-187 · Hard · SPR

$$\frac{2x-1}{3}-\frac{x+2}{4}=\frac{1}{2}$$

What is the value of $5x$?

Answer:

43. A-189 · Hard

Which of the following equations has no solution?

A) $5x-1=2(x+4)+x$ **B)** $4x-7=4(x-2)+3$

C) $3(x+2)=3x+6$ **D)** $2x+5=2(x+3)-1$

44. A-190 · Hard

$$3x+kx=2x+12$$

In the equation above, k is a constant. For which of the following values of k does the equation NOT have exactly one solution?

A) 0 **B)** 1

C) -2 **D)** -1

45. A-191 · Hard · SPR

A hiker walks $\frac{2}{5}$ of a trail on the first day and $\frac{1}{3}$ of the remaining distance on the second day. After the second day, 12 kilometers of the trail remain. How long, in kilometers, is the whole trail?

Answer:

46. A-245 · Hard · SPR

Pump A can fill an empty tank in 6 hours working alone. Pump B can fill the same tank in 9 hours working alone. How many hours will it take the two pumps, working together, to fill the empty tank? (Enter as a decimal.)

Answer:

47. A-294 · Hard · SPR

Cyclist A leaves a town riding at a constant 12 kilometers per hour. One hour later, cyclist B leaves the same town on the same road at a constant 18 kilometers per hour. How many hours after cyclist B starts does B catch up with A?

Answer:

48. A-309 · Hard · SPR

$$\frac{x+k}{3}=\frac{2x-1}{4}$$

In the equation above, k is a constant. If $x=5$ is a solution, what is the value of k? (Enter as a decimal.)

Answer:

49. A-312 · Hard · SPR

A group of friends splits a restaurant bill equally. If 2 more friends had joined, every person would have paid $\$6$ less. The group originally had 4 friends. What was the total bill, in dollars?

Answer:

50. A-313 · Hard · SPR

What is the sum of all solutions of the equation $|x-4|=2x-5$?

Answer:

51. A-333 · Hard · No calculator needed

A wall is $9x$ meters long, where x is a positive integer. Lamps are placed at both ends of the wall and every $\frac{9}{7}$ meters along it. Which expression gives the number of lamps?

A) $\frac{81x}{7}+1$ **B)** $7x$

C) $7x+1$ **D)** $9x+1$

Linear equations in two variables (66)

52. A-010 · Medium

What is the slope of the line represented by the equation $4x - 2y = 10$?

A) -5 **B)** -2

C) 2 **D)** 4

53. A-011 · Medium

Saudi Arabia's annual date exports increased at a constant rate between 2021 and 2024. In 2021, exports were 210 thousand metric tons; in 2024, exports were 288 thousand metric tons. Which equation models the exports y (in thousand metric tons) t years after 2021?

A) $y = 26t + 210$ **B)** $y = 78t + 210$

C) $y = 26t + 288$ **D)** $y = 210t + 26$

54. A-028 · Medium

A line contains the points $(2, 5)$ and $(6, 17)$. What is the x-intercept of this line?

A) $-\frac{1}{3}$ **B)** $\frac{1}{3}$

C) 1 **D)** 3

55. A-036 · Medium

The graph of the equation $\frac{x}{4} + \frac{y}{6} = 1$ in the xy-plane is a line. What is the slope of this line?

A) $-\frac{3}{2}$ **B)** $-\frac{2}{3}$

C) $\frac{2}{3}$ **D)** $\frac{3}{2}$

56. A-049 · Medium · SPR

A NEOM production line starts each day with a number of units already completed and then produces additional units at a constant rate per hour. After 3 hours of operation, a total of 460 units have been completed. After 7 hours, a total of 780 units have been completed. What is the constant hourly production rate, in units per hour?

Answer:

57. A-062 · Medium · SPR

Line k is defined by $y = -\frac{21}{4}x + 9$. Line j is perpendicular to line k in the xy-plane. What is the slope of line j?

Answer:

58. A-063 · Medium · SPR

Layla ordered soil and gravel for her garden in Jeddah, which cost a total of 518 SAR. The given equation represents the relationship between the number of cubic meters of soil, x, and the number of tons of gravel, y, that Layla ordered: $28.5x + 32.75y = 518$. How much more, in SAR, did a ton of gravel cost Layla than a cubic meter of soil?

Answer:

59. A-064 · Medium

Noura made 48 cups of date paste. She then filled x small jars and y large jars with all the paste she made. The equation $3x + 5y = 48$ represents this situation. Which is the best interpretation of $5y$ in this context?

A) The number of large jars Noura filled
B) The number of small jars Noura filled
C) The total number of cups of paste in the large jars
D) The total number of cups of paste in the small jars

60. A-083 · Medium

What is the x-intercept of the line given by the equation $3x + 4y = 24$?

A) $(0, 6)$ **B)** $(6, 0)$

C) $(8, 0)$ **D)** $(0, 8)$

61. A-084 · Medium

Line k is parallel to the line given by $y = -2x + 7$ and passes through the point $(4, 5)$. Which of the following is an equation of line k?

A) $y = -2x + 13$ **B)** $y = -2x - 3$

C) $y = 2x - 3$ **D)** $y = 2x + 13$

62. A-094 · Medium

A line has a slope of -3 and passes through the point $(5, -2)$. What is the y-intercept of the line?

A) 7 **B)** 10

C) 13 **D)** 15

63. A-095 · Medium

Which of the following equations, when graphed in the xy-plane, is a line with the same slope as the line $5x - 2y = 8$?

A) $10x - 4y = 6$ **B)** $2x - 5y = 10$

C) $5x + 2y = 8$ **D)** $2x + 5y = 10$

64. A-107 · Medium

A line representing monthly electricity savings, in SAR, from installing solar panels has a slope of 45 and passes through the point $(4, 260)$, where x is the number of months since installation. Which of the following is an equation of this line?

A) $y = 45x + 80$ **B)** $y = 45x + 180$

C) $y = 80x + 45$ **D)** $y = 45x - 80$

65. A-112 · Medium

A recycling program in Dammam pays SAR 0.50 for every kilogram of paper collected and gives a SAR 60 bonus to every school team that meets its monthly collection target. In one month, the total amount paid, in SAR, to all participating teams is modeled by

$$0.50k + 60t = 900$$

where k is the total kilograms of paper collected and t is the number of teams that met their target. What does the term $60t$ represent in this equation?

A) The total number of teams participating in the program
B) The total bonus amount, in SAR, paid to the teams that met their target
C) The average number of kilograms collected per team
D) The total number of kilograms collected by teams that met their target

66. A-121 · Medium

M is a set of five consecutive odd integers. When the least two integers in set M are summed, the result is x. When the greatest two integers in set M are summed, the result is y. Which of the following is true?

A) $y = x + 8$ **B)** $y = x + 10$

C) $y = x + 12$ **D)** $y = 2x$

67. A-193 · Medium · SPR

The table shows three values of x and their corresponding values of y for the linear function f.

x	2	5	9
$f(x)$	7	13	k

What is the value of k?

Answer:

68. A-197 · Medium · SPR

The graph of $3x + 4y = 24$ in the xy-plane and the two coordinate axes bound a triangle. What is the area of the triangle?

Answer:

69. A-201 · Medium · SPR

A line in the xy-plane passes through the points $(a, 5)$ and $(7, 11)$ and has slope $\frac{3}{2}$. What is the value of a?

Answer:

70. A-257 · Medium · SPR

A print shop charges a per-copy price plus a flat setup fee. An order of 3 copies costs $\$19$ and an order of 8 copies costs $\$39$. What is the cost, in dollars, of an order of 12 copies?

Answer:

71. A-295 · Medium · SPR

The points $(-2, 1)$, $(1, k)$ and $(7, -11)$ lie on the same line in the xy-plane. What is the value of k?

Answer:

72. A-296 · Medium

The graph of $5x + 12y = 60$ in the xy-plane crosses the x-axis at point P and the y-axis at point Q. What is the distance between P and Q?

A) $\sqrt{119}$ **B)** 17

C) 13 **D)** 60

73. A-311 · Medium · SPR

The average (arithmetic mean) of two numbers x and y is 12, and $2x - y = 3$. What is the value of x?

Answer:

74. A-319 · Medium · No calculator needed

In the xy-plane, what is the y-intercept of the graph of $x = 4y - 12$?

A) $(0, -12)$ **B)** $(0, -3)$

C) $(-12, 0)$ **D)** $(0, 3)$

75. A-324 · Medium · No calculator needed

A farm has 40 hectares of date palms and 25 hectares of olive trees. The equation $40x + 25y = 3{,}600$ represents the farm's total harvest, in tonnes, last season. What does x represent?

A) The average date harvest per hectare, in tonnes
B) The number of tonnes harvested per date palm tree
C) The number of hectares of date palms
D) The total date harvest, in tonnes

76. A-325 · Medium · No calculator needed

A line in the xy-plane has slope 3 and passes through the points $(2, k)$ and $(k, 14)$. What is the value of k?

A) 8 **B)** 4

C) 11 **D)** 5

77. A-330 · Medium · SPR · No calculator needed

A shopper spends exactly 600 dollars on x kilograms of oranges at 4 dollars per kilogram and y kilograms of apples at 5 dollars per kilogram, so $4x + 5y = 600$. If the shopper buys 10 more kilograms of oranges and still spends 600 dollars, by how many kilograms does y decrease?

Answer:

78. A-012 · Hard

Line ℓ passes through the point $(3, -2)$ and is parallel to the line with equation $6x - 3y = 9$. What is the y-intercept of line ℓ?

A) -8 **B)** -6

C) -3 **D)** 2

79. A-013 · Hard · SPR

Line m is perpendicular to the line represented by the equation $2x + 5y = 20$ and passes through the point $(4, 1)$. What is the x-intercept of line m?

Answer:

80. A-020 · Hard

Line ℓ passes through the point $(-1, 4)$ and is parallel to the line with equation $4x + 2y = 10$. What is the y-intercept of line ℓ?

A) -2 **B)** 0

C) 2 **D)** 6

81. A-029 · Hard · SPR

Line k is perpendicular to the line represented by the equation $3x - 4y = 8$ and passes through the point $(2, -1)$. What is the y-intercept of line k?

Answer:

82. A-065 · Hard

The table shows four values of x and their corresponding values of y. There is a linear relationship between x and y.

x	y
-8	73
-4	61
4	37
8	25

Which equation represents this relationship?

A) $3x + y = 49$ **B)** $3x + y = 41$

C) $x + 3y = 49$ **D)** $x + 3y = 41$

83. A-085 · Hard · SPR

For what value of k does the system of equations below have no solution?

$$4x - 6y = 10$$

$$kx - 9y = 20$$

Answer:

84. A-086 · Hard

A straight irrigation canal in an agricultural project near Al-Kharj is modeled by the equation $2x - 5y = 20$, where x and y are measured in meters. A new access road is to be built perpendicular to the canal, crossing it at the point $(10, 0)$. Which of the following is an equation of the access road?

A) $y = -\frac{5}{2}x + 25$ **B)** $y = \frac{5}{2}x - 25$

C) $y = -\frac{2}{5}x + 4$ **D)** $y = \frac{2}{5}x - 4$

85. A-096 · Hard · SPR

The lines given by $3x + y = 15$ and $x - 2y = -2$ intersect at the point (x, y). What is the value of $x + y$?

Answer:

86. A-108 · Hard

A line segment has endpoints $(2, 3)$ and $(10, 11)$. What is the equation of the line containing this segment?

A) $y = x + 1$ **B)** $y = x - 1$

C) $y = 2x - 1$ **D)** $y = 2x + 1$

87. A-134 · Hard · SPR

Line ℓ passes through the point $(-2, 5)$ and is perpendicular to the line $3x - 4y = 12$. What is the y-intercept of line ℓ?

Answer:

88. A-136 · Hard

In the xy-plane, line ℓ passes through the point $(-3, 8)$. The y-coordinate of the y-intercept of line ℓ is positive, and the x-coordinate of the x-intercept of line ℓ is between 0 and 1 (not inclusive). Which of the following could be the slope of line ℓ?

A) $-\frac{3}{2}$ **B)** $-\frac{5}{2}$

C) $-\frac{8}{3}$ **D)** -3

89. A-146 · Hard

Solution X contains 1 millimole (mmol) of chloride in every 8.0 mL, and solution Y contains 1 mmol of chloride in every 2.5 mL. A mixture of x mL of X and y mL of Y contains a total of 7.6 mmol of chloride. Which equation represents this situation?

A) $8.0x + 2.5y = 7.6$ **B)** $0.125x + 0.40y = 7.6$

C) $8.0x + 2.5y = 1$ **D)** $0.125x + 0.40y = 1$

90. A-157 · Hard · No calculator needed

Line p has equation $3x - ky = 15$, where k is a nonzero constant. Line q is perpendicular to line p and passes through the y-intercept of line p. Which of the following is an equation of line q?

A) $y = \frac{k}{3}x - \frac{15}{k}$ **B)** $y = -\frac{3}{k}x - \frac{15}{k}$

C) $y = -\frac{k}{3}x - \frac{15}{k}$ **D)** $y = -\frac{k}{3}x + \frac{15}{k}$

91. A-159 · Hard · Calculator

A contractor paints the walls of a room with total wall area x square feet. Primer is applied to all of the wall area in one coat; one gallon of primer covers 400 square feet and costs $\$20$. Then paint is applied to $\frac{3}{5}$ of the wall area in two coats; one gallon of paint covers 300 square feet per coat and costs $\$30$. The contractor buys 10% more paint than the calculated amount, but buys exactly the calculated amount of primer. Gallons may be purchased in fractions. Which of the following functions gives the total cost $C(x)$, in dollars, of the primer and paint?

A) $C(x) = 0.182x$ **B)** $C(x) = 0.187x$

C) $C(x) = 0.232x$ **D)** $C(x) = 0.170x$

92. A-161 · Hard · SPR · No calculator needed

In the xy-plane, line ℓ has equation $5x - 2y = 11$ and line m has equation $ax + by = d$, where a, b and d are positive constants and $a + b = 21$. Lines ℓ and m are perpendicular, and line m passes through the point $(4, 3)$. What is the value of d?

Answer:

93. A-167 · Hard · Desmos

A coffee shop blends Ethiopian beans, which cost $\$18$ per kilogram, with Colombian beans, which cost $\$11$ per 500 grams. The blend has a total mass of 36 kilograms, and its cost works out to $\$20.50$ per kilogram. How many kilograms of Ethiopian beans are in the blend?

A) 18 **B)** 27

C) 13.5 **D)** 22.5

94. A-175 · Hard

Line ℓ has equation $3x - 5y = 20$. Line m is perpendicular to ℓ and passes through $(6, -1)$. What is the y-intercept of line m?

A) $\left(0, -\frac{23}{5}\right)$ **B)** $(0, 1)$

C) $(0, 9)$ **D)** $(0, -11)$

95. A-176 · Hard · SPR

The graph of $y = -3x + 12$ is translated 4 units to the right and 3 units up to form the graph of line n. What is the x-coordinate of the x-intercept of line n?

Answer:

96. A-194 · Hard · SPR

A line in the xy-plane passes through the points $(-2, 8)$ and $(4, -4)$. What is the x-coordinate of the x-intercept of the line?

Answer:

97. A-195 · Hard · SPR

$$ax + 3y = 12$$

In the equation above, a is a constant. The graph of the equation in the xy-plane is parallel to the graph of $y = -\frac{2}{3}x + 5$. What is the value of a?

Answer:

98. A-196 · Hard · SPR

$$kx - 4y = 10$$

In the equation above, k is a constant. The graph of the equation in the xy-plane is perpendicular to the graph of $2x + y = 7$. What is the value of k?

Answer:

99. A-198 · Hard

A line in the xy-plane has x-intercept -4 and y-intercept 6. Which of the following is an equation of the line?

A) $3x - 2y = -12$ **B)** $2x + 3y = 12$

C) $2x - 3y = -12$ **D)** $3x + 2y = 12$

100. A-199 · Hard · SPR

The graph of $y = 2x - 3$ is translated 2 units to the left and 5 units down to form the graph of line q. What is the y-intercept of line q?

Answer:

101. A-200 · Hard · SPR

A gym charges a regular sign-up fee of $\$45$ plus $\$12$ per month. A student plan has a different sign-up fee and a monthly price that is 15% less than the regular monthly price. After 7 months, a student pays the same total as a regular member. What is the student sign-up fee, in dollars? (Enter as a decimal.)

Answer:

102. A-202 · Hard · SPR

A line passes through the point $(0, -4)$ and is parallel to the graph of $6x - 2y = 9$. What is the x-coordinate of the x-intercept of the line? (Enter as a fraction.)

Answer:

103. A-203 · Hard

Maya spends exactly $\$60$ on notebooks at $\$5$ each and pens at $\$3$ each. She buys x notebooks and y pens, so $5x + 3y = 60$. Which of the following could be (x, y)?

A) $(4, 12)$ **B)** $(9, 6)$

C) $(6, 10)$ **D)** $(7, 8)$

104. A-251 · Hard · SPR

The point $(k, 2k)$ lies on the line $3x - 2y = 14$ in the xy-plane. What is the value of k?

Answer:

105. A-252 · Hard · SPR

A line in the xy-plane passes through $(1, 2)$, and its x-intercept is twice its y-intercept. What is the y-intercept of the line? (Enter as a fraction or decimal.)

Answer:

106. A-253 · Hard · SPR

The lines $2x+3y = 12$ and $x-y = 1$ intersect at a point. The two lines and the y-axis form a triangle. What is the area of the triangle? (Enter as a decimal.)

Answer:

107. A-254 · Hard · SPR

The segment with endpoints $(2, -3)$ and $(8, 9)$ has its midpoint on the line $y = mx - 1$ in the xy-plane. What is the value of m? (Enter as a fraction.)

Answer:

108. A-255 · Hard

Points $A(1, 2)$ and $B(7, 6)$ are given in the xy-plane. Which of the following is an equation of the line that is perpendicular to $\overline{AB}$ and passes through its midpoint?

A) $2x + 3y = 20$ **B)** $3x - 2y = 4$

C) $3x + 2y = 20$ **D)** $2x - 3y = -4$

109. A-258 · Hard · SPR

The lines $y = 2x - 3$ and $y = -x + 6$ intersect at the point P. A third line passes through P and is perpendicular to $y = 2x - 3$. What is the y-intercept of the third line? (Enter as a fraction or decimal.)

Answer:

110. A-260 · Hard · SPR

$$3x - 2y = 12$$

Line ℓ in the xy-plane is defined by the equation above. Line m has the same y-intercept as ℓ and is perpendicular to ℓ. What is the x-coordinate of the x-intercept of m?

Answer:

111. A-308 · Hard · SPR

The lines $y = 2x + a$ and $y = -3x + b$ intersect at the point $(4, 0)$ on the x-axis. What is the value of $a - b$?

Answer:

112. A-320 · Hard · No calculator needed

In the equation $ax+by = c$, a, b and c are constants with $a > 0$, $b < 0$ and $c > 0$. The graph of the equation in the xy-plane does NOT pass through which quadrant?

A) Quadrant II **B)** Quadrant III

C) Quadrant I **D)** Quadrant IV

113. A-326 · Hard · SPR · No calculator needed

In the xy-plane, the region with $x \geq 0$, $y \geq 0$ and $2x + y \leq 12$ is a triangle. The line $y = mx$ divides this triangle into two regions of equal area. What is the value of m?

Answer:

114. A-331 · Hard · No calculator needed

If $7x - 2y = 3x + 6y$ and x and y are nonzero, what is the value of $\dfrac{x + y}{x - y}$?

A) The value cannot be determined from the information given.

B) $\dfrac{1}{3}$

C) 2

D) 3

115. A-338 · Hard · No calculator needed

In the xy-plane, the graph of $3x - ky = 12$, where k is a nonzero constant, has an x-intercept equal to its y-intercept. What is the value of k?

A) -4 **B)** 3

C) 4 **D)** -3

116. A-339 · Hard · No calculator needed

For every real number m, the graph of $y = mx + 3m - 2$ in the xy-plane passes through the point P. What are the coordinates of P?

A) $(-3, -2)$ **B)** $(0, -2)$

C) $(3, -2)$ **D)** $(-2, 3)$

117. A-340 · Hard · No calculator needed

A cafe sells juice for 4 dollars each and sandwiches for 7 dollars each. A customer spends exactly 112 dollars and buys at least one juice and at least one sandwich. How many different combinations of juices and sandwiches are possible?

A) 5 **B)** 4

C) 3 **D)** 28

Linear functions (77)

118. A-001 · Medium

The function f is linear. Selected values of x and $f(x)$ are shown in the table below.

x	$f(x)$
2	11
5	20

What is the value of $f(0)$?

A) 2 **B)** 5

C) 8 **D)** 11

119. A-002 · Medium

As part of a NEOM sustainable housing initiative aligned with Vision 2030, the number of homes connected to a renewable microgrid increases at a constant rate. At the start of construction ($m = 0$ months), 1200 homes were connected. After 8 months, 4400 homes were connected. Which function $h(m)$ best models the number of homes connected after m months?

A) $h(m) = 1200m + 400$ **B)** $h(m) = 400m + 1200$

C) $h(m) = 1200m + 4400$ **D)** $h(m) = 550m + 1200$

120. A-003 · Medium · SPR

If $f(x) = 3x - 7$ and $g(x) = f(x + 2)$ for all values of x, what is the value of $g(1)$?

Answer:

121. A-018 · Medium

The function f is linear. Selected values of x and $f(x)$ are shown in the table below.

x	$f(x)$
2	9
6	21

What is the value of $f(10)$?

A) 27 **B)** 30

C) 33 **D)** 36

122. A-024 · Medium

The function f is linear. Selected values of x and $f(x)$ are shown in the table below.

x	$f(x)$
1	7
4	19

What is the value of $f(10)$?

A) 37 **B)** 40

C) 43 **D)** 46

123. A-042 · Medium

Which of the following lines is perpendicular to the line $y = -\frac{2}{3}x + 5$?

A) $y = \frac{3}{2}x - 1$ **B)** $y = -\frac{3}{2}x + 4$

C) $y = \frac{2}{3}x + 2$ **D)** $y = -\frac{2}{3}x - 7$

124. A-044 · Medium

The function f is linear, with $f(1) = 6$ and $f(5) = 18$. If $g(x) = f(x - 4) + 2$, what is the value of $g(10)$?

A) 20 **B)** 21

C) 23 **D)** 25

125. A-066 · Medium · SPR

According to a model, the wingspan, in millimeters, of a desert locust can be estimated by adding 1.2 to five times the body length of the locust, in centimeters. According to the model, what would be the wingspan, in millimeters, of a desert locust that has a body length of 4.6 centimeters?

Answer:

126. A-067 · Medium

In the xy-plane, the graph of the linear function f contains the points $(0, 5)$ and $(6, 41)$. Which equation defines f, where $y = f(x)$?

A) $f(x) = 6x + 5$
B) $f(x) = 5x + 6$
C) $f(x) = 41x + 5$
D) $f(x) = 36x + 41$

127. A-079 · Medium

A linear function models the total cost, in SAR, of a school field trip as $C(n) = 25n + 400$, where n is the number of students attending. What does the value 400 represent in this context?

A) The cost per student
B) The total cost if exactly 25 students attend
C) A fixed cost that does not depend on the number of students attending
D) The number of students who can attend for free

128. A-080 · Medium

A line passes through the points $(2, 11)$ and $(6, 27)$. What is the equation of the line?

A) $y = 4x + 3$
B) $y = 4x + 5$
C) $y = 3x + 5$
D) $y = 3x + 11$

129. A-087 · Medium

A ride-hailing fare in Jeddah is modeled by the linear function $F(m) = 2.25m + 8$, where m is the number of kilometers traveled and $F(m)$ is the fare in SAR. What does the value 2.25 represent in this context?

A) The fare for a trip of 0 kilometers
B) The total fare for a 1-kilometer trip
C) The additional fare, in SAR, for each additional kilometer traveled
D) The maximum possible fare

130. A-088 · Medium

The table shows three values of the linear function g.

x	2	5	8
$g(x)$	13	22	31

Which of the following defines g?

A) $g(x) = 3x + 7$
B) $g(x) = 3x + 4$
C) $g(x) = 2x + 9$
D) $g(x) = 4x + 5$

131. A-097 · Medium

A school fundraiser sells raffle tickets. The relationship between the number of tickets sold, t, and the total funds raised, $R(t)$, in SAR, is $R(t) = 15t + 200$. What is the meaning of the value 15 in this context?

A) The fixed amount raised before any tickets are sold
B) The total number of tickets available
C) The amount raised, in SAR, per ticket sold
D) The number of tickets needed to break even

132. A-098 · Medium

A hiking trail rises steadily in elevation. At a distance of 2 km from the trailhead, the elevation is 340 meters. At a distance of 6 km, the elevation is 620 meters. What is the rate of elevation gain, in meters per kilometer?

A) 46
B) 60
C) 70
D) 80

133. A-099 · Medium · SPR

A car rental company charges a linear fee based on the number of days rented. Renting for 3 days costs SAR 450, and renting for 7 days costs SAR 850. What is the daily rate, in SAR, charged by the company?

Answer:

134. A-125 · Medium

Which of the following is true of $y = 5$?

A) It is not a function.
B) It has a range of 5.
C) It has no domain.
D) It has a slope of 5.

135. A-135 · Medium · SPR

In an arithmetic sequence, the 3rd term is 17 and the 10th term is 52. What is the 20th term?

Answer:

136. A-143 · Medium · SPR

The function $L(w) = 1.7729w + 8.4$ models a quantity L as a function of w. According to the model, by how much does L increase for a 120.5-point increase in w? (Round to the nearest tenth.)

Answer:

137. A-148 · Medium

The linear function f satisfies $f(2) = 7$ and $f(6) = 19$. What is the value of $f(10)$?

A) 22
B) 28
C) 31
D) 34

138. A-204 · Medium · SPR

The linear function f satisfies $f(2) = 9$ and $f(-1) = -3$. What is the value of $f(10)$?

Answer:

139. A-205 · Medium · SPR

The function f is defined by $f(x) = 5x + 1$. The function g is defined by $g(x) = f(2x) - 3$. What is the value of $g(4)$?

Answer:

140. A-211 · Medium · SPR

The linear function f is defined by $f(x) = -2.5x + 20$. The graph of $y = f(x)$ in the xy-plane has x-intercept $(a, 0)$ and y-intercept $(0, b)$. What is the value of $a + b$?

Answer:

141. A-212 · Medium · SPR

The linear function f satisfies $f(4) = 10$ and $f(10) = -2$. For what value of x does $f(x) = 0$?

Answer:

142. A-213 · Medium

The table gives three values of the linear function f.

x	1	3	7
$f(x)$	5	11	23

Which of the following defines f?

A) $f(x) = 6x - 1$
B) $f(x) = 2x + 3$
C) $f(x) = 3x + 5$
D) $f(x) = 3x + 2$

143. A-215 · Medium · SPR

The function f is defined by $f(x) = -3x + 7$. The function g is defined by $g(x) = f(x + 2) - 5$. What is the value of $g(0)$?

Answer:

144. A-216 · Medium · SPR

A town's population grew linearly from $4{,}200$ in 2015 to $5{,}000$ in 2023. According to this model, in what year will the population reach $6{,}000$?

Answer:

145. A-217 · Medium · SPR

A line in the xy-plane has slope $-\frac{3}{4}$ and passes through the point $(8, -1)$. What is the y-coordinate of the y-intercept of the line?

Answer:

146. A-265 · Medium

A tank holds 480 liters of water and drains at a constant rate. The function $N(t) = 480 - 12t$ gives the number of liters in the tank t minutes after draining begins. Which of the following is the best interpretation of the t-intercept of the graph of $y = N(t)$?

A) The tank is completely empty 40 minutes after draining begins.
B) The tank loses 12 liters every minute.
C) The tank is half empty 20 minutes after draining begins.
D) The tank held 480 liters when draining began.

147. A-297 · Medium · SPR

A ride-share charges a base fare of $\$3$ plus $\$0.40$ for every $\frac{1}{5}$ of a mile traveled. What is the fare, in dollars, for a trip of 4.5 miles? (Ignore the dollar sign when entering your answer.)

Answer:

148. A-298 · Medium · SPR

The functions $f(x) = 3x + 2$ and $g(x) = -x + 14$ are graphed in the xy-plane and intersect at the point (a, b). What is the value of $a + b$?

Answer:

149. A-300 · Medium · SPR

A plant's height is recorded each week. The height was 14 cm at the end of week 2 and 26 cm at the end of week 6. If the height grows at a constant rate, in which week will the height first reach 41 cm?

Answer:

150. A-307 · Medium · SPR

The linear function f is defined by $f(x) = \frac{2}{3}x + 4$. For what value of x is $f(x) = x$?

Answer:

151. A-327 · Medium · No calculator needed

The function $V(t) = 1{,}200 - 0.75t$ gives the volume, in liters, of water in a tank t minutes after it starts draining. Which function gives the volume, in liters, h hours after the tank starts draining?

A) $1{,}200 - 0.75(h + 60)$ **B)** $1{,}200 - 45h$

C) $20 - 0.75h$ **D)** $1{,}200 - 0.0125h$

152. A-004 · Hard

For the linear function h, $h(k) = c$ and $h(k+5) = c+15$, where c and k are constants. What is the value of $h(k + 2) - h(k)$?

A) 3 **B)** 6

C) 9 **D)** 15

153. A-005 · Hard · SPR

A contractor building a section of the Jeddah Corniche waterfront walkway charges a fixed inspection fee plus a constant cost per meter of walkway constructed. Building 150 meters costs a total of SAR $97{,}500$, and building 250 meters costs a total of SAR $152{,}500$. Based on this linear relationship, what is the total cost, in SAR, to build 400 meters of walkway?

Answer:

154. A-025 · Hard · SPR

The function g is linear, and $g(a + 3) = g(a) + 15$ for all values of a. If $g(0) = -4$, what is the value of $g(6)$?

Answer:

155. A-034 · Hard

If $f(x) = 4x-3$ and $f(2a) = 3f(a)+7$, what is the value of a?

A) $-\frac{1}{4}$ **B)** $-\frac{1}{2}$

C) $\frac{1}{4}$ **D)** $\frac{1}{2}$

156. A-068 · Hard

A crew at Jeddah Islamic Port has been unloading containers from a ship. The equation below models the approximate number of containers, y, that remains to be unloaded x hours after the crew started working.

$$y = 340 - 28x$$

The graph of this equation in the xy-plane is a line. What is the best interpretation of the x-intercept in this context?

A) The crew will have unloaded all the containers in about 12.1 hours.
B) The crew has been unloading about 12.1 containers per hour.
C) The crew has been unloading about 28 containers per hour.
D) The crew started with 340 containers to unload.

157. A-069 · Hard · SPR

Some values of the linear function g are shown in the table below.

x	$g(x)$
0	-5
3	13
9	49

What is the value of $g(7)$?

Answer:

158. A-081 · Hard · SPR

For the linear function h, $h(3) = 20$ and $h(7) = 44$. What is the value of x for which $h(x) = 62$?

Answer:

159. A-082 · Hard

Two mobile data plans are modeled by linear functions of x, the number of gigabytes used in a month. Plan A costs $A(x) = 8x + 60$ SAR, and Plan B costs $B(x) = 12x + 20$ SAR. For what value of x do the two plans cost the same amount?

A) 5 **B)** 8

C) 10 **D)** 15

160. A-089 · Hard · SPR

For the linear function p, $p(-2) = 5$ and $p(4) = -13$. What is the value of $p(0)$?

Answer:

161. A-100 · Hard

Function p is defined by $p(x) = 4x + 6$. Function q is defined by the table below.

x	0	3	6
$q(x)$	10	25	40

Which statement is true about the rates of change of p and q?

A) p has a greater rate of change than q.
B) q has a greater rate of change than p.
C) p and q have the same rate of change.
D) The rates of change cannot be compared.

162. A-101 · Hard · SPR

For the linear function h, $h(2) = 9$ and the slope of the graph of $y = h(x)$ is 4. What is the value of x for which $h(x) = 33$?

Answer:

163. A-120 · Hard

For a particular number c, the sequence above begins $c,\ 2c+3,\ 3c+6,\ 4c+9,\ \ldots$, and each term thereafter continues the same pattern. What is the value of the 12th term?

A) $11c+30$ **B)** $12c+30$

C) $12c+33$ **D)** $13c+33$

164. A-147 · Hard · SPR

For the linear function f, the table gives $f(5)=52$, $f(6)=78$, $f(7)=104$. The function g is defined by $f(x)=g(x+1)+5x$. What is the value of $g(12)$?

Answer:

165. A-150 · Hard · Calculator

A utility charges a $\$15$ service fee each month. Usage is billed at $\$2.40$ per unit for the first 20 units and $\$3.60$ for each unit beyond 20. A 10% fuel surcharge is added to the usage charge only (not to the service fee). Then 5% tax is applied to the sum of the service fee and the surcharged usage charge. For $u>20$ units, which function gives the total monthly bill $B(u)$, in dollars?

A) $B(u)=3.78u-9.45$ **B)** $B(u)=4.158u-11.97$

C) $B(u)=4.158u-10.395$ **D)** $B(u)=4.158u-12.72$

166. A-153 · Hard · Calculator

A utility computes a customer's monthly water bill B, in dollars, for u units of water in this order: a service fee of $\$18$ plus $\$2.40$ for each unit used; then a one-time credit of $\$12$ is subtracted; then a 5% surcharge is added to the amount at that point; and finally a fixed $\$3.50$ fee is added. Which of the following defines B for u units?

A) $B=2.52u+10.4$ **B)** $B=2.52u+9.975$

C) $B=2.52u+22.4$ **D)** $B=2.52u+9.8$

167. A-162 · Hard · SPR · No calculator needed

The function f is linear. The function g is defined by $g(x)=-2f(x-1)+5$, and the graph of $y=g(x)$ passes through the points $(0,3)$ and $(2,-5)$. The graph of $y=h(x)$ is the graph of $y=f(x)$ translated 2 units to the left and 6 units up. What is the y-coordinate of the y-intercept of the graph of h?

Answer:

168. A-165 · Hard · Calculator

A ride service charges a fixed pickup fee plus a constant rate for each mile traveled. A 5-mile ride costs $\$14.50$ and a 12-mile ride costs $\$29.20$. The service then raises its per-mile rate by 20% and leaves the pickup fee unchanged. Using 1 mile $=1.6$ kilometers, what will a 20-kilometer ride cost after the change?

A) $\$36.30$ **B)** $\$54.40$

C) $\$84.64$ **D)** $\$35.50$

169. A-168 · Hard · SPR · Calculator

Plan P costs $\$25$ per month for up to 4 gigabytes (GB) of data, plus $\$6$ for each additional GB. Plan Q costs $\$14$ per month for up to 2 GB of data, plus $\$0.008$ for each additional megabyte (MB). Charges for additional data are prorated, and 1 GB $=1000$ MB. For what amount of data, in MB, do the two plans cost the same?

Answer:

170. A-169 · Hard · SPR · No calculator needed

The linear function f satisfies $f\left(\frac{2}{3}\right)=\frac{1}{4}$ and $f\left(\frac{8}{3}\right)=\frac{13}{4}$. What is the value of $f\left(\frac{1}{2}+\frac{3}{5}\right)$?

Answer:

171. A-170 · Hard · Calculator

Plan A charges a one-time enrollment fee of $\$85$ plus $\$18.50$ per month. Plan B has no enrollment fee and charges $\$28$ per month. For a whole number of months m, which of the following describes all values of m for which Plan A costs strictly less than Plan B in total?

A) $m\le 8$ **B)** $m\ge 10$

C) $m\ge 8$ **D)** $m\ge 9$

172. A-177 · Hard

A rental company charges $\$30$ for the first day of a rental and $\$22$ for each additional day. A customer who rents for d days, where $d\ge 1$, uses a coupon that takes 10% off the total rental charge. Which function gives the customer's total cost, in dollars?

A) $0.9(22d+8)$ **B)** $22d+8-0.1$

C) $0.9(22d)+8$ **D)** $0.1(22d+8)$

173. A-178 · Hard · SPR

The linear function f satisfies $f(2)=7$ and $f(6)=19$. The function g is defined by $g(x)=f(x)+f(2x)$. What is the value of $g(5)$?

Answer:

174. A-179 · Hard · SPR

The linear function f satisfies $f(3)=5$. For every increase of 4 in x, the value of $f(x)$ decreases by 10. What is the x-intercept of the graph of $y=f(x)$ in the xy-plane? (Enter only the x-coordinate.)

Answer:

175. A-206 · Hard · SPR

The linear functions f and g are defined by $f(x) = 3x - 2$ and $g(x) = -x + 14$. The function h is defined by $h(x) = f(x) + g(x)$. What is the x-coordinate of the x-intercept of the graph of $y = h(x)$?

Answer:

176. A-207 · Hard

A company's profit, in dollars, from selling n items is modeled by $P(n) = 18n - 540$. Which of the following is the best interpretation of the n-intercept of the graph of $y = P(n)$?

A) The company makes a profit of $\$540$ after selling 18 items.
B) The company breaks even when it sells 30 items.
C) The company earns $\$18$ on each item it sells.
D) The company loses $\$540$ when it sells no items, and $\$18$ more for each item.

177. A-208 · Hard · SPR

A garage charges $\$5$ for the first hour and $\$3$ for each additional hour (whole hours only). Mr. Haddad pays $\$41$. For how many hours did he park?

Answer:

178. A-209 · Hard · SPR

Candle A is 12 inches tall and burns down at 0.6 inch per hour. Candle B is 10 inches tall and burns down at 0.4 inch per hour. After how many hours will the two candles have the same height?

Answer:

179. A-210 · Hard

The function $g(m) = 15 - 0.04m$ models the amount of fuel, in gallons, left in a car's tank after the car has driven m miles. Which of the following is the best interpretation of the slope of the graph of $y = g(m)$?

A) The car uses 0.04 gallon of fuel per mile driven.
B) The car travels 0.04 mile on each gallon.
C) The tank holds 15 gallons when full of fuel.
D) The tank holds 0.04 gallon when it is empty.

180. A-214 · Hard · SPR

The function f is defined by $f(x) = mx + 4$, where m is a positive constant. If $f(3) = m^2 - 6$, what is the value of m?

Answer:

181. A-261 · Hard · SPR

The function f is defined by $f(x) = 3x - 7$. For what value of x is $f(x) = 5$?

Answer:

182. A-262 · Hard · SPR

The function g is defined by $g(x) = ax + b$, where a and b are constants and $a > 0$. If $g(g(x)) = 9x + 8$ for all x, what is the value of $g(1)$?

Answer:

183. A-263 · Hard · SPR

The function f satisfies $f(x + 3) = 2x + 1$ for all real numbers x. What is the value of $f(5)$?

Answer:

184. A-264 · Hard · SPR

The function f satisfies $f(2x - 1) = 6x + 4$ for all real numbers x. What is the value of $f(7)$?

Answer:

185. A-266 · Hard · SPR

An employee earns $\$18$ per hour for the first 40 hours worked in a week and 1.5 times that rate for each hour beyond 40. How much, in dollars, does the employee earn in a week in which 46 hours are worked?

Answer:

186. A-267 · Hard · SPR

The formula $F = 1.8C + 32$ converts a temperature of C degrees Celsius to F degrees Fahrenheit. At what temperature, in degrees, are the two scales equal?

Answer:

187. A-268 · Hard · SPR

A machine is worth $\$24{,}000$ when new. Its value decreases linearly, and after 3 years it is worth $\$16{,}500$. After how many years from new will the machine be worth $\$4{,}000$?

Answer:

188. A-269 · Hard · SPR

The functions f and g are defined by $f(x) = 2x - 8$ and $g(x) = -x + k$, where k is a constant. The graphs of $y = f(x)$ and $y = g(x)$ have the same x-intercept. What is the value of k?

Answer:

189. A-270 · Hard · SPR

The functions f and g are defined by $f(x) = 3x + 1$ and $g(x) = ax - 7$, where a is a constant. The function $h(x) = f(x) + g(x)$ has the same value for every x. What is the value of a?

Answer:

190. A-299 · Hard · SPR

The linear function $g(x) = ax + b$ has a slope equal to $\frac{1}{3}$ of the slope of the line through the points $(0, 0)$ and $(3, 6)$. The graph of g passes through $(2, 3)$. What is the value of $g(0)$? (Enter as a fraction or a decimal rounded to the nearest hundredth.)

Answer:

191. A-316 · Hard · SPR · No calculator needed

The function f is linear, $f(3) = k$ and $f(k) = 3$, where k is a constant and $k \neq 3$. If $f(10) = -1$, what is the value of k?

Answer:

192. A-317 · Hard · No calculator needed

The function f is linear, $f(5) + f(-5) = 14$, and $f(2) = 13$. What is the value of $f(-2)$?

A) -13 **B)** 1

C) -1 **D)** 7

193. A-318 · Hard · SPR · No calculator needed

The function f is linear, $f(2) + f(8) = 30$, and $f(11) = 33$. What is the value of $f(0)$?

Answer:

194. A-335 · Hard · SPR · No calculator needed

An oven's temperature is $70°$F when it is switched on at $t = 0$. For $3 \leq t \leq 12$, where t is the time in minutes, the temperature is modeled by $T = a + bt$, where a and b are constants. The model gives $T = 200$ at $t = 3$ and $T = 380$ at $t = 7$. What is the value of a?

Answer:

Systems of two linear equations in two variables (79)

195. A-006 · Medium

If $2x + 3y = 12$ and $x - y = 1$, what is the value of $x + y$?

A) 3 **B)** 4

C) 5 **D)** 6

196. A-007 · Medium

During Riyadh Season, 3 standard tickets and 2 VIP tickets to a concert cost a total of SAR 850. 5 standard tickets and 1 VIP ticket cost a total of SAR 950. What is the price, in SAR, of one VIP ticket?

A) 150 **B)** 175

C) 200 **D)** 225

197. A-019 · Medium

At a juice stand in Jeddah, 2 small cups and 3 large cups of juice cost a total of SAR 39. 4 small cups and 1 large cup cost a total of SAR 33. What is the price, in SAR, of one large cup?

A) 6 **B)** 7

C) 8 **D)** 9

198. A-026 · Medium

If $3x + 2y = 18$ and $x + y = 7$, what is the value of $x - y$?

A) -1 **B)** 1

C) 3 **D)** 7

199. A-043 · Medium

A school orders tables at SAR 400 each and chairs at SAR 100 each for a new STEM lab, purchasing 40 items total for SAR $7{,}000$. How many tables did the school order?

A) 8 **B)** 10

C) 12 **D)** 15

200. A-051 · Medium · SPR

A date-packing facility near Al-Madinah packs dates into 5 kg bags and 20 kg bags. A shipment contains 24 bags in total, weighing 285 kg altogether. How many 20 kg bags are in the shipment?

Answer:

201. A-052 · Medium · SPR

Layla's total daily round-trip commute in Jeddah averages 84 minutes. Her average morning commute is 6 minutes less than her average evening commute. What is her average evening commute, in minutes?

Answer:

202. A-074 · Medium · SPR

$$y = 4x$$

$$3x + y = 21$$

The solution to the given system of equations is (x, y). What is the value of $7x$?

Answer:

203. A-075 · Medium

A cultural heritage exhibition in Jeddah sells two types of tickets: adult tickets for SAR 45 and student tickets for SAR 25. If the exhibition sold 80 tickets for a total of SAR $3{,}000$, how many student tickets were sold?

A) 20 **B)** 30

C) 50 **D)** 60

204. A-078 · Medium

At how many points do the graphs of the equations $y = 3x - 11$ and $y = 12x$ intersect in the xy-plane?

A) 0 **B)** 1

C) 2 **D)** Infinitely many

205. A-090 · Medium

At a school bookstore, 3 notebooks and 2 pens cost SAR 29, while 5 notebooks and 2 pens cost SAR 43. What is the cost, in SAR, of one notebook?

A) 5 **B)** 6

C) 7 **D)** 8

206. A-091 · Medium

If $2x + 3y = 16$ and $x - y = 3$, what is the value of x?

A) 3 **B)** 4

C) 5 **D)** 6

207. A-102 · Medium

A school orders a combination of tablets and laptops for a new digital-learning lab. Tablets cost SAR $1{,}200$ each and laptops cost SAR $3{,}000$ each. The school orders 40 devices in total for SAR $66{,}000$. How many laptops did the school order?

A) 8 **B)** 10

C) 12 **D)** 15

208. A-103 · Medium

If $4x - y = 9$ and $2x + y = 15$, what is the value of x?

A) 2 **B)** 3

C) 4 **D)** 5

209. A-104 · Medium · SPR

At a school play, adult tickets cost SAR 25 and student tickets cost SAR 10. A total of 180 tickets were sold for SAR $3{,}150$. How many adult tickets were sold?

Answer:

210. A-113 · Medium · SPR

Sara trains along the Jeddah Corniche by walking part of the distance at 5 kilometers per hour and jogging the rest at 9 kilometers per hour. She exercises for a total of 2 hours and covers a total distance of 14 kilometers. How many hours did she spend walking?

Answer:

211. A-131 · Medium · SPR

$$3x + 2y = 19$$

$$x + 4y = 13$$

If (x, y) satisfies the system above, what is the value of $4x + 6y$?

Answer:

212. A-133 · Medium · SPR

$$ax + 6y = 15$$

$$4x + 8y = 20$$

In the system above, a is a constant. For what value of a does the system have infinitely many solutions?

Answer:

213. A-137 · Medium

In the given system of equations, k is a constant. If the system has no solution, what is the value of k?

$$\frac{2}{3}x - \frac{1}{2}y = 5$$

$$kx - 3y = 7$$

A) -4 **B)** $\frac{1}{9}$

C) $\frac{14}{15}$ **D)** 4

214. A-138 · Medium · SPR

A technician mixed a solution that is 0.4% sodium chloride by mass with a solution that is 0.1% sodium chloride by mass. The mixture weighs a total of 60 grams and contains a total of 0.12 grams of sodium chloride. How many grams of the 0.4% solution were used?

Answer:

215. A-139 · Medium

A chemist mixes a 2% acid solution with an 8% acid solution to make 40 milliliters of mixture that contains 1.4 milliliters of pure acid. How many milliliters of the 8% solution were used?

A) 10 **B)** 15

C) 25 **D)** 30

216. A-231 · Medium · SPR

$$ax + 4y = 14$$

$$3x + 2y = 7$$

In the system of equations above, a is a constant. If the system has infinitely many solutions, what is the value of a?

Answer:

217. A-234 · Medium · SPR

$$y = 2x + k$$

$$y = -x + 7$$

In the system of equations above, k is a constant. The graphs of the two equations intersect at the point with x-coordinate 2. What is the value of k?

Answer:

218. A-241 · Medium · SPR

A theater sold 150 tickets for a performance. Adult tickets cost $\$12$ and child tickets cost $\$7$. The total ticket revenue was $\$1{,}380$. How many child tickets were sold?

Answer:

219. A-284 · Medium · SPR

A jar contains 35 coins, all nickels and dimes. The total value of the coins is $\$2.60$. How many dimes are in the jar?

Answer:

220. A-303 · Medium · SPR

Three books and two notebooks weigh 1.9 kilograms. Two books and three notebooks weigh 1.6 kilograms. What is the combined weight, in kilograms, of one book and one notebook?

Answer:

221. A-304 · Medium · SPR

The ratio of x to y is 3 to 2, and $x + 2y = 28$. What is the value of x?

Answer:

222. A-305 · Medium · SPR

Car rental company A charges a $\$30$ fee plus $\$0.20$ per mile. Company B charges an $\$18$ fee plus $\$0.35$ per mile. For how many miles driven do the two companies charge the same total?

Answer:

223. A-306 · Medium · SPR

The sum of two numbers is 46. One of the numbers is 6 more than 3 times the other number. What is the larger number?

Answer:

224. A-334 · Medium · No calculator needed

How many solutions does the system $2x + y = 11$, $x - y = 1$, $x + 3y = -13$ have?

A) Exactly one **B)** Exactly three

C) Zero **D)** Exactly two

225. A-008 · Hard

For what value of k does the system of equations below have no solution?

$3x - ky = 12$ $6x - 8y = 9$

A) 2 **B)** 4

C) 6 **D)** 8

226. A-009 · Hard · SPR

If $5x + 4y = 21$ and $3x + 2y = 13$, what is the value of $x + y$?

Answer:

227. A-023 · Hard

For what value of k does the system of equations below have infinitely many solutions?

$2x + ky = 10$ $6x + 12y = 30$

A) 2 **B)** 3

C) 4 **D)** 6

228. A-027 · Hard

For what value of c does the system of equations below have no solution?

$cx + 6y = 9$ $2x + 4y = 5$

A) 1.5 **B)** 3

C) 4 **D)** 6

229. A-035 · Hard · SPR

If $2x + 3y = 16$ and $4x - y = 18$, what is the value of $x + y$?

Answer:

230. A-046 · Hard · SPR

A café blends two types of coffee beans: Type A costs SAR 30 per kilogram and Type B costs SAR 45 per kilogram. The café wants to make 20 kilograms of a blend that costs SAR 36 per kilogram. How many kilograms of Type B should be used?

Answer:

231. A-053 · Hard · SPR

A vending machine in a Dammam school cafeteria stocks 18 juice boxes, each either 200 mL or 350 mL. The total volume of the 200 mL boxes is 1,400 mL more than the total volume of the 350 mL boxes. How many 350 mL boxes are in the machine?

Answer:

232. A-076 · Hard · SPR

$$4x - 6y = 10y + 2$$

$$ty = \frac{1}{2} + 2x$$

In the given system of equations, t is a constant. If the system has no solution, what is the value of t?

Answer:

233. A-077 · Hard · SPR

$$\frac{1}{3}x + \frac{1}{2}y = \frac{1}{4}$$

$$ax + y = c$$

In the given system of equations, a and c are constants. If the system has infinitely many solutions (x, y), what is the value of a?

Answer:

234. A-092 · Hard · SPR

Ahmad invests a total of SAR $12{,}000$ in two accounts. Account A earns 4% annual interest, and Account B earns 7% annual interest. After one year, the total interest earned from both accounts is SAR 690. How much money, in SAR, was invested in Account B?

Answer:

235. A-093 · Hard

For what value of k does the system of equations below have infinitely many solutions?

$$6x + 9y = 21$$

$$kx + 6y = 14$$

A) 3 **B)** 4

C) 6 **D)** 9

236. A-105 · Hard

A jar contains only 25-halala and 50-halala coins, worth a total of SAR 21.50. There are 52 coins in total. How many 50-halala coins are in the jar?

A) 18 **B)** 26

C) 34 **D)** 40

237. A-106 · Hard · SPR

The solution to the system of equations below is (x, y). What is the value of $2x - y$?

$$5x + 2y = 38$$

$$3x - 2y = 10$$

Answer:

238. A-130 · Hard · SPR

$$3x - ky = 7$$

$$12x - 20y = 5$$

In the system above, k is a constant. For what value of k does the system have no solution?

Answer:

239. A-132 · Hard · SPR

$$2x + 5y = 23$$
$$5x + 2y = 26$$

If (x, y) is the solution to the system above, what is the value of $(x - y)^2$?

Answer:

240. A-140 · Hard · SPR

One solution is 0.5% salt by mass and another solution is 0.1% salt by mass. A mixture of the two weighs 80 grams in total and contains 0.2 grams of salt. How many grams of the 0.1% solution are in the mixture?

Answer:

241. A-145 · Hard

The line $4x + y = k$ contains the point (a, b), and the line $2x + k = 3y$ contains the point $(3a, 5b)$, where k is a constant. Which of the following expresses a in terms of b?

A) $\frac{5b}{7}$ **B)** $\frac{7b}{5}$

C) $\frac{14b}{5}$ **D)** $\frac{5b}{14}$

242. A-152 · Hard · No calculator needed

$$3x + 2y = 12$$
$$9x + 6y = 36$$

For the system above, which of the following ordered pairs (x, y), where t is any real number, is a solution to the system?

A) $(2t + 4,\ -3t)$ **B)** $(2t + 4,\ 3t)$

C) $(3t + 4,\ -2t)$ **D)** $(4 - 2t,\ -3t)$

243. A-154 · Hard · SPR · Calculator

A crew spreads fertilizer on two kinds of turf. Type A turf receives 30 grams per square meter, and Type B turf receives 4 kilograms per 100 square meters. In all, the crew treats $70{,}000$ square meters and uses 2.6 metric tons of fertilizer (1 metric ton $= 1000$ kilograms). How many square meters of Type A turf are treated?

Answer:

244. A-155 · Hard · No calculator needed

$$kx + 3y = 12$$
$$4x + (k - 1)y = 12$$

In the system of equations above, k is a constant. For which value of k does the system have no solution?

A) 3 **B)** -4

C) -3 **D)** 4

245. A-156 · Hard · SPR · No calculator needed

$$6x - 2(y - 5) = 2x + 3y + 1$$
$$ax + 20y = b$$

In the system above, a and b are constants. If the system has infinitely many solutions, what is the value of $\frac{b}{a}$?

Answer:

246. A-158 · Hard · SPR · No calculator needed

$$ax + ay = 15$$
$$3ax - 2ay = 20$$

In the system of equations above, a is a nonzero constant. The ordered pair $(2, y)$ is a solution to the system. What is the value of a?

Answer:

247. A-163 · Hard · No calculator needed

$$\frac{x}{6} + \frac{y}{4} = 2$$

Which of the following equations, when paired with the equation above, forms a system of two linear equations in x and y that has no solution?

A) $4x - 6y = 40$ **B)** $4x + 6y = 48$

C) $4x + 6y = 40$ **D)** $6x + 4y = 48$

248. A-164 · Hard · SPR · Calculator

A contractor finishes a 96-square-meter floor using two products. Primer P is applied in two coats, and each coat uses 125 milliliters per square meter. Sealant Q is applied in one coat and covers 5 square meters per liter. Every square meter receives exactly one of the two products, and the job used a total of 21.2 liters. How many square meters did the contractor cover with primer P?

Answer:

249. A-166 · Hard · SPR · No calculator needed

$$3x + ky = 12$$
$$21x + 14y = 6m$$

In the system of equations above, k and m are constants. If the system has infinitely many solutions (x, y), what is the value of km?

Answer:

250. A-182 · Hard · SPR

$$5x + 3y = 19$$

$$2x - y = 3$$

If (x, y) is the solution to the system of equations above, what is the value of $7x + 2y$?

Answer:

251. A-183 · Hard

$$2x + ky = 8$$

$$kx + 8y = 16$$

In the system of equations above, k is a constant. For which value of k does the system have infinitely many solutions?

A) -4 **B)** 2

C) 4 **D)** 8

252. A-230 · Hard

$$3x + ky = 7$$

$$9x + 12y = m$$

In the system of equations above, k and m are constants. For which of the following values of k and m does the system have no solution?

A) $k = 12,\ m = 7$ **B)** $k = 4,\ m = 7$

C) $k = 3,\ m = 21$ **D)** $k = 4,\ m = 21$

253. A-232 · Hard · SPR

$$3x + 5y = 17$$

$$5x + 3y = 23$$

If (x, y) is the solution to the system of equations above, what is the value of $x + y$?

Answer:

254. A-233 · Hard · SPR

$$2x - 3y = 7$$

$$4x + y = 21$$

If (x, y) is the solution to the system of equations above, what is the value of $6x - 2y$?

Answer:

255. A-235 · Hard · SPR

$$ax + 3y = 7$$

$$2x - by = 13$$

In the system of equations above, a and b are constants. If $(2, -1)$ is the solution to the system, what is the value of ab?

Answer:

256. A-236 · Hard · SPR

A chemist mixes x liters of a 12% saline solution with y liters of a 30% saline solution to make 18 liters of a 24% saline solution. How many liters of the 12% solution are used?

Answer:

257. A-237 · Hard · SPR

A boat travels 18 kilometers downstream in 2 hours and returns the same 18 kilometers upstream in 3 hours. What is the speed of the boat in still water, in kilometers per hour? (Enter as a decimal.)

Answer:

258. A-238 · Hard · SPR

Sara is 4 years older than twice Omar's age. In 6 years, the sum of their ages will be 40. How old is Sara now?

Answer:

259. A-239 · Hard

$$4x - 6y = 9$$

One equation of a system of two linear equations is given above. Which of the following could be the second equation so that the system has no solution?

A) $2x - 3y = 10$ **B)** $6x + 4y = 9$

C) $3x - 2y = 9$ **D)** $2x - 3y = \dfrac{9}{2}$

260. A-240 · Hard · SPR

$$\frac{x}{2} + \frac{y}{3} = 5$$

$$\frac{x}{4} - \frac{y}{6} = \frac{1}{2}$$

If (x, y) is the solution to the system of equations above, what is the value of xy?

Answer:

261. A-242 · Hard · SPR

The lines $2x + y = 7$, $x - y = 2$ and $ax + 3y = 12$ all pass through the same point in the xy-plane. What is the value of a?

Answer:

262. A-243 · Hard · SPR

$$kx + 4y = 10$$

$$2x + (k + 2)y = 5$$

In the system of equations above, k is a constant. What is the sum of all values of k for which the system has no solution?

Answer:

263. A-244 · Hard · SPR

The graphs of $y = 2x - 4$ and $y = -x + 8$ and the x-axis bound a triangular region in the xy-plane. What is the area of the triangle?

Answer:

264. A-282 · Hard · SPR

$$x + y = 10$$

$$y + z = 14$$

$$x + z = 12$$

The system of equations above has solution (x, y, z). What is the value of $x + y + z$?

Answer:

265. A-283 · Hard · SPR

The digits of a two-digit positive integer add up to 11. If the digits are reversed, the new number is 27 greater than the original number. What is the original number?

Answer:

266. A-285 · Hard · SPR

A person invests a total of $\$8{,}000$ in two accounts. One earns 5% simple interest per year and the other earns 7% simple interest per year. After one year the total interest is $\$490$. How much, in dollars, was invested at 5%?

Answer:

267. A-286 · Hard · SPR

Two trains start at the same time from stations 360 kilometers apart and travel toward each other on parallel tracks, meeting after 2.5 hours. One train is 20 kilometers per hour faster than the other. What is the speed, in kilometers per hour, of the faster train?

Answer:

268. A-288 · Hard · SPR

A shop mixes nuts that cost $\$12$ per kilogram with nuts that cost $\$18$ per kilogram to make 30 kilograms of a mixture that costs $\$14$ per kilogram. How many kilograms of the $\$18$ nuts are in the mixture?

Answer:

269. A-289 · Hard

$$y = 3x + c$$

$$y = -2x + 8$$

In the system of equations above, c is a constant. For which of the following values of c is the solution (x, y) in the second quadrant (where $x < 0$ and $y > 0$)?

A) $c = -2$ **B)** $c = 4$

C) $c = 12$ **D)** $c = 8$

270. A-302 · Hard · SPR

$$3x + 2y = 20$$

$$5x - ky = 12$$

In the system above, k is a constant. The solution (x, y) to the system has $x = y$. What is the value of k?

Answer:

271. A-328 · Hard · SPR · No calculator needed

In the system $y = 2x + k$ and $y = -x + 10$, k is a constant. For how many integer values of k is the solution (x, y) in Quadrant I (that is, $x > 0$ and $y > 0$)?

Answer:

272. A-332 · Hard · No calculator needed

Consider the system $3x - 4y = 10$ and $-6x + 8y = -20$. Which point lies on the graphs of both equations for every real number t?

A) $\left(t, \dfrac{3t - 10}{4}\right)$ **B)** $\left(t, \dfrac{10 - 3t}{4}\right)$

C) $\left(t, \dfrac{3t + 10}{4}\right)$ **D)** $\left(\dfrac{4t - 10}{3}, t\right)$

273. A-337 · Hard · No calculator needed

The solution to the system $ax + by = 31$ and $bx + ay = 19$, where a and b are constants, is $(4, 1)$. What is the value of $a^2 - b^2$?

A) 16 **B)** 40

C) 10 **D)** 58

Linear inequalities in one or two variables (57)

274. A-014 · Medium

A truck transporting cargo between Jeddah Islamic Port and Riyadh Dry Port can carry at most 5000 kilograms. The truck is already loaded with 60 boxes of textiles, each weighing 25 kilograms. Each additional box of electronics weighs 35 kilograms. What is the maximum number of boxes of electronics the truck can carry in addition to the textiles?

A) 85 **B)** 90

C) 100 **D)** 105

275. A-021 · Medium

A delivery van moving goods between Jeddah and Makkah can carry at most 3600 kilograms. The van is already loaded with 40 boxes of packaged dates, each weighing 45 kilograms. Each additional box of bottled water weighs 60 kilograms. What is the maximum number of boxes of bottled water the van can carry in addition to the dates?

A) 25 **B)** 28

C) 30 **D)** 32

276. A-030 · Medium

If $4x - 7 \leq 21$ and x is a positive integer, what is the greatest possible value of x?

A) 5 **B)** 6

C) 7 **D)** 8

277. A-032 · Medium

If $Q = 24m$, where m is an integer, and Q is greater than 270 and less than 290, what is the value of m?

A) 10 **B)** 11

C) 12 **D)** 13

278. A-040 · Medium

Which of the following describes the solution set of $|x - 4| \leq 9$?

A) $-5 \leq x \leq 13$ **B)** $x \leq -5$ or $x \geq 13$

C) $-13 \leq x \leq 5$ **D)** $x \leq -13$ or $x \geq 5$

279. A-054 · Medium

King Fahd International Stadium has a capacity of 60,000 spectators. At kickoff, 30,000 spectators are already seated, and fans continue entering through the gates at a constant rate of 200 per minute. Let t represent the number of minutes after kickoff. Which inequality can be solved to find the values of t for which the number of spectators in the stadium is at or above capacity?

A) $30{,}000 + 200t \leq 60{,}000$ **B)** $30{,}000 + 200t \geq 60{,}000$

C) $60{,}000 + 200t \geq 30{,}000$ **D)** $200(30{,}000 + t) \geq 60{,}000$

280. A-055 · Medium

A weather station's forecast temperature f, in degrees Celsius, must be within $2°C$ of the actual recorded temperature a for the forecast to be considered accurate. Which inequality represents this condition?

A) $|f - a| \leq 2$ **B)** $|f - a| \geq 2$

C) $|f + a| \leq 2$ **D)** $f - a \leq 2$

281. A-070 · Medium

A logistics company in Dammam needs to hire at least 12 delivery drivers for a new route. The drivers will be part-time, paid 45 SAR per hour, or full-time, paid 70 SAR per hour. The company's budget allows no more than 720 SAR per hour in driver wages. The company must hire at least 4 part-time drivers and at least 2 full-time drivers. Which system of inequalities represents the conditions described, if x is the number of part-time drivers and y is the number of full-time drivers?

A) $45x + 70y \geq 720$
$x + y \leq 12$
$x \geq 4$
$y \geq 2$

B) $45x + 70y \leq 720$
$x + y \geq 12$
$x \geq 4$
$y \geq 2$

C) $45x + 70y \geq 720$
$x + y \geq 12$
$x \leq 4$
$y \leq 2$

D) $45x + 70y \leq 720$
$x + y \leq 12$
$x \leq 4$
$y \leq 2$

282. A-071 · Medium

In Saudi Arabia, the standard width of a car parking space is at least 2.4 meters and no more than 2.9 meters. A mosque recently paved its parking area and wants to determine the number of parking spaces, n, that could be placed perpendicular to a curb that is 87 meters long, based on the standard width of a parking space. Which of the following describes all the possible values of n?

A) $30 \leq n \leq 36$ **B)** $2.4 \leq n \leq 2.9$

C) $24 \leq n \leq 87$ **D)** $30 \leq n \leq 87$

283. A-109 · Medium

Which of the following ordered pairs (x, y) is a solution to the system of inequalities below?

$$y \geq 3x - 4$$

$$y < -2x + 6$$

A) $(1, 2)$ **B)** $(2, 5)$

C) $(0, -5)$ **D)** $(3, -2)$

284. A-123 · Medium

Let a and b be numbers such that $20 < a < 30$ and $45 < b < 60$. Which of the following represents all possible values of $a - b$?

A) $-40 < a - b < -15$ **B)** $-40 < a - b < -25$

C) $-15 < a - b < 15$ **D)** $-25 < a - b < -15$

285. A-126 · Medium

The inequality $-5(x - 1) \leq 3(x + 3)$ is equivalent to

A) $x \geq -\frac{1}{2}$ **B)** $x \leq -\frac{1}{2}$

C) $x \geq \frac{1}{2}$ **D)** $x \leq \frac{1}{2}$

286. A-181 · Medium

$$x + 2y \leq 8$$

$$3x - y > 5$$

$$y \geq 1$$

Which of the following ordered pairs (x, y) is a solution to the system of inequalities above?

A) $(3, 1)$ **B)** $(2, 2)$

C) $(4, 3)$ **D)** $(1, 3)$

287. A-220 · Medium · SPR

$$-4 < 3 - 2x \leq 9$$

How many integers x satisfy the compound inequality above?

Answer:

288. A-222 · Medium · SPR

$$3x - a < 2x + 5$$

In the inequality above, a is a constant. If the solution set of the inequality is all x with $x < 9$, what is the value of a?

Answer:

289. A-223 · Medium · SPR

$$-3x + 7 > k$$

In the inequality above, k is a constant. If the solution set is all x with $x < 2$, what is the value of k?

Answer:

290. A-229 · Medium · SPR

$$5 < 2x - 1 < k$$

In the compound inequality above, k is a constant. If the solution set is all x with $3 < x < 7$, what is the value of k?

Answer:

291. A-273 · Medium

Which of the following describes all values of x that are within 4 units of 9 on the number line?

A) $x \leq 5$ or $x \geq 13$ **B)** $-5 \leq x \leq 13$

C) $4 \leq x \leq 9$ **D)** $5 \leq x \leq 13$

292. A-274 · Medium · SPR

What is the greatest integer x that satisfies $5 - 2x \geq 3x - 10$?

Answer:

293. A-278 · Medium · SPR

What is the sum of all integers x that satisfy both $3x - 7 > 2$ and $2x + 5 \leq 19$?

Answer:

294. A-301 · Medium · SPR

Dana's scores on her first four tests are 78, 90, 82 and 88. To get an average of at least 85 on all five tests, what is the least score she can earn on the fifth test?

Answer:

295. A-310 · Medium · SPR

A rectangular garden has a perimeter of at most 60 meters, and its length is at least twice its width. What is the greatest possible whole-number width of the garden, in meters?

Answer:

296. A-329 · Medium · No calculator needed

In the inequality $kx + 4 < 10$, k is a constant and $k < 0$. Which describes the solution set?

A) $x < \frac{6}{k}$ **B)** $x > \frac{6}{k}$

C) $x > -\frac{6}{k}$ **D)** $x < -\frac{6}{k}$

297. A-341 · Medium · No calculator needed

Which inequality has the solution set $x \leq -3$ or $x \geq 11$?

A) $|x + 4| \geq 7$ **B)** $|x - 4| \leq 7$

C) $|x - 4| \geq 7$ **D)** $|x - 7| \geq 4$

298. A-015 · Hard

Which ordered pair (x, y) satisfies the system of inequalities below?

$y > 3x - 4$ $y \leq -2x + 10$

A) $(1, 2)$ **B)** $(4, 3)$

C) $(0, -5)$ **D)** $(5, 1)$

299. A-037 · Hard

If $20 < a < 35$ and $40 < b < 55$, which of the following represents all possible values of $a - b$?

A) $-35 < a - b < -20$ **B)** $-35 < a - b < -5$

C) $-15 < a - b < -5$ **D)** $-20 < a - b < 5$

300. A-041 · Hard

Which of the following describes the solution set of $|2x + 3| > 7$?

A) $x < -5$ or $x > 2$ **B)** $-5 < x < 2$

C) $x < 2$ **D)** $x > -5$

301. A-045 · Hard

A gym in Jeddah charges a SAR 150 monthly membership fee plus SAR 20 per personal training session. A member wants their total monthly cost to be at least SAR 150 but no more than SAR 350. Which compound inequality represents the possible number of sessions s the member can book in a month?

A) $0 \leq s \leq 10$ **B)** $0 \leq s \leq 200$

C) $7.5 \leq s \leq 17.5$ **D)** $150 \leq s \leq 350$

302. A-056 · Hard · SPR

An STC mobile plan costs SAR 60 per month plus SAR 8 for each additional GB of data used beyond the included allotment, where any portion of an additional GB is billed as a full GB. If Noura's monthly budget for this plan is SAR 100, what is the maximum number of additional whole GB she can use and still stay within budget?

Answer:

303. A-072 · Hard · SPR

A number p is at most 7 less than 4 times the value of q. If the value of q is -3, what is the greatest possible value of p?

Answer:

304. A-073 · Hard

$$y < -6x + 9$$

Which point (x, y) is a solution to the given inequality in the xy-plane?

A) $(-2, 21)$ **B)** $(0, 10)$

C) $(1, 2)$ **D)** $(3, -8)$

305. A-110 · Hard

Which table below shows only points (x, y) that satisfy the system of inequalities?

$$y > -x + 2$$

$$y \leq 3x - 1$$

A)

x	2	4
y	3	6

B)

x	2	0
y	3	1

C)

x	1	4
y	-3	6

D)

x	2	4
y	3	15

306. A-111 · Hard

A charity bazaar in Jeddah is preparing gift baskets for a Vision 2030 community outreach event. A small basket costs SAR 8 to prepare and a large basket costs SAR 20. The organizers have a materials budget of SAR $2{,}500$ and must prepare at least 150 baskets in total. What is the maximum number of large baskets the organizers can prepare while meeting both requirements?

A) 100 **B)** 104

C) 108 **D)** 125

307. A-114 · Hard · SPR

In the xy-plane, the point $(4, k)$ satisfies the system of inequalities below, where k is the least possible integer value.

$$y > 2x - 9$$

$$y \geq -x + 6$$

What is the value of k?

Answer:

308. A-149 · Hard · SPR · Calculator

A science department orders microscopes at $\$240$ each and slide kits at $\$15$ each. The order must include at least 6 slide kits for every microscope, the total cost may not exceed $\$2{,}500$, and the storage cabinet holds at most 46 items (microscopes and kits combined). What is the greatest number of microscopes the department can order?

Answer:

309. A-171 · Hard · SPR · Desmos

$$x \geq 0, \quad y \geq 0, \quad 2x+y \leq 10, \quad x+3y \leq 15, \quad 3x+2y > k$$

In the system of inequalities above, k is a constant. What is the least value of k for which the system has no solution (x, y)?

Answer:

310. A-180 · Hard · SPR

At a school fair, ride tickets cost $\$3$ each and food coupons cost $\$5$ each. Noura buys at least 3 food coupons, buys at least twice as many ride tickets as food coupons, and spends at most $\$60$ in total. What is the greatest total number of tickets and coupons she can buy?

Answer:

311. A-218 · Hard · SPR

$$x \geq 0, \quad y \geq 0, \quad 2x + y \leq 10, \quad x + 2y \leq 8$$

For the ordered pairs (x, y) that satisfy the system of inequalities above, what is the greatest possible value of $3x + 4y$?

Answer:

312. A-219 · Hard · SPR

A phone plan costs $\$20$ per month plus $\$0.05$ per minute of calls. An 8% tax is applied to the whole monthly bill. A customer wants the bill, including tax, to be at most $\$50$. What is the greatest whole number of minutes the customer can use in a month?

Answer:

313. A-221 · Hard

The system of inequalities $y > 2x + 1$ and $y < -x + 9$ is graphed in the xy-plane. For which of the following sets of values of k is the point $(k, 5)$ a solution to the system?

A) $2 < k < 4$ **B)** $k > 2$

C) $k < 2$ **D)** $k < 4$

314. A-224 · Hard · SPR

Gym A charges a monthly fee of $\$35$ plus $\$4$ per class. Gym B charges a monthly fee of $\$15$ plus $\$9$ per class. What is the least whole number of classes in a month for which Gym A costs less than Gym B?

Answer:

315. A-225 · Hard · SPR

How many ordered pairs (x, y) of positive integers satisfy $x + 2y \leq 9$?

Answer:

316. A-226 · Hard · SPR

A store takes 20% off the price p of a jacket, and then a customer uses a $\$5$ coupon. The customer pays less than $\$60$. What is the greatest whole-dollar value of p?

Answer:

317. A-227 · Hard · SPR

In the xy-plane, the graph of $3x + 2y \leq 18$ together with $x \geq 0$ and $y \geq 0$ is a triangular region. What is the area of this region?

Answer:

318. A-228 · Hard · SPR

A shipping container can hold at most 24 tons. Crates weigh either 1.5 tons or 2 tons. A shipment includes at least 4 crates of the 2-ton type. What is the greatest number of 1.5-ton crates the shipment can include?

Answer:

319. A-275 · Hard · SPR

How many integers x satisfy $|2x - 6| < 8$?

Answer:

320. A-276 · Hard · SPR

$$y \leq -x + 6$$

$$y \geq x - 2$$

$$x \geq 0$$

The system of inequalities above is graphed in the xy-plane. What is the area of the region of solutions?

Answer:

321. A-277 · Hard · SPR

A bakery makes cakes, which earn a profit of $\$10$ each, and pies, which earn a profit of $\$8$ each. A cake needs 3 hours of oven time and a pie needs 2 hours. The oven is available for at most 24 hours, and the bakery can make at most 8 pies. If the bakery makes only whole cakes and pies, what is the greatest possible total profit, in dollars?

Answer:

322. A-279 · Hard

In the xy-plane, the point $(2, 3)$ is a solution to the inequality $y > ax - 1$, where a is a constant. Which of the following could be the value of a?

A) 4 **B)** 1

C) 2 **D)** 3

323. A-280 · Hard · SPR

What is the least positive integer x that satisfies $|3x - 1| \geq 11$?

Answer:

324. A-281 · Hard · SPR

$$\frac{x}{3} - 1 < \frac{x + 2}{4}$$

What is the greatest integer x that satisfies the inequality above?

Answer:

325. A-314 · Hard

Which of the following sets of points (x, y) contains only solutions to the inequality $y < 2x - 3$?

A) $(0, -4),\ (1, -2),\ (2, 0)$ **B)** $(0, -3),\ (1, -2),\ (2, 0)$

C) $(1, -3),\ (2, 0),\ (3, 4)$ **D)** $(0, -5),\ (1, -1),\ (2, 2)$

326. A-315 · Hard · SPR

A freelancer keeps 85% of every invoice after fees. She invoices $\$40$ plus $\$12$ for each hour worked, and she wants to keep at least $\$300$ in total. What is the least whole number of hours she must work?

Answer:

327. A-321 · Hard · No calculator needed

The system $2y - x > 4$ and $y < 5$ is graphed in the xy-plane. Which inequality describes all possible x-coordinates of points in the solution region?

A) $x > 6$ **B)** $x < 14$

C) $x < 6$ **D)** $x < -4$

328. A-322 · Hard · No calculator needed

For real numbers x and y, $x + y < 12$ and $x - y > 4$. Which of the following must be true?

A) $y > 4$ **B)** $x > 8$

C) $y < 4$ **D)** $x < 8$

329. A-323 · Hard · SPR · No calculator needed

If $-3 \leq x \leq 2$ and $-5 \leq y \leq 1$, what is the greatest possible value of xy?

Answer:

330. A-342 · Hard · No calculator needed

For how many integer values of k does the point (k, k) satisfy both $y \geq 2x - 5$ and $y > -x + 1$?

A) 5 **B)** 4

C) Infinitely many **D)** 6

Advanced Math (332)

Equivalent expressions (77)

331. AM-011 · Medium

Which of the following is equivalent to $\frac{x^2-9}{x-3}$, for $x \neq 3$?

A) $x+3$ **B)** $x-3$

C) x^2-3 **D)** $x+9$

332. AM-015 · Medium

Which of the following is equivalent to $\frac{x^2-25}{x+5}$, for $x \neq -5$?

A) $x-5$ **B)** $x+5$

C) x^2-5 **D)** $x-25$

333. AM-025 · Medium

If a and b are positive numbers such that $a^3 = b^2$, which of the following is equivalent to $b\sqrt{a}$?

A) $b^{2/3}$ **B)** $b^{4/3}$

C) b^2 **D)** b^3

334. AM-038 · Medium

What is the quotient when $2x^3-3x^2-11x+6$ is divided by $x-3$ (the division has no remainder)?

A) $2x^2+3x-2$ **B)** $2x^2-3x-2$

C) $2x^2+3x+2$ **D)** $2x^2-9x+16$

335. AM-055 · Medium

Which expression is equivalent to $4(2x+7)-5x$?

A) $3x+28$ **B)** $3x+2$

C) $13x+7$ **D)** $13x+28$

336. AM-056 · Medium

Which expression is equivalent to $(5x^2+3x-9)-(2x^2-7x+4)$?

A) $3x^2+10x-13$ **B)** $3x^2-4x-13$

C) $7x^2-4x-5$ **D)** $7x^2+10x-5$

337. AM-057 · Medium

An urban planner in Dammam models the area of a rectangular section of a park as $3x^2+ax$ square meters, where a is a constant. This expression is equivalent to $x(3x+11)$. What is the value of a?

A) 3 **B)** 8

C) 11 **D)** 14

338. AM-059 · Medium · SPR

The sum of $-3x^2+5x+18$ and $6x^2-2x-9$ can be written in the form ax^2+bx+c, where a, b, and c are constants. What is the value of $a+b+c$?

Answer:

339. AM-078 · Medium

Which of the following is equivalent to $\frac{2x^2+8x}{2x}$, for $x \neq 0$?

A) $x+4$ **B)** $x+8$

C) $2x+4$ **D)** x^2+4

340. AM-082 · Medium · SPR

If $x > 0$ and $\sqrt{x\sqrt{x}} = x^a$, what is the value of a?

Answer:

341. AM-083 · Medium · SPR

If $a^{b/5} = 32$ for positive integers a and b, where a is a prime number, what is the value of b?

Answer:

342. AM-085 · Medium

Which of the following is a factor of the polynomial $(x^2-9)(x+7)$?

A) $x-2$ **B)** $x+3$

C) $x-4$ **D)** $x+5$

343. AM-095 · Medium

If $k = m^2n^2$, k is not 0, and the values of m and n are both tripled, then the value of k increases by which of the following factors?

A) 3 **B)** 9

C) 27 **D)** 81

344. AM-100 · Medium · SPR

In the equation $\sqrt{200}+\sqrt{200} = m\sqrt{n}$, m and n are integers such that $m > n$. What is the value of m?

Answer:

345. AM-103 · Medium · SPR

The value of y decreased by 5 is directly proportional to the value of x increased by 3. If $y = 7$ when $x = 1$, what is the value of x when $y = 19$?

Answer:

346. AM-111 · Medium

The expression $\dfrac{11p^2 + 11q^2}{9p^2 + 30p + 9}$ is undefined when

A) $p = -\dfrac{1}{3}$ or $p = -3$

B) $p = 0$ or $q = 0$

C) $q = -\dfrac{1}{3}$ or $q = -3$

D) $p = -3$ or $q = -\dfrac{1}{3}$

347. AM-117 · Medium

For $x > 0$, which expression is equivalent to $\sqrt{(9x^2)^3 \cdot x^6}$?

A) $27x^6$

B) $27x^9$

C) $81x^6$

D) $3x^6$

348. AM-119 · Medium · SPR

If $\left(\sqrt{p}\right)^6 = q^{-3}$ and $q = \dfrac{1}{4}$, what is the value of p?

Answer:

349. AM-135 · Medium · SPR

$\dfrac{\sqrt[3]{x^7}}{\sqrt{x}} = x^{a/b}$ for $x > 1$, where a and b are positive integers with no common factor. What is the value of $\dfrac{a}{b}$?

Answer:

350. AM-137 · Medium · SPR

If $x + \dfrac{1}{x} = 5$, what is the value of $x^2 + \dfrac{1}{x^2}$?

Answer:

351. AM-139 · Medium

$\dfrac{2}{x-2} + \dfrac{3}{x+5} = \dfrac{rx+t}{(x-2)(x+5)}$ The equation above is true for all $x \neq 2, -5$, where r and t are constants. What is the value of rt?

A) -20

B) 15

C) 20

D) 60

352. AM-143 · Medium · SPR

$\dfrac{\sqrt{x^5}}{\sqrt[3]{x^4}} = x^{a/b}$ for $x > 1$, where a and b are positive integers with no common factor. What is the value of $\dfrac{a}{b}$?

Answer:

353. AM-331 · Medium · No calculator needed

If $2a - 3b = 7$, what is the value of $\dfrac{4^a}{8^b}$?

A) $\dfrac{1}{128}$

B) 2^{14}

C) 7

D) 128

354. AM-012 · Hard

The expression $\dfrac{3}{x-4} - \dfrac{2}{x+1}$ is equivalent to which of the following, for $x \neq 4$ and $x \neq -1$?

A) $\dfrac{x+11}{x^2-3x-4}$

B) $\dfrac{x-11}{x^2-3x-4}$

C) $\dfrac{5x-5}{x^2-3x-4}$

D) $\dfrac{x+11}{x^2+3x-4}$

355. AM-023 · Hard

The expression $\dfrac{5}{x+2} + \dfrac{3}{x-1}$ is equivalent to which of the following, for $x \neq -2$ and $x \neq 1$?

A) $\dfrac{8x+1}{x^2+x-2}$

B) $\dfrac{8x-1}{x^2+x-2}$

C) $\dfrac{2x+11}{x^2+x-2}$

D) $\dfrac{8x+1}{x^2-x-2}$

356. AM-024 · Hard

The operation $\odot$ is defined for all real numbers $x \neq 1$ by $x^{\odot} = \dfrac{x+3}{x-1}$. Which of the following is equivalent to $x^{\odot} - 1$?

A) $\dfrac{x+2}{x-1}$

B) $\dfrac{4}{x-1}$

C) $\dfrac{2x+4}{x-1}$

D) $\dfrac{2}{x-1}$

357. AM-043 · Hard

If $a(x-2)(x+6) = a(x-h)^2 + k$ for all values of x, where $a \neq 0$, what is k in terms of a?

A) $-16a$

B) $-12a$

C) $4a$

D) $16a$

358. AM-058 · Hard

Which expression is equivalent to $x^4 - 2x^2 - 15$?

A) $(x^2-5)(x^2+3)$

B) $(x^2+5)(x^2-3)$

C) $(x^2-15)(x^2+1)$

D) $(x^2-1)(x^2+15)$

359. AM-081 · Hard · SPR

If $\dfrac{x^{a^2}}{x^{b^2}} = x^{20}$, $x > 1$, and $a + b = 4$, what is the value of $a - b$?

Answer:

360. AM-089 · Hard · SPR

If $x > 0$ and $\sqrt{x} = x\sqrt{x} = x^2 = \sqrt{x^2}$, what is the value of x?

Answer:

361. AM-090 · Hard

If $s^2 = t^{-1}$ and $t^{1/3} = u^{-1/2}$, then what is the value of s in terms of u, for $s > 0$?

A) $u^{-3/4}$ **B)** $u^{-1/2}$

C) $u^{1/2}$ **D)** $u^{3/4}$

362. AM-097 · Hard

Let $\langle x \rangle$ be defined by $\langle x \rangle = \dfrac{x+5}{x-2}$ for any x such that $x \neq 2$. Which of the following is equivalent to $\langle x \rangle - 1$?

A) $\dfrac{x+3}{x-2}$ **B)** $\dfrac{7}{x-2}$

C) $\dfrac{2x+3}{x-2}$ **D)** $\dfrac{7}{x-2} - 1$

363. AM-098 · Hard

Let a and b be numbers such that $a^4 = b^3$. Which of the following is equivalent to $a\sqrt{b}$?

A) $b^{5/4}$ **B)** $b^{7/6}$

C) $b^{4/3}$ **D)** $b^{3/2}$

364. AM-106 · Hard

What is the value of $\left(\dfrac{3x}{y}\right)\left(\dfrac{x^{-1}}{(3y)^{-1}}\right)$?

A) 1 **B)** 3

C) 6 **D)** 9

365. AM-109 · Hard · SPR

Find the value of $\dfrac{\sqrt{c+7}}{c} \cdot c\sqrt{c}$ if $c = 9$.

Answer:

366. AM-110 · Hard · SPR

Evaluate $\left(\dfrac{5}{a^2} + \dfrac{4}{3a}\right)^a$ for $a = -3$.

Answer:

367. AM-115 · Hard · SPR

For $x \neq -3$ and $x \neq 2$, the expression $\dfrac{x^2 - 2x - 15}{x^2 + x - 6}$ is equivalent to $\dfrac{x+a}{x+b}$, where a and b are constants. What is the value of $a - b$?

Answer:

368. AM-118 · Hard · SPR

If $a^{\frac{3}{4}} = 5$ and $a > 0$, what is the value of $a^{-\frac{3}{2}}$?

Answer:

369. AM-127 · Hard

If $a^2 + b^2 = 4ab$, where $a \neq b$ and $b \neq 0$, what is the value of $\dfrac{a+b}{a-b}$?

A) 2 **B)** 3

C) 6 **D)** $\pm\sqrt{3}$

370. AM-131 · Hard · SPR

The equation $\dfrac{ax+b}{x+2} = 3 - \dfrac{11}{x+2}$ is true for all values of x not equal to -2, where a and b are constants. What is the value of ab?

Answer:

371. AM-133 · Hard · SPR

The expression $10x^4 + 33x^2 + 20$ can be rewritten as $(5x^2 + a)(2x^2 + b)$, where a and b are positive integers, or as $(5x^2 + c)(2x^2 + d)$, where c and d are positive nonintegers. What is the value of $a + d$?

Answer:

372. AM-149 · Hard

$\dfrac{40x+100}{2x^2-5x-25} + \dfrac{x^2y-5xy}{4xy}$ Which of the following is equivalent to the expression above for $x > 5$, $y > 0$?

A) 5 **B)** $\dfrac{x-1}{x+15}$

C) $\dfrac{x^2-10x+105}{4(x-5)}$ **D)** $\dfrac{x^2-10x+105}{4(2x+5)}$

373. AM-167 · Hard · SPR · No calculator needed

For real numbers p and q with $p^2+4q^2 = 72$, the product $(2x+p)(x+q)$ is equivalent to $2x^2+bx+c$, where b and c are constants. What is the greatest possible value of c?

Answer:

374. AM-170 · Hard · SPR · No calculator needed

For constants p, q and c, the equation

$$(2x-3)(x^2+px+q) = 2x^3 - 5x^2 + cx + 9$$

is true for all real values of x. What is the value of c?

Answer:

375. **AM-172** · Hard · No calculator needed

For a positive real number x, $\frac{\sqrt[3]{x} \cdot \sqrt[4]{x^3}}{\sqrt[6]{x}} = \sqrt[3]{k}$, where $k = 2^{11}$. Which of the following is equal to x?

A) 2^4 **B)** $2^{\frac{11}{3}}$

C) $2^{\frac{121}{36}}$ **D)** $2^{\frac{44}{3}}$

376. **AM-175** · Hard · No calculator needed

For $x > 0$ and any positive integer n, which of the following is equivalent to $\frac{\left(x^{2n+1}\right)^3 + 4x^{6n+3} - 2x\left(x^{3n+1}\right)^2}{x^{6n+1}}$?

A) $5x^2$ **B)** $3x^4$

C) $3x^{12n+4}$ **D)** $3x^2$

377. **AM-176** · Hard · SPR · No calculator needed

For real numbers $k \neq 3$ and $x \neq 0$:

$$\frac{4(k-3)}{x(k-3)} + \frac{2}{x} - \frac{3}{4} = \frac{3}{8}$$

What is the value of x?

Answer:

378. **AM-178** · Hard · Calculator

The function f is defined by $f(x) = \frac{9^x \cdot 12^{x+1}}{8^x}$. Which of the following expressions is equivalent to $f(x)$ and displays the value of $f(2)$ as a coefficient?

A) $\frac{729}{4}\left(\frac{27}{2}\right)^{x-2}$

B) $2187\left(\frac{27}{2}\right)^{x-2}$

C) $2187\left(\frac{27}{2}\right)^{x+2}$

D) $162\left(\frac{27}{2}\right)^{x-1}$

379. **AM-182** · Hard · No calculator needed

For integers a, b, c, and d, $(ax+b)(cx+d) = 6x^2+kx+5$ for all values of x, where k is a constant. Which of the following must be true?

I. k is odd

II. When k is divided by 4, the remainder is 1

III. k is not divisible by 5

A) II and III only **B)** I, II, and III

C) I only **D)** I and III only

380. **AM-193** · Hard · SPR · No calculator needed

For all real values of x,

$$\frac{6^{2x+3}}{2^{2x-1}} = k \cdot 9^x$$

where k is a constant. What is the value of k?

Answer:

381. **AM-194** · Hard · No calculator needed

$$(x+p)(x+q) = x^2 + bx + c$$

In the equation above, p and q are real constants and b and c are integers. Which of the following must be true?

I. $p+q$ is an integer

II. p^2+q^2 is an integer

III. $\frac{p^2+q^2}{2}$ is an integer

A) I only **B)** I and II only

C) II and III only **D)** I, II, and III

382. **AM-195** · Hard · SPR · No calculator needed

For all $x > 1$,

$$\frac{a}{x-1} - \frac{bx+c}{x^2+x+1} = \frac{3x^2+8x+4}{x^3-1}$$

where a, b, and c are constants. What is the value of c?

Answer:

383. **AM-197** · Hard · No calculator needed

For $y \neq 0$ and $y \neq \frac{1}{3}$, which of the following is equivalent to

$$\frac{\frac{1}{y^3} - 27}{\frac{1}{y} - 3}\ ?$$

A) $\frac{1}{y^2} - \frac{3}{y} + 9$ **B)** $\frac{1}{y^2} + 9$

C) $\frac{1}{y^2} + \frac{6}{y} + 9$ **D)** $\frac{1}{y^2} + \frac{3}{y} + 9$

384. **AM-198** · Hard · SPR · No calculator needed

The polynomial $p(x) = x^3 - 9x^2 + kx - 12$ has a positive integer constant k. For some positive integer n, $x-n$ is a factor of $p(x)$. What is the sum of all possible values of k?

Answer:

385. **AM-203** · Hard · Calculator

$$P = \frac{a}{\sqrt[5]{w^2}}$$

The formula above relates the positive quantities P and w, where a is a positive constant. Which of the following expresses w in terms of P and a?

A) $w = \left(\frac{P}{a}\right)^{2/5}$

B) $w = \left(\frac{a}{P}\right)^{5/2}$

C) $w = \left(\frac{P}{a}\right)^{5/2}$

D) $w = \left(\frac{a}{P}\right)^{2/5}$

386. **AM-207** · Hard · Calculator

For all real numbers x,

$$8^x \cdot 4^{x/3} \cdot \left(\sqrt{2}\right)^x = b^x,$$

where b is a positive constant. Which of the following is equal to 2?

A) b^{25} **B)** $b^{-6/25}$

C) $b^{6/25}$ **D)** $b^{25/6}$

387. **AM-212** · Hard · No calculator needed

$$\frac{2}{a} - \frac{3}{b} = \frac{5}{c}$$

In the equation above, a, b and c are nonzero constants and $2c \neq 5a$. Which of the following expresses b in terms of a and c?

A) $b = \frac{3ac}{5a - 2c}$ **B)** $b = \frac{2c - 5a}{3ac}$

C) $b = \frac{3ac}{2c + 5a}$ **D)** $b = \frac{3ac}{2c - 5a}$

388. **AM-215** · Hard · SPR · No calculator needed

The polynomial $p(x) = x^3 + bx^2 + cx + 12$ has integer coefficients b and c, and $x + 3$ is a factor of $p(x)$. If p has exactly one real zero, what is the greatest possible value of c?

Answer:

389. **AM-217** · Hard · SPR · No calculator needed

The polynomial $6x^4 + kx^2 + 24$, where k is a constant, can be written as $(2x^2 + a)(3x^2 + b)$, where a and b are integers. What is the greatest possible value of k?

Answer:

390. **AM-218** · Hard · No calculator needed

For a real number a with $|a| \neq 1$,

$$\frac{1}{1-a} + \frac{1}{1+a} + \frac{2}{1+a^2} + \frac{4}{1+a^4} = 10.$$

What is the value of a^8?

A) $\frac{5}{4}$ **B)** $\frac{9}{5}$

C) $\frac{1}{5}$ **D)** $\frac{4}{5}$

391. **AM-224** · Hard · SPR · No calculator needed

For real numbers a and b, $a^2 - b^2 = 36$. What is the value of $(3a + b)^2 - (a + 3b)^2$?

Answer:

392. **AM-225** · Hard · No calculator needed

If $2x + 3y = 8$, which of the following is equivalent to $2x^2 + 3xy - 4x - 6y$?

A) $8x - 4y$ **B)** $8x - 16$

C) $8x - 8$ **D)** $16 - 8x$

393. **AM-226** · Hard · SPR · No calculator needed

Let t be a positive real number with $t \neq 1$. Define $m = \sqrt[3]{t^2}$ and $n = \sqrt{t}$. If

$$\frac{m^p \cdot n^3}{\sqrt[6]{t}} = t^5,$$

what is the value of p?

Answer:

394. **AM-229** · Hard · No calculator needed

The expression $\frac{(2x-3)^4(x+2) - 7}{5}$ is equivalent to $ax^5 + bx^4 + cx^3 + dx^2 + ex + f$, where a, b, c, d, e and f are constants. What is the value of $a+b+c+d+e+f$?

A) 31 **B)** $-\frac{4}{5}$

C) $\frac{3}{5}$ **D)** -4

395. **AM-235** · Hard · No calculator needed

The polynomial $12x^4 - 84x^2 + 144$ can be written in the form $(ax^2 + b)(cx^2 + d)$, where a, b, c and d are integers with $a > 0$ and $c > 0$. What is the least possible value of $a + b + c + d$?

A) -27 **B)** -22

C) -18 **D)** -38

396. **AM-237** · Hard · Calculator

The function f is defined by $f(x) = \frac{6^{x+1} \cdot 4^x}{3^{x-2}}$. It can be written in the form $f(x) = a \cdot b^{x-1}$, where a and b are positive constants. What is the value of $a + b$?

A) 62 **B)** 440

C) 456 **D)** 3456

397. AM-239 · Hard · SPR · No calculator needed

For a positive real number m, let $a = \sqrt[3]{m}$ and $b = \sqrt[4]{m}$. If $ab = 128$, what is the value of ab^2?

Answer:

398. AM-243 · Hard · SPR · No calculator needed

For all real numbers x, the function g is defined by $g(x) = \dfrac{9^{x+2} \cdot 10^{x+1}}{6^{x-1}}$. The function can be written in the form $g(x) = k \cdot 15^{x+1}$, where k is a constant. What is the value of k?

Answer:

399. AM-248 · Hard · No calculator needed

Positive real numbers a and b, both greater than 1, satisfy $a^3 = b^2$. If $\dfrac{a^9}{\sqrt[3]{b}} = b^k$, what is the value of k?

A) $\dfrac{17}{3}$ **B)** 6

C) $\dfrac{19}{3}$ **D)** $\dfrac{17}{2}$

400. AM-250 · Hard · No calculator needed

Positive real numbers x and y satisfy $x \neq y$, $x + y = 15$ and $xy = 36$. What is the value of

$$\frac{\dfrac{1}{x} - \dfrac{1}{y}}{\dfrac{1}{x^2} - \dfrac{1}{y^2}} \; ?$$

A) 1 **B)** 4

C) $\dfrac{12}{5}$ **D)** $\dfrac{5}{12}$

401. AM-273 · Hard · SPR

$\dfrac{7+i}{1+i} = a + bi$, where a and b are real numbers and $i = \sqrt{-1}$. What is the value of $a - b$?

Answer:

402. AM-278 · Hard · SPR

The polynomial $x^3 - 9x^2 + bx - c$ has zeros 2, 3 and a third zero. What is the value of $b + c$?

Answer:

403. AM-282 · Hard · SPR

If $x + y = 5$ and $xy = 6$, what is the value of $x^3 + y^3$?

Answer:

404. AM-283 · Hard · SPR

If $(x + y)^2 = 49$ and $x^2 + y^2 = 29$, what is the value of $(x - y)^2$?

Answer:

405. AM-287 · Hard · SPR

$(2 + 3i)(4 - i) - (1 + i)^2 = a + bi$, where a and b are real numbers and $i = \sqrt{-1}$. What is the value of ab?

Answer:

406. AM-312 · Hard · No calculator needed

How many real solutions does the equation $\sqrt{x^2 - 10x + 25} = 5 - x$ have?

A) Exactly two **B)** Zero

C) Infinitely many **D)** Exactly one

407. AM-314 · Hard · No calculator needed

The expression $\dfrac{6x^2 + 7x - 5}{2x + 3}$ is equivalent to $Ax + B + \dfrac{C}{2x + 3}$, where A, B and C are constants. What is the value of $A + B + C$?

A) -4 **B)** 2

C) 4 **D)** 0

Nonlinear equations in one variable and systems of equations in two variables (96)

408. AM-009 · Medium

As part of a heritage-tourism marketing campaign in AlUla, aligned with Vision 2030, the number of visitors (in thousands) x weeks after the campaign begins is modeled by $v = x^2 - 2x - 3$. A separate baseline estimate predicts $v = -3$ for the same period. For how many nonnegative values of x are the two estimates equal?

A) 0 **B)** 1

C) 2 **D)** 3

409. AM-010 · Medium · SPR

What is the positive solution to the equation $x^2 - 5x - 24 = 0$?

Answer:

410. AM-017 · Medium

What are the solutions to the equation $2x^2 - 5x - 3 = 0$?

A) $x = -\frac{1}{2}$ and $x = 3$

B) $x = \frac{1}{2}$ and $x = -3$

C) $x = -\frac{3}{2}$ and $x = 1$

D) $x = \frac{3}{2}$ and $x = -1$

411. AM-021 · Medium

If x and y are positive integers such that $x^2 - y^2 = 45$ and $x - y = 5$, what is the value of $x + y$?

A) 7

B) 9

C) 11

D) 13

412. AM-029 · Medium

An integer n is subtracted from its square. Which of the following could NOT be the result?

A) 0

B) An even integer

C) An odd integer

D) A positive integer

413. AM-030 · Medium

For which value of k does the equation $x^2 + 6x + k = 0$ have exactly one real solution?

A) 3

B) 6

C) 9

D) 12

414. AM-034 · Medium

What is the solution set of the equation $\frac{x}{x-3} = \frac{3}{x-3} + 2$?

A) $x = 3$ only

B) $x = 9$ only

C) $x = 3$ and $x = 9$

D) There is no solution.

415. AM-050 · Medium

Which inequality is equivalent to $8x - 12y > 20$?

A) $2x - 3y > 5$

B) $3x - 2y > 5$

C) $4x - 3y > 5$

D) $2y - 3x > 5$

416. AM-051 · Medium

How many distinct real solutions does the equation $4x^2 - 20x + 25 = 0$ have?

A) Exactly one

B) Exactly two

C) Infinitely many

D) Zero

417. AM-052 · Medium

In the system of equations below, x and y satisfy both equations.

$$y = 5x + 2$$

$$y = 5x^2 + 2$$

Which ordered pair (x, y) is a solution to this system?

A) $(0, 0)$

B) $(0, 2)$

C) $(1, 5)$

D) $(1, 7)$

418. AM-053 · Medium · SPR

$7|x| = 63$ What is the positive solution to the given equation?

Answer:

419. AM-068 · Medium

What is the sum of the y-coordinates of the solutions to the system below?

$$y = x^2 - 4x$$

$$y = 2x - 5$$

A) -2

B) 0

C) 2

D) 4

420. AM-069 · Medium · SPR

What is the solution to $\sqrt{2x + 7} = 5$?

Answer:

421. AM-075 · Medium

The height of a triangle is 4 cm more than its base. If the area of the triangle is 30 cm^2, what is the length of the base, in cm?

A) 4

B) 5

C) 6

D) 8

422. AM-076 · Medium · SPR

What is the solution to the equation $\frac{x+5}{x-1} = 3$, for $x \neq 1$?

Answer:

423. AM-104 · Medium · SPR

The curve $y = \frac{x^2}{3}$ and the line $y = \frac{2x}{3}$ intersect at the origin and at the point (a, b). What is the value of b?

Answer:

424. AM-107 · Medium

For what values of x is the expression $2x^2 - 2x - 24$ equal to 0?

A) $x = -4,\ x = 3$

B) $x = 4,\ x = -3$

C) $x = 4,\ x = 3$

D) $x = -4,\ x = -3$

425. **AM-113** · Medium · SPR

If $(x-3)(x+7)=-25$, what is the value of x?

Answer:

426. **AM-114** · Medium · SPR

The square of a positive number is 12 more than 4 times the number. What is the number?

Answer:

427. **AM-122** · Medium · SPR

$$xy = 48$$

$$3x - y = 0$$

If (x, y) is a solution to the system above and $x > 0$, what is the value of y?

Answer:

428. **A-124** · Medium · SPR

Layla is 4 years older than Noura. The square of Layla's age, in years, is 64 greater than the square of Noura's age. What is the sum, in years, of Layla's and Noura's ages?

Answer:

429. **AM-128** · Medium · SPR

If $2^{x+3} - 2^x = 56$, what is the value of x?

Answer:

430. **AM-130** · Medium · SPR

What is the product of the positive solutions to the equation $x^4 - 13x^2 + 36 = 0$?

Answer:

431. **AM-134** · Medium · SPR

$57x^2 + (57b + a)x + ab = 0$ In the equation above, a and b are positive constants. The product of the solutions is kab, where k is a constant. What is the value of k?

Answer:

432. **AM-142** · Medium

$\frac{1}{x^2+10x+25} = 4$ Which of the following could be the value of $x+5$?

A) $\frac{1}{2}$ **B)** $\frac{5}{2}$

C) $\frac{9}{2}$ **D)** $\frac{11}{2}$

433. **AM-311** · Medium · SPR · No calculator needed

What is the sum of all solutions to the equation $(x-3)^2 = (2x+1)^2$?

Answer:

434. **AM-318** · Medium · SPR · No calculator needed

How many integers x satisfy the inequality $x(x-4) < 2x$?

Answer:

435. **AM-007** · Hard · SPR

In the xy-plane, the graphs of $y = x^2 - 6x + 8$ and $y = 2x - 4$ intersect at two points. What is the sum of the x-coordinates of these two points?

Answer:

436. **AM-008** · Hard

What is the solution set of the equation

$$\sqrt{x+7} = x - 5?$$

A) $x = 2$ only **B)** $x = 9$ only

C) $x = 2$ and $x = 9$ **D)** There is no solution.

437. **AM-014** · Hard · SPR

In the xy-plane, the graphs of $y = x^2 - 4x - 5$ and $y = 3x + 1$ intersect at two points. What is the sum of the x-coordinates of these two points?

Answer:

438. **AM-022** · Hard · SPR

In the xy-plane, the graphs of $y = x^2 - x - 14$ and $y = 2x - 4$ intersect at two points. What is the positive difference between the x-coordinates of these two points?

Answer:

439. AM-027 · Hard · SPR

Suppose m and n are positive integers such that $mn > 20$ and $12mn$ is a perfect square. What is the least possible value of mn?

Answer:

440. AM-031 · Hard · SPR

If $x^2 - 8x + 5 = 0$ has roots r and s, what is the value of $r^2 + s^2$?

Answer:

441. AM-039 · Hard · SPR

What is the positive solution to the equation $x^2 - 6x - 7 = 0$, found by completing the square?

Answer:

442. AM-044 · Hard · SPR

If $y = x^2 - 7$ and $x + y = 5$, what is the value of x^2y^2 for the solution (x, y) of this system with $x > 0$?

Answer:

443. AM-054 · Hard

A transportation company serving pilgrims in Makkah uses the formula $10p = q + 25r$, where p is the fare per passenger, in SAR; q is a fixed service charge, in SAR; and r is the distance traveled, in kilometers. Which equation correctly expresses p in terms of q and r?

A) $p = \frac{q}{10} + 25r$

B) $p = q + \frac{25r}{10}$

C) $p = 10(q + 25r)$

D) $p = \frac{q + 25r}{10}$

444. AM-070 · Hard · SPR

A company's monthly profit, in thousands of SAR, is modeled by $P(x) = -2x^2 + 40x - 150$, where x is the price of the product in SAR. For what price x, other than $x = 5$, does the company break even (that is, $P(x) = 0$)?

Answer:

445. AM-077 · Hard

A system of equations is given below, where x represents the number of years since a company's founding and y represents its annual revenue, in millions of SAR.

$$y = x^2 - 5x + 20$$

$$y = 3x + 8$$

For how many values of x, with $x \geq 0$, are the two models equal?

A) 0

B) 1

C) 2

D) 3

446. AM-087 · Hard · SPR

$x^2(x + 5)(x - b) = 0$

In the given equation, b is a positive constant. The sum of the solutions of the equation is 7. What is the value of b?

Answer:

447. AM-091 · Hard · SPR

a and b are positive consecutive integers, and $b > a$. If b is thirteen more than $\sqrt{a}$, what is the value of a?

Answer:

448. AM-093 · Hard · SPR

If the graphs of $y = x^2 + 1$ and $y = -5x + p$ intersect at the point $(2, r)$, what is the value of $\frac{p}{r}$?

Answer:

449. AM-094 · Hard · SPR

The functions f and h are defined by $f(x) = x^2 + 3x - 40$ and $h(x) = x^2$. If $3h(x) = f(2x)$, what is the positive value of x?

Answer:

450. AM-096 · Hard · SPR

Let m be an even integer. How many possible values of m satisfy $\sqrt{m + 11} \leq 4$?

Answer:

451. AM-099 · Hard

Let $f(x) = ax^3$ and $g(x) = bx^5$ for any value of x. If a and b are positive constants, for how many values of x is $f(x) = g(x)$?

A) None

B) One

C) Two

D) Three

452. AM-102 · Hard · SPR

If $(x+3)^2 = 1$ and $(x-4)^2 = 36$, what is the value of x?

Answer:

453. AM-112 · Hard · SPR

What value of x satisfies the equation below?

$$\frac{x}{x-3} + \frac{2}{x} = \frac{3}{x-3}$$

Answer:

454. AM-120 · Hard · SPR

What value of x satisfies the equation $2^{3x} \cdot 4^{x+1} = 8^{x+3}$?

Answer:

455. AM-123 · Hard

$$y = x^2 - 4x + 1$$

$$y = 2x - 8$$

How many solutions does the system above have?

A) Zero **B)** Exactly one

C) Exactly two **D)** Infinitely many

456. AM-129 · Hard · SPR

The equation $2x^2 - 12x + k = 0$, where k is a constant, has two real solutions. One solution is twice the other solution. What is the value of k?

Answer:

457. AM-141 · Hard

$(ax+3)(5x^2-bx+4) = 20x^3-9x^2-2x+12$ In the equation above, a and b are constants. What is the value of ab?

A) 18 **B)** 20

C) 24 **D)** 40

458. AM-166 · Hard · SPR · Calculator

Real numbers x and y satisfy

$$\sqrt[3]{2^x} \cdot 4^y = 256 \quad \text{and} \quad \frac{16^x}{\sqrt[3]{4^y}} = 2^{22}.$$

What is the value of xy?

Answer:

459. AM-169 · Hard · No calculator needed

For a real number x, $4^x - 2^{x+2} - 32 = 0$. What is the value of 2^{x+1}?

A) -8 **B)** 8

C) 16 **D)** 32

460. AM-171 · Hard · No calculator needed

The equation $|x+6| - 7 = \frac{x}{3} + c$, where c is a constant, has exactly one solution x. What is the value of $2c + x$?

A) -11 **B)** -6

C) -5 **D)** -16

461. AM-174 · Hard · No calculator needed

In the xy-plane, the line $y = mx - 5$ intersects the parabola $y = x^2 - 6x + 11$ at exactly one point, where m is a negative constant. What is the value of m?

A) 14 **B)** -14

C) -2 **D)** 2

462. AM-177 · Hard · SPR · Desmos

$$x^2 + y^2 = 65$$

$$x - y = 7$$

The solution (a, b) to the system of equations above satisfies $a > 0$. What is the value of $a + 3b$?

Answer:

463. AM-189 · Hard · SPR · Desmos

In the xy-plane, the graphs of $y = x^2 + 6x + 4$ and $y = mx - 4$ have no point in common, where m is a constant. What is the sum of all integer values of m?

Answer:

464. AM-196 · Hard · Desmos

$$2^x = 3x + k$$

In the equation above, k is a constant. What is the smallest integer value of k for which the equation has two distinct real solutions?

A) -2 **B)** -1

C) 0 **D)** -3

465. AM-202 · Hard · SPR · Desmos

For an integer k, the equation

$$9 - 3\,|2x - 8| = 1 - \frac{5k}{2}$$

has no real solution x. What is the greatest possible value of k?

Answer:

466. **AM-204** · Hard · Desmos

For a constant k, the equation

$$2\sqrt{x+1}+2\sqrt{15-x}=k$$

has exactly one real solution. What is the value of k?

A) 16 **B)** $2+2\sqrt{15}$

C) 8 **D)** $8\sqrt{2}$

467. **AM-206** · Hard · SPR · No calculator needed

Positive real numbers x and y satisfy

$$\sqrt{x}-\sqrt{y}=2 \quad \text{and} \quad x+y=26.$$

What is the value of xy?

Answer:

468. **AM-208** · Hard · No calculator needed

$$x^2+y^2=45$$
$$y=mx+15$$

In the system above, m is a negative constant. If the system has exactly one solution (x,y), what is the value of x?

A) -3 **B)** 3

C) 6 **D)** -6

469. **AM-211** · Hard · No calculator needed

$$y=3x^2+4$$
$$y=1-kx$$

In the system above, k is a constant. For which of the following values of k does the system have two distinct solutions (x,y) whose x-coordinates are both negative?

A) 6 **B)** 7

C) -7 **D)** 2

470. **AM-214** · Hard · SPR · No calculator needed

For how many integer values of k with $-10 \le k \le 10$ does the equation

$$\sqrt{x+k}=x-2$$

have exactly one real solution?

Answer:

471. **AM-220** · Hard · No calculator needed

In the xy-plane, the line $y=kx+4$, where k is a constant, is tangent to the parabola $y=x^2+4x+13$ for exactly two values of k. For each of these values the line touches the parabola at one point. What is the greatest y-coordinate of these two points of tangency?

A) 10 **B)** 13

C) 34 **D)** 3

472. **AM-222** · Hard · SPR · No calculator needed

For a real number k, consider the equation

$$x-4\sqrt{x}=k.$$

For how many integer values of k from -10 to 10, inclusive, does the equation have exactly one real solution x?

Answer:

473. **AM-231** · Hard · SPR · No calculator needed

$$x^3+(3-k)x^2-(10+3k)x+10k=0$$

In the equation above, k is a constant. The equation has exactly two distinct real solutions for exactly two values of k. What is the sum of these two values of k?

Answer:

474. **AM-232** · Hard · SPR · Desmos

$$|2x-k|+k=x+6$$

In the equation above, k is a constant. The equation has exactly one solution. What is the value of k?

Answer:

475. **AM-245** · Hard · No calculator needed

For nonzero real numbers p and q, the equation $px^2+3qx+4p=0$ has exactly one real solution, and $pq=-48$. If $p>0$, what is the value of $p+q$?

A) 14 **B)** -14

C) -2 **D)** 2

476. **AM-263** · Hard · SPR

The equation $x^2-9x+k=0$, where k is a constant, has two real solutions that differ by 3. What is the value of k?

Answer:

477. **AM-264** · Hard

The equation $2x^2-10x+c=0$, where c is a constant, has a product of solutions equal to 6. What is the sum of the reciprocals of the two solutions?

A) $\frac{5}{6}$ **B)** $\frac{6}{5}$

C) $\frac{1}{5}$ **D)** $\frac{1}{30}$

478. AM-265 · Hard · SPR

Let r and s be the two solutions of $x^2-7x+3=0$. What is the value of r^2+s^2?

Answer:

479. AM-266 · Hard

The equation $x^2-5x+6=0$ has solutions p and q. Which of the following equations has solutions $2p$ and $2q$?

A) $x^2-10x+24=0$ **B)** $x^2-10x+12=0$

C) $x^2-5x+12=0$ **D)** $2x^2-5x+6=0$

480. AM-267 · Hard · SPR

The equation $x^2-bx+27=0$, where b is a positive constant, has two real solutions, and one solution is 3 times the other. What is the value of b?

Answer:

481. AM-268 · Hard

$$x^2-6x=2|x-3|$$

What is the sum of all real solutions of the equation above?

A) 0 **B)** 3

C) 6 **D)** 9

482. AM-269 · Hard

$$\frac{x}{x-4}=2+\frac{4}{x-4}$$

How many solutions does the equation above have?

A) Zero **B)** Exactly one

C) Exactly two **D)** Infinitely many

483. AM-274 · Hard · SPR

For what value of k does the equation $|x^2-4|=k$ have exactly three distinct real solutions?

Answer:

484. AM-275 · Hard · SPR

The equation $-2x^2+12x-5=k$, where k is a constant, has no real solutions. What is the least integer value of k?

Answer:

485. AM-277 · Hard · SPR

The solutions of $x^2-bx+12=0$, where $b>0$, differ by 4. What is the value of b?

Answer:

486. AM-279 · Hard · SPR

$$\left(\frac{x+3}{x-1}\right)^2-3\left(\frac{x+3}{x-1}\right)-10=0$$

What is the sum of all real solutions of the equation above?

Answer:

487. AM-295 · Hard · SPR

The equation $2x^2-7x+k=0$, where k is a constant, has two real solutions r and s with $r^2+s^2=9$. What is the value of k?

Answer:

488. AM-296 · Hard · SPR

What is the product of all real solutions of $x^4-5x^2+4=0$?

Answer:

489. AM-297 · Hard · SPR

What is the sum of all real solutions of $2^{2x}-6\cdot2^x+8=0$?

Answer:

490. AM-301 · Hard · SPR

The solutions of $x^2-5x+2=0$ are r and s. What is the value of $\frac{r}{s}+\frac{s}{r}$?

Answer:

491. AM-302 · Hard · SPR

The polynomial $p(x)=x^3-7x^2+14x-8$ has three real zeros. What is the sum of the squares of the zeros?

Answer:

492. AM-303 · Hard · No calculator needed

In the equation $(k-2)x^2+4x+1=0$, k is a constant. The equation has exactly one real solution for exactly two values of k. What is the sum of these two values of k?

A) 8 **B)** 4

C) 2 **D)** 6

493. AM-307 · Hard · No calculator needed

In the equation $\frac{x^2}{x-a} = \frac{9}{x-a}$, a is a constant. For how many real values of a does the equation have exactly one real solution?

A) Infinitely many
B) Exactly two
C) Exactly one
D) Zero

494. AM-309 · Hard · Desmos

How many solutions (x, y) does the system $x^2+y^2 = 25$ and $y = x^2 - 5$ have?

A) 3
B) 4
C) 1
D) 2

495. AM-310 · Hard · Desmos

How many real solutions does the equation $x^2 = 2^x$ have?

A) 1
B) 2
C) 3
D) 4

496. AM-317 · Hard · Desmos

In the xy-plane, the graphs of $y = |x-2|$ and $y = mx+1$, where m is a constant, intersect at exactly two points. Which of the following could be the value of m?

A) -2
B) $-\frac{3}{4}$
C) 2
D) $\frac{1}{2}$

497. AM-320 · Hard · SPR · No calculator needed

In the equation $\frac{x+a}{x^2-a^2} = -\frac{3}{20x}$, a is a constant. If $x = 6$ is a solution, what is the value of a?

Answer:

498. AM-321 · Hard · No calculator needed

In the equation $\frac{x^2}{\sqrt{x^2-k^2}} = \frac{k^2}{\sqrt{x^2-k^2}} + 24$, k is a positive constant. Which of the following is a solution?

A) $-\sqrt{k^2+576}$
B) $\sqrt{576-k^2}$
C) $\sqrt{k^2+24}$
D) $k+24$

499. AM-322 · Hard · No calculator needed

In the equation $\frac{ax}{x-2} = 3 + \frac{4}{x-2}$, a is a constant. For which values of a does the equation have no solution?

A) 3 and 4 only
B) 3 only
C) 2 only
D) 2 and 3 only

500. AM-324 · Hard · SPR · No calculator needed

For how many integer values of b does the equation $x^2 + bx + 36 = 0$ have two distinct integer solutions?

Answer:

501. AM-325 · Hard · No calculator needed

The equations $x^2-x-12 = 0$ and $x^2+kx+8 = 0$ have a common solution, where k is a positive constant. What is the value of k?

A) -6
B) $\frac{17}{3}$
C) $-\frac{17}{3}$
D) 6

502. AM-326 · Hard · Desmos

What is the sum of all distinct real solutions of $x^3-2x^2-15x+36 = 0$?

A) 2
B) 7
C) -2
D) -1

503. AM-327 · Hard · No calculator needed

What is the solution set of the equation $2x^{2/3} - 3 = 15$?

A) $\{-27, 27\}$
B) $\{27\}$
C) $\{-9^{2/3}, 9^{2/3}\}$
D) $\{-6, 6\}$

Nonlinear functions (159)

504. AM-002 · Medium

As part of a Red Sea coastal development aligned with Vision 2030, the estimated resident population t years after 2024 is modeled by

$$P(t) = 5000\,(1.15)^t\,.$$

According to the model, by approximately what factor does the population increase every 5 years?

A) 1.15
B) 1.75
C) 2.01
D) 2.99

505. AM-004 · Medium

A car is purchased in Riyadh for $120{,}000$ SAR. Its value depreciates according to

$$V(t) = 120000\,(0.88)^t\,,$$

where t is the number of years since purchase. To the nearest thousand SAR, what is the value of the car after 4 years?

A) $61{,}000$ SAR
B) $72{,}000$ SAR
C) $84{,}000$ SAR
D) $94{,}000$ SAR

506. AM-006 · Medium

The function h is defined by $h(x) = 2x^2 - 12x + 7$. What is the x-coordinate of the vertex of the graph of $y = h(x)$ in the xy-plane?

A) -3 **B)** 2

C) 3 **D)** 6

507. AM-013 · Medium

The function f is defined by $f(x) = a \cdot 3^x$, where a is a constant. If $f(2) = 45$, what is the value of $f(4)$?

A) 135 **B)** 225

C) 405 **D)** 450

508. AM-018 · Medium

If $f(x) = x^2 - 4x + k$ and $f(1) = 9$, what is the value of k?

A) 6 **B)** 9

C) 12 **D)** 14

509. AM-026 · Medium

Let $f(x) = ax^2$ and $g(x) = bx^4$ for any value of x, where a and b are positive constants. For how many values of x is $f(x) = g(x)$?

A) None **B)** One

C) Two **D)** Three

510. AM-032 · Medium

The variable y varies inversely with x. When $x = 6$, $y = 15$. What is the value of y when $x = 10$?

A) 5 **B)** 6

C) 9 **D)** 25

511. AM-036 · Medium

If $f(x) = x^2 + 1$ and $g(x) = 2x - 3$, what is the value of $f(g(2))$?

A) 1 **B)** 2

C) 3 **D)** 5

512. AM-040 · Medium

The function C gives the Celsius equivalent of a temperature reading x, and the function P gives the predicted crop yield, in tons, at temperature $x°$C for a NEOM greenhouse. Suppose $C(30) = 30$ and $P(30) = 18$. What is the value of $P(C(30))$?

A) 30 **B)** 18

C) 48 **D)** 540

513. AM-042 · Medium

For which value of x is the function $f(x) = \dfrac{2x - 1}{x^2 - 2x - 24}$ undefined?

A) -6 **B)** -4

C) 2 **D)** 5

514. AM-045 · Medium

As part of a NEOM smart-city energy initiative, the storage capacity of a battery facility is modeled by

$$E(t) = 1450\,(1.08)^t,$$

where $E(t)$ is the capacity, in megawatt-hours, and t is the number of years after the facility opened, for $0 \le t \le 15$. Which is the best interpretation of 1450 in this context?

A) The predicted capacity, in megawatt-hours, 15 years after the facility opened
B) The estimated capacity, in megawatt-hours, when the facility opened
C) The annual percent increase in capacity
D) The number of years it takes the capacity to double

515. AM-047 · Medium

The function h is defined by $h(x) = 3x^2 - 24x + 10$. What is the x-coordinate of the vertex of the graph of $y = h(x)$ in the xy-plane?

A) -4 **B)** 3

C) 4 **D)** 8

516. AM-049 · Medium

During a scheduled maintenance check, the water level change in a Jeddah desalination reservoir is modeled by $f(x) = x^3 - 5x + 12$, where $f(x)$ is the change in level, in centimeters, and x is the number of hours after maintenance begins, for $-3 \le x \le 3$. What is the value of $f(-2)$?

A) 6 **B)** 10

C) 14 **D)** 18

517. AM-060 · Medium

A decorative fountain shoots a stream of water whose height, in meters, is modeled by $h(t) = -5t^2 + 20t$, where t is the time in seconds after the water leaves the nozzle. What is the maximum height, in meters, reached by the stream of water?

A) 10 **B)** 15

C) 20 **D)** 25

518. AM-061 · Medium

A population of bacteria in a lab culture is modeled by $P(t) = 300(1.5)^t$, where t is the time in hours. What is the population after 3 hours?

A) 675 **B)** 900

C) $1{,}012.5$ **D)** $1{,}350$

519. AM-064 · Medium

If $f(x) = 2x^2 - 5x + 3$, what is the value of $f(4)$?

A) 11 **B)** 13

C) 15 **D)** 17

520. AM-065 · Medium

The number of subscribers to an online tutoring platform grows according to $N(t) = 500(1.2)^t$, where t is measured in months since launch. Which of the following is closest to the number of subscribers after 5 months?

A) $1{,}000$ **B)** $1{,}100$

C) $1{,}244$ **D)** $1{,}500$

521. AM-071 · Medium

A farmer has 80 meters of fencing to enclose a rectangular plot, with one side against an existing wall (so fencing is needed only on the other three sides). If x represents the width of the plot (perpendicular to the wall), which function $A(x)$ models the area of the plot, in square meters?

A) $A(x) = 80x - 2x^2$ **B)** $A(x) = 80x - x^2$

C) $A(x) = 40x - 2x^2$ **D)** $A(x) = 80 - 2x^2$

522. AM-072 · Medium

The amount of a medication remaining in a patient's bloodstream is modeled by $M(t) = 200(0.85)^t$, where t is the number of hours since the dose was taken and $M(t)$ is measured in milligrams. Which of the following is closest to the amount remaining after 6 hours?

A) 65.5 **B)** 75.5

C) 85.5 **D)** 95.5

523. AM-079 · Medium

The function f is defined by $f(x) = (-5)(3)^x + 7$. What is the y-intercept of the graph of $y = f(x)$ in the xy-plane?

A) $(0, -5)$ **B)** $(0, 2)$

C) $(0, 7)$ **D)** $(0, -12)$

524. AM-080 · Medium

The population of a coastal town near Yanbu has grown at an average rate of 2.4% per year since 2010. There were approximately $60{,}000$ residents in 2010. Which of the following functions best models the town's population P, in thousands of residents, t years after 2010?

A) $P(t) = 60(1.024)^t$
B) $P(t) = 60(2.4)^t$
C) $P(t) = 1.024t + 60$
D) $P(t) = 2.4t + 60$

525. AM-092 · Medium · SPR

x and y^2 are directly proportional. If $x = 6$ when $y^2 = 8$, what is the value of y^2 when $x = 15$?

Answer:

526. AM-105 · Medium

Each term in the sequence below is 6 times the previous term. What is the 9th term in the sequence?

$$3,\ 18,\ 108,\ 648,\ \ldots$$

A) 648×9 **B)** 3^9

C) 3×6^8 **D)** 6×3^8

527. AM-108 · Medium

Which of the following parabolas has its turning point in the second quadrant of the coordinate plane?

A) $y = (x + 2)^2 - 3$ **B)** $y = (x - 3)^2 - 1$

C) $y = -(x + 3)^2 - 1$ **D)** $y = -(x + 1)^2 + 2$

528. AM-116 · Medium

The graph of $y = 2x^2 + 12x + 11$ in the xy-plane is a parabola. What are the coordinates of its vertex?

A) $(-3, -7)$ **B)** $(3, 65)$

C) $(-3, 11)$ **D)** $(-6, 11)$

529. AM-121 · Medium · SPR

The first term of a geometric sequence is 2 and the fourth term is 54. What is the sum of the first five terms of the sequence?

Answer:

530. AM-124 · Medium

The functions f and g are defined by $f(x) = 2x - 3$ and $g(x) = x^2 + 1$. Which expression is equivalent to $g(f(x))$?

A) $4x^2 - 12x + 10$ **B)** $4x^2 - 12x + 9$

C) $2x^2 - 1$ **D)** $4x^2 + 10$

531. AM-126 · Medium

The function g is defined by $g(x) = \dfrac{5}{(x-2)^2 - 9}$. For which values of x is $g(x)$ undefined?

A) $x = 2$ only **B)** $x = -1$ and $x = 5$

C) $x = -5$ and $x = 1$ **D)** $x = 11$ only

532. AM-138 · Medium

The function $h(x) = a^x + b$, where $a > 0$ and b is a constant, passes through $(0, 10)$ and $(2, 13)$. What is the value of ab?

A) 13 **B)** 18

C) 20 **D)** 26

533. AM-140 · Medium · SPR

The function $h(x) = 2(x - 4)^2 - 32$ has x-intercepts at $(0, 0)$ and $(t, 0)$. What is the value of t?

Answer:

534. AM-146 · Medium · SPR

$g(x) = \dfrac{2x-4}{(x+11)(x-6)}$ If $g(a+5) = 0$, what is the value of a?

Answer:

535. AM-291 · Medium · SPR

The function f is defined by $f(x) = (x-1)(x-5)$. The graph of g is the reflection of the graph of f across the y-axis, shifted up 4 units. What is the value of $g(2)$?

Answer:

536. AM-001 · Hard

A decorative fountain in a public park in Riyadh shoots a jet of water whose height above the ground, in meters, is modeled by

$$h(t) = -4\,(t-3)^2 + 45,$$

where t is the time, in seconds, after the jet is activated. For which values of t does the water reach a height of 29 meters?

A) $t = 1$ only **B)** $t = 5$ only

C) $t = 1$ and $t = 5$ **D)** $t = 2$ and $t = 4$

537. AM-003 · Hard · SPR

The function f is defined by

$$f(x) = a\,(x-2)\,(x+6)\,,$$

where a is a constant. The graph of $y = f(x)$ in the xy-plane passes through the point $(0, -24)$. What is the value of $f(5)$?

Answer:

538. AM-005 · Hard

The function f is defined by $f(x) = a \cdot 2^x$, where a is a constant. If $f(3) = 40$, what is the value of $f(5)$?

A) 64 **B)** 128

C) 160 **D)** 200

539. AM-016 · Hard

A ball is thrown upward, and its height above the ground, in meters, is modeled by

$$h(t) = -5\,(t-2)^2 + 30,$$

where t is the time, in seconds, after the ball is thrown. For which values of t does the ball reach a height of 10 meters?

A) $t = 0$ only **B)** $t = 4$ only

C) $t = 0$ and $t = 4$ **D)** $t = 1$ and $t = 3$

540. AM-019 · Hard

The function h is defined by $h(x) = (x-a)\,(x-b)$, where a and b are constants and $a < b$. The graph of $y = h(x)$ crosses the x-axis at $x = -3$ and $x = 7$. What is the minimum value of h?

A) -25 **B)** -20

C) -10 **D)** 25

541. AM-020 · Hard · SPR

The function p is defined by $p(x) = a \cdot x^3$, where a is a nonzero constant. If $p(2) = 40$, what is the value of $p(-2)$?

Answer:

542. AM-028 · Hard · SPR

The function N is defined by $N(t) = k \cdot 2^{-at}$ for positive constants k and a. The table shows values of N for select values of t.

t	0	1	2
$N(t)$	200	50	12.5

What is the value of a?

Answer:

543. AM-033 · Hard

The variable z varies jointly with x and the square of y. If $z = 48$ when $x = 3$ and $y = 4$, what is the value of z when $x = 5$ and $y = 2$?

A) 10 **B)** 16

C) 20 **D)** 25

544. AM-035 · Hard · SPR

A sequence is defined recursively by $a_1 = 5$ and $a_n = 2a_{n-1} - 3$ for $n \geq 2$. What is the value of a_4?

Answer:

545. AM-037 · Hard

The graph of $y = h(x)$ is the graph of $y = x^2$ shifted right 3 units and down 5 units. What is the value of $h(7)$?

A) 6 **B)** 11

C) 14 **D)** 16

546. AM-041 · Hard · SPR

A ball's height above the ground, in meters, t seconds after being launched from a platform in Al-Ula is modeled by $h(t) = -5t^2 + 20t + c$, where c is the platform height. If $h(1) = 35$, what is the value of $h(3)$?

Answer:

547. AM-046 · Hard · SPR

The function g is defined by

$$g(x) = a\,(x-3)\,(x+4)\,,$$

where a is a constant. The graph of $y = g(x)$ in the xy-plane passes through the point $(0, -36)$. What is the value of $g(6)$?

Answer:

548. AM-048 · Hard

A cooperative in Al-Ahsa tracks its annual date export tonnage with the function $f(x) = a \cdot 3^x$, where a is a constant and x is the number of years since record-keeping began. If $f(2) = 54$, what is the value of $f(4)$?

A) 108 **B)** 162

C) 324 **D)** 486

549. AM-062 · Hard

The height, in meters, of a ball above the ground t seconds after being thrown is given by $h(t) = -4.9t^2 + 14.7t + 2$. Which of the following is closest to the number of seconds it takes for the ball to hit the ground ($h(t) = 0$)?

A) 2.1 **B)** 2.6

C) 3.1 **D)** 3.6

550. AM-063 · Hard

A water-treatment filter reduces the concentration of a contaminant by 12% every hour it operates. If the initial concentration is 250 parts per million (ppm), which of the following functions models the concentration $C(t)$, in ppm, after t hours of operation?

A) $C(t) = 250(0.12)^t$ **B)** $C(t) = 250(0.88)^t$

C) $C(t) = 250(1.12)^t$ **D)** $C(t) = 250 - 12t$

551. AM-066 · Hard

The graph of the function $g(x) = (x-3)^2 - 7$ is the same as the graph of $y = x^2$ after which of the following transformations?

A) Shifted left 3 and up 7 **B)** Shifted right 3 and down 7

C) Shifted right 3 and up 7 **D)** Shifted left 3 and down 7

552. AM-067 · Hard · SPR

If $x = 5$ is one solution to the equation $x^2 - kx + 20 = 0$, what is the value of k?

Answer:

553. AM-073 · Hard

A company's weekly revenue, in thousands of SAR, from selling a product priced at x SAR is modeled by $R(x) = -3x^2 + 150x$. What price x maximizes the weekly revenue?

A) 15 **B)** 20

C) 25 **D)** 30

554. AM-074 · Hard · SPR

The equation $x^2 - 12x + c = 0$ has two equal (repeated) roots. What is the value of c?

Answer:

555. AM-084 · Hard

The function f is defined by $f(x) = \dfrac{12-x}{2+x}$, where $x \neq -2$. As the value of x increases without bound, the value of $f(x)$ approaches which of the following?

A) -2 **B)** -1

C) 1 **D)** 12

556. AM-086 · Hard

$y = (x-p)^2\,(x+p)\,(x-q)$, where p and q are positive constants and $p \neq q$. How many distinct x-intercepts does the graph of this equation have in the xy-plane?

A) 1 **B)** 2

C) 3 **D)** 4

557. AM-088 · Hard

In a sequence, the first term is -4. Each term after the first is three less than double the preceding term. Which term is the first one whose absolute value is greater than 150?

A) 3 **B)** 4

C) 5 **D)** 6

558. AM-101 · Hard · SPR

The table below shows some values of the function N.

t	0	1	2
$N(t)$	100	25	$\frac{25}{4}$

If $N(t) = k \cdot 2^{-at}$ for positive constants k and a, what is the value of a?

Answer:

559. AM-125 · Hard · SPR

The function f is defined by $f(x) = 3x + k$, where k is a constant. If $f(f(2)) = 38$, what is the value of k?

Answer:

560. AM-132 · Hard

The function f is defined by $f(x) = a(x+4)(x-10)$, where a is a constant. The minimum value of f is -98. What is the value of $f(0)$?

A) -98 **B)** -80

C) -40 **D)** 80

561. AM-136 · Hard · SPR

The function g satisfies $g(kn) = n + 10$ for every real number n, where k is a constant. If $g(4) = 82$, what is the value of k?

Answer:

562. AM-144 · Hard

A theater's revenue, in SAR, from selling t tickets is $R(t) = t(320 - 10t)$. For which of the following values of t is the revenue $2{,}400$ SAR?

A) 14 **B)** 18

C) 20 **D)** 26

563. AM-145 · Hard · SPR

The function $m(z) = -\sqrt{z^2 + pz + q}$ has graph passing through $(6, 0)$ and $\left(0, -\sqrt{252}\right)$, where p and q are constants. What is the greatest value of t for which $m(t) = 0$?

Answer:

564. AM-147 · Hard

$y = \dfrac{5(x+3)(x-9)(x+2)}{(x-5)(x-5c)}$ In the equation above, c is a positive constant. The expression is undefined at $x = 10$ (and $x = 5$). What is the value of c?

A) -3 **B)** -2

C) 2 **D)** 3

565. AM-148 · Hard · SPR

The function $h(x) = -\sqrt{x^2 + bx + c}$ has a graph passing through $(2, 0)$ and $\left(0, -\sqrt{314}\right)$, where b and c are constants. What is the greatest value of m for which $h(m) = 0$?

Answer:

566. AM-150 · Hard

The function $f(x) = ax^2 + bx + c$ has $a > 0$ and $c < 0$. Which of the following must be true? I. f has two distinct real zeros II. The zeros of f have opposite signs III. The vertex of the graph of f lies in the fourth quadrant

A) I only **B)** I and III only

C) I and II only **D)** I, II and III

567. AM-151 · Hard · SPR

x	$g(x)$
-12	$-\frac{17}{4}$
-9	$-\frac{10}{3}$
3	258

The table above shows three values of x and their corresponding values of $g(x)$. If $g(x) = \dfrac{x^2 + a}{bx} + 2^{(cx-1)}$, what is the value of abc?

Answer:

568. AM-152 · Hard

The function f is defined by $f(x) = 5(3)^{x+2}$. The functions g and h are equivalent to f. Which of the following equations displays, as a constant or coefficient, the y-coordinate of the y-intercept of the graph of the function it defines in the xy-plane?

I. $g(x) = 45(3)^x$ II. $h(x) = 15(3)^{x+1}$

A) I only **B)** II only

C) I and II **D)** Neither I nor II

569. AM-153 · Hard

The function f is defined by $f(x) = 96(2)^{x-3}$. The functions g and h are equivalent to f. Which of the following equations displays, as a constant or coefficient, the y-coordinate of the y-intercept of the graph of the function it defines in the xy-plane?

I. $g(x) = 48(2)^{x-2}$ II. $h(x) = 12(2)^x$

A) I only **B)** II only

C) I and II **D)** Neither I nor II

570. AM-154 · Hard

The function f is defined by $f(x) = 3(4)^{x+2}$, where $x \geq 0$. The functions g and h are equivalent to f. Which of the following equations displays, as a constant or coefficient, the minimum value of the function it defines, where $x \geq 0$?

I. $g(x) = 48(4)^x$ II. $h(x) = 48(2)^{2x}$

A) I only **B)** II only

C) I and II **D)** Neither I nor II

571. AM-155 · Hard

The function f is defined by $f(x) = 6(2)^{x+3}$, where $x \geq 0$. The functions g and h are equivalent to f. Which of the following equations displays, as a constant or coefficient, the minimum value of the function it defines, where $x \geq 0$?

I. $g(x) = 12(2)^{x+2}$ II. $h(x) = 24(2)^{x+1}$

A) I only **B)** II only

C) I and II **D)** Neither I nor II

572. AM-156 · Hard · SPR

The function $f(x) = 200(1.5)^{\frac{x}{2}}$ is defined for all x. For every increase of x by 4, the value of $f(x)$ increases by $p\%$. What is the value of p?

Answer:

573. AM-157 · Hard

The function $f(t) = 2400(0.8)^{\frac{t}{5}}$ models the value, in dollars, of a machine t years after it was bought. For every increase of t by 10, the value of $f(t)$ decreases by $p\%$. What is the value of p?

A) 20 **B)** 36

C) 40 **D)** 64

574. AM-158 · Hard

The function $N(t) = 500(2)^{\frac{t}{3}}$ models the number of bacteria in a culture t hours after the start. Which of the following gives the number of bacteria m minutes after the start?

A) $N(m) = 500(2)^{\frac{m}{180}}$ **B)** $N(m) = 500(2)^{180m}$

C) $N(m) = 500(2)^{20m}$ **D)** $N(m) = 500(2)^{\frac{m}{20}}$

575. AM-159 · Hard

The function $f(x) = 12{,}000(0.85)^x$ models the number of visitors a museum had each year, where x is the number of years since the start of 2018 and $0 \le x \le 6$. If $y = f(x)$ is graphed in the xy-plane, which of the following is the best interpretation of the y-intercept of the graph in this context?

A) The minimum estimated number of visitors during the 6 years was $4{,}526$.
B) The minimum estimated number of visitors during the 6 years was $12{,}000$.
C) The estimated number of visitors in 2018 was $4{,}526$.
D) The estimated number of visitors in 2018 was $12{,}000$.

576. AM-160 · Hard · SPR

The exponential function $f(x) = ab^x$, where a and b are positive constants, satisfies $f(2) = 18$ and $f(5) = 486$. What is the value of $f(4)$?

Answer:

577. AM-161 · Hard · SPR

A town's population is modeled by $P(t) = 50(3)^{\frac{t}{4}}$, where t is the number of years since 2010. According to the model, in how many years after 2010 will the population be $1{,}350$?

Answer:

578. AM-162 · Hard · SPR

The amount of a medicine in a patient's body is modeled by $A(t) = 640\left(\frac{1}{2}\right)^{\frac{t}{8}}$ milligrams, where t is the number of hours after the dose. After how many hours does the model estimate exactly 40 milligrams remain?

Answer:

579. AM-163 · Hard

The function $h(t) = -16t^2 + 96t + 4$ models the height, in feet, of a ball t seconds after it is launched. The graph of $y = h(t)$ in the ty-plane has a vertex at $(3, 148)$. Which of the following is the best interpretation of the vertex in this context?

A) The ball was launched from a height of 148 feet.
B) The ball reached its maximum height of 148 feet 3 seconds after it was launched.
C) The ball hit the ground 148 seconds after it was launched.
D) The ball was 3 feet above the ground 148 seconds after it was launched.

580. AM-164 · Hard

A theater sells 600 tickets when the price is $\$20$. For each $\$1$ increase in price, 20 fewer tickets are sold. Which ticket price, in dollars, gives the greatest revenue?

A) 5 **B)** 25

C) $12{,}000$ **D)** $12{,}500$

581. AM-165 · Hard · SPR

The cost, in dollars, of making each unit when a factory produces n units is modeled by $C(n) = 0.02n^2 - 3.2n + 210$. What is the minimum cost per unit, in dollars, according to the model?

Answer:

582. AM-168 · Hard · No calculator needed

The quadratic function $f(x) = ax^2 + bx + c$, where a, b and c are constants, has two distinct real zeros. It is known that $f(2) = f(8)$, $f(0) < 0$ and $f(5) > 0$. Consider the following statements.

I. $b > 0$

II. The product of the two zeros is greater than 25.

III. Both zeros are positive.

Which of the statements must be true?

A) I only **B)** II and III only

C) I and III only **D)** I, II, and III

583. **AM-173** · Hard · SPR · Calculator

The size of a bacterial colony, in thousands of cells, is modeled by $V(t) = 500 \cdot 2^{t/2} \cdot 4^{t/3} \cdot 16^{-t/12}$, where t is the number of hours since the start of observation. The size of the colony is multiplied by a constant factor c every 6 hours. What is the value of c?

Answer:

584. **AM-179** · Hard · No calculator needed

The function f is defined by $f(x) = ax^2 + bx + c$, where a, b, and c are constants. The graph of $y = f(x)$ in the xy-plane passes through $(1, 8)$ and $(-1, 2)$ and has exactly one x-intercept. What is the greatest possible value of a?

A) $\frac{5}{2}$ **B)** 9

C) $\frac{9}{2}$ **D)** $\frac{1}{2}$

585. **AM-180** · Hard · No calculator needed

The quadratic function f has its vertex at $(3, -2)$, and $f(5) = 6$. The function g is defined by $g(x) = \frac{3}{4} f\left(\frac{1}{2}x - 1\right)$. The graph of $y = f(x)$ has y-intercept $(0, p)$ and the graph of $y = g(x)$ has y-intercept $(0, q)$. What is the value of $q - p$?

A) $-\frac{23}{2}$ **B)** $\frac{13}{2}$

C) -4 **D)** $\frac{45}{2}$

586. **AM-181** · Hard · SPR · No calculator needed

The function f is defined by $f(x) = 2^x + k$, where k is a constant. For a real number p, $f(p) = 9$ and $f(-p) = \frac{21}{4}$. What is the value of p?

Answer:

587. **AM-183** · Hard · SPR · Calculator

The altitude h, in meters, of a small rocket t seconds after launch is modeled by $h(t) = at^2 + bt + c$, where a, b, and c are constants. At launch the altitude is 3 meters; 2 seconds after launch it is 43 meters; and 5 seconds after launch it is 28 meters. According to the model, what is the altitude, in meters, 3 seconds after launch?

Answer:

588. **AM-184** · Hard · SPR · No calculator needed

The quadratic function f has zeros at $x = -2$ and $x = 5$, and $f(1) = -24$. The function h is defined by $h(x) = f(x + 2) - 3f(-x)$. What is the value of $h(1)$?

Answer:

589. **AM-185** · Hard · SPR · No calculator needed

The quadratic function f has an axis of symmetry at $x = 2$, and its graph passes through $(3, -3)$ and $(5, 13)$. The functions g and h are defined by $g(x) = \frac{x - 6}{2}$ and $h(x) = 3 - f(g(x))$. What is the y-coordinate of the y-intercept of the graph of $y = h(x)$?

Answer:

590. **AM-186** · Hard · SPR · No calculator needed

The function f is defined by $f(x) = c \cdot b^{x+3}$, where c and b are positive constants. If $f(1) = 1296$ and $f(3) = 729$, then for each increase of 1 in the value of x, $f(x)$ decreases by $p\%$. What is the value of p?

Answer:

591. **AM-187** · Hard · Calculator

The value V, in dollars, of a delivery van x years after it is purchased is modeled by $V(x) = 18{,}000(0.49)^{x/2}$. Which of the following best describes how the van's value changes each year?

A) It decreases by 24.5% each year.
B) It decreases by 30% each year.
C) It decreases by 70% each year.
D) It decreases by 51% each year.

592. **AM-188** · Hard · SPR · No calculator needed

In the xy-plane, the graph of the quadratic function p and the graph of the line $y = 5 - x$ intersect at exactly two points, one with x-coordinate 1 and one with x-coordinate 4. If $p(0) = -3$, what is the value of $p(6)$?

Answer:

593. **AM-190** · Hard · SPR · Calculator

The function f is defined by $f(x) = ab^x + c$, where a, b, and c are constants and $b > 1$. The values $f(2) = 14$, $f(4) = 20$, and $f(6) = 32$ are known. What is the value of c?

Answer:

594. **AM-191** · Hard · No calculator needed

For $x \geq 0$, let $f(x) = 120 - 90(0.8)^x$ and $g(x) = -3(x - 4)^2 + 125$. Which of the following statements is true?

A) Only g attains a maximum, 125 **B)** Neither attains a maximum

C) Both f and g attain a maximum **D)** Only f attains a maximum, 120

595. **AM-192** · Hard · No calculator needed

$$-2(x-5)^2+k=m(x-1)(x-r)$$

In the equation above, k, m, and r are constants. The equation is true for all real values of x. What is the value of k?

A) -32 **B)** 18

C) 32 **D)** 50

596. **AM-199** · Hard · Calculator

A patient takes a 100-milligram dose of a medication at 8:00 a.m. and again at 8:00 p.m. every day, beginning at 8:00 a.m. on Monday. The dose is absorbed instantly, and between any two consecutive doses exactly half of the medication present in the body is eliminated. To the nearest tenth of a milligram, how much medication is in the body immediately after the 8:00 a.m. dose on Thursday?

A) 187.5 **B)** 196.9

C) 198.4 **D)** 199.2

597. **AM-200** · Hard · Calculator

The function $P(t)=\dfrac{120}{1+5(0.8)^t}$ models the number of active users of an app, in thousands, t weeks after launch, for $0\le t\le 10$. Consider the following statements.

I. P is increasing on the interval $0\le t\le 10$.

II. There is a time t in the interval at which $P(t)=60$.

III. There is a time t in the interval at which $P(t)=80$.

Which of the statements must be true?

A) I only **B)** I and II only

C) II and III only **D)** I, II, and III

598. **AM-201** · Hard · Calculator

The function f is defined by $f(x)=a\cdot b^x$, where a and b are positive constants. For every value of x, $f(x+2)$ is 44% greater than $f(x)$. For every value of x, $f(x+3)$ is $p\%$ greater than $f(x)$. What is the value of p?

A) 132 **B)** 60

C) 66 **D)** 72.8

599. **AM-205** · Hard · No calculator needed

The value V, in dollars, of a piece of lab equipment t years after it was purchased is modeled by

$$V(t)=2400\,(0.64)^{t/2.5}.$$

Which of the following is the best interpretation of the model?

A) It falls 36% every 2.5 years.
B) It falls 36% every 5 years.
C) It falls 64% every 2.5 years.
D) It falls 2.5% every year.

600. **AM-209** · Hard · No calculator needed

The quadratic function f has a zero at $x=2$. The graph of $y=f(x)$ in the xy-plane passes through the points $(0,-16)$ and $(5,54)$. What is the minimum value of f?

A) -18 **B)** -16

C) -9 **D)** -1

601. **AM-210** · Hard · No calculator needed

The graph of the quadratic function f in the xy-plane passes through the points $(0,1)$, $(1,-2)$ and $(3,4)$. The graph of $y=g(x)$ is obtained by translating the graph of $y=f(x)$ 3 units to the right and 5 units down. What is the y-coordinate of the y-intercept of the graph of g?

A) 34 **B)** 39

C) -1 **D)** 29

602. **AM-213** · Hard · No calculator needed

The function f is linear. The function g is defined by $g(x)=\dfrac{f(x+2)-x}{x-3}$ for $x\neq 3$. Some values of g are shown in the table below.

x	1	5
$g(x)$	3	-11

What is the y-intercept of the graph of $y=f(x)$ in the xy-plane?

A) -5 **B)** -4

C) 4 **D)** -3

603. **AM-216** · Hard · No calculator needed

The function f is defined for $x\ge 0$ by

$$f(x)=48\left(\frac{1}{8}\right)^{\frac{2-x}{3}}.$$

Which of the following is equivalent to $f(x)$ for all $x\ge 0$ and displays the value of $f(0)$ as a constant factor?

A) $f(x)=192\cdot\left(\dfrac{1}{2}\right)^x$
B) $f(x)=12\cdot 2^x$
C) $f(x)=48\cdot 2^x$
D) $f(x)=12\cdot\left(\dfrac{1}{2}\right)^x$

604. **AM-219** · Hard · No calculator needed

The number of bacteria in a colony t years after January 1, 2020 is modeled by $A(t)=300\cdot 4^{t/3}$. Which of the following gives the number of bacteria m months after January 1, 2023?

A) $300\cdot 2^{m/18}$ **B)** $1200\cdot 2^{m/36}$

C) $900\cdot 2^{m/18}$ **D)** $1200\cdot 2^{m/18}$

605. **AM-221** · Hard · No calculator needed

The function $g(x) = 2x^2 + bx + c$, where b and c are constants, has zeros $k + 1$ and $3k - 5$, where k is a constant. If $g(2) = g(10)$, what is the value of c?

A) 70 **B)** 140

C) 24 **D)** 35

606. **AM-223** · Hard · Desmos

Let $g(x) = 2^x$ and $h(x) = 3x - 1$. For which values of x is $g(x) > h(x)$?

A) $x < 1$ or $x > 3$ **B)** $1 < x < 3$

C) $x > 3$ only **D)** $x < 3$

607. **AM-227** · Hard · SPR · No calculator needed

The function $f(x) = 2\left(3x + 1 - \sqrt{5}\right)\left(3x + 1 + \sqrt{5}\right)$ can be written as $f(x) = ax^2 + bx + c$, where a, b and c are constants. What is the value of $\dfrac{b + c}{a}$?

Answer:

608. **AM-228** · Hard · No calculator needed

Let $g(x) = 7 + 9\left(\dfrac{1}{4}\right)^x$ for all real numbers x. Consider the following statements.

I. $g(x) > 7$ for every real number x.

II. The minimum value of g is 7.

III. g is a decreasing function.

Which of the statements must be true?

A) I and III only **B)** I and II only

C) II and III only **D)** I, II, and III

609. **AM-230** · Hard · No calculator needed

The temperature T, in degrees Fahrenheit, of a sample t minutes after it is placed in a chamber is modeled by one of the functions below for $t \geq 0$. For which model is 85 the greatest temperature the sample actually reaches?

A) $T = 25 + 60(0.8)^t$ **B)** $T = 85 - 60(0.8)^t$

C) $T = 40 + 45(1.05)^t$ **D)** $T = 25 + 85(0.8)^t$

610. **AM-233** · Hard · No calculator needed

The polynomial function p is defined by $p(x) = -(x + 3)^2(x - 1)^3(x - 4)^2$. Consider the following statements.

I. $p(x) \geq 0$ for all $x < 1$.

II. The graph of $y = p(x)$ crosses the x-axis at $x = 4$.

III. $p(x) \to \infty$ as $x \to -\infty$.

Which of the statements must be true?

A) I only **B)** I and II only

C) I and III only **D)** II and III only

611. **AM-234** · Hard · No calculator needed

The quadratic function $h(x) = ax^2 + bx + c$, where a, b and c are constants, satisfies $h(-3) = h(1)$, $h(0) > h(-1)$ and $h(-1) > 0$. Consider the following statements.

I. $b > 0$

II. $c > 0$

III. The equation $h(x) = 0$ has two distinct real solutions.

Which of the statements must be true?

A) I only **B)** I and II only

C) II and III only **D)** I, II, and III

612. **AM-236** · Hard · SPR · No calculator needed

The quadratic function f has its vertex at $(3, 4)$. The function g is defined by $g(x) = -2f(x + 1) + 7$, and the graph of $y = g(x)$ passes through the point $(4, 2)$. What is the value of $f(9) - f(2)$?

Answer:

613. **AM-238** · Hard · SPR · No calculator needed

A rocket's height, in feet, t seconds after launch is modeled by $h(t) = -16t^2 + bt + c$, where b and c are constants. The rocket is launched from a height of at most 94 feet, and the maximum height it reaches is 130 feet. For how many seconds is the rocket at a height of at least 94 feet?

Answer:

614. **AM-240** · Hard · SPR · Calculator

An investment grows by 6% every half-year. If V_0 is its initial value, the value after t years is $V(t)$. After how many years will the value first reach three times the initial value? Round your answer to the nearest hundredth.

Answer:

615. **AM-241** · Hard · SPR · No calculator needed

The quadratic function $f(x) = 2x^2 + bx + c$ has $(x - 9)$ as a factor. The function g is defined by $g(x) = f(x+3)$, and the graph of g in the xy-plane has its vertex at $x = 1$. What is the other zero of f?

Answer:

616. AM-242 · Hard · No calculator needed

$$T(t) = 24 + 66(0.85)^t$$

The function T models the temperature, in degrees Celsius, of a cup of tea t minutes after it is placed in a room. Which of the following is the best interpretation of the number 0.85 in this context?

A) T falls 15% per minute
B) the gap $T - 24$ falls 15% per minute
C) $T(1) = 0.85\,T(0)$
D) T levels off at 85% of $T(0)$

617. AM-244 · Hard · No calculator needed

The quadratic function f satisfies $f(-1) = f(7)$, $f(0) = -4$ and $f(4) = 12$. What is the value of $f(10)$?

A) -98 **B)** -84
C) 0 **D)** 98

618. AM-246 · Hard · No calculator needed

The table gives values of a quadratic function g at equally spaced inputs.

x	2	5	8	11
$g(x)$	44	95	200	359

What is the value of $g(17)$?

A) 677 **B)** 572
C) 785 **D)** 839

619. AM-247 · Hard · Calculator

The concentration of a medication in the bloodstream, in milligrams per liter, is modeled by $C(t) = 18\left(\frac{1}{3}\right)^{\frac{t-4}{2}}$, where $t \geq 0$ is the number of hours after the first measurement. Which of the following is an equivalent form of $C(t)$ that displays the maximum value of C as a constant or coefficient?

A) $486 \cdot 3^{-\frac{t+2}{2}}$ **B)** $18 \cdot 3^{\frac{4-t}{2}}$
C) $162 \cdot 3^{-\frac{t}{2}}$ **D)** $54 \cdot 3^{\frac{2-t}{2}}$

620. AM-249 · Hard · No calculator needed

A water balloon is launched upward from a rooftop. Its height above the ground, in feet, t seconds after launch is modeled by a quadratic function h. The maximum height of 125 feet is reached at a time between $t = 1$ and $t = 6$, and $h(1) = 80$ and $h(6) = 105$. What is the height, in feet, of the balloon at the moment of launch?

A) 80 **B)** 45
C) 75 **D)** 37

621. AM-251 · Hard

The function f is defined by $f(x) = x^2 - 4x + 1$. The function g is defined by $g(x) = f(x + 3)$. What are the coordinates of the vertex of the graph of g in the xy-plane?

A) $(5, -3)$ **B)** $(-1, -3)$
C) $(2, 0)$ **D)** $(-1, 0)$

622. AM-252 · Hard · SPR

The functions f and g are defined by $f(x) = 3x - 2$ and $g(x) = x^2 + 1$. What is the value of $f(g(2)) - g(f(2))$?

Answer:

623. AM-253 · Hard

The graph of $y = f(x)$ in the xy-plane has x-intercepts at $x = -2$ and $x = 6$. Which of the following could be the x-intercepts of the graph of $y = f(2x)$?

A) $x = -4$ and $x = 12$ **B)** $x = -1$ and $x = 3$
C) $x = 0$ and $x = 8$ **D)** $x = -2$ and $x = 6$

624. AM-254 · Hard · SPR

The quadratic function f has a minimum value of -5 at $x = 4$. The function g is defined by $g(x) = -2f(x - 3) + 1$. What is the maximum value of g?

Answer:

625. AM-255 · Hard

The function f is defined by $f(x) = |x - 3|$ and the function g is defined by $g(x) = f(x + 1) - 2$. What are the coordinates of the vertex of the graph of g?

A) $(4, -2)$ **B)** $(2, -2)$
C) $(2, 2)$ **D)** $(4, 2)$

626. AM-256 · Hard · SPR

The function f is defined by $f(x) = 2x + 1$. The function g is defined by $g(x) = f(f(x)) - f(x)$. What is the value of $g(3)$?

Answer:

627. AM-257 · Hard

The function h is defined by $h(x) = \sqrt{x - 2} + 1$. The function g is defined by $g(x) = h(x + 5)$. What is the domain of g?

A) $x \geq -3$ **B)** $x \geq 2$
C) $x \geq 7$ **D)** $x \geq -5$

628. AM-258 · Hard · SPR

The function f is defined by $f(x) = (x-1)^2 + 4$. The function g is defined by $g(x) = f(x-k) + m$, where k and m are constants. The vertex of the graph of g is $(6, -3)$. What is the value of $k + m$?

Answer:

629. AM-259 · Hard

The functions f and g are defined by $f(x) = x + 3$ and $g(x) = 2x - 1$. For which values of x is $f(g(x)) = g(f(x))$?

A) Only $x = 0$

B) Only $x = 3$

C) All real numbers

D) No real numbers

630. AM-260 · Hard · SPR

The function f satisfies $f(1) = 3$ and $f(3) = 9$. The functions g and h are defined by $g(x) = f(x) + 4$ and $h(x) = f(x-2)$. What is the value of $g(3) + h(3)$?

Answer:

631. AM-261 · Hard

The graph of $y = f(x)$ in the xy-plane has its vertex at $(-2, 5)$. What are the coordinates of the vertex of the graph of $y = -f(x+1) + 3$?

A) $(-3, -2)$

B) $(-1, -2)$

C) $(-3, 8)$

D) $(-1, 8)$

632. AM-262 · Hard · SPR

The graph of $g(x) = (x-h)^2 + k$ in the xy-plane has a minimum value of 2 and passes through the points $(1, 6)$ and $(5, 6)$. What is the value of $h + k$?

Answer:

633. AM-270 · Hard · SPR

The function f satisfies $f(x+3) = -f(x)$ for all real numbers x, and $f(2) = 7$. What is the value of $f(23)$?

Answer:

634. AM-271 · Hard · SPR

The function g is defined for all integers and satisfies $g(x+5) = g(x)$. The table gives some values of g.

x	1	2	3	4	5
$g(x)$	4	-1	6	0	2

What is the value of $g(87) + g(103) + g(55)$?

Answer:

635. AM-272 · Hard · SPR

Let $f(x) = x^3 - 12x$. For how many integer values of k does the equation $f(x) = k$ have exactly three distinct real solutions?

Answer:

636. AM-276 · Hard · SPR

The function f is defined by $f(x) = 4x^2 - 12x + 5$. For a certain real number a, $f(a) = f(a+4)$. What is the value of a?

Answer:

637. AM-280 · Hard · SPR

The function f is linear and $g(x) = \dfrac{f(x)}{x+3}$. The table gives two values of g.

x	1	3
$g(x)$	3	$\frac{5}{2}$

What is the y-intercept of the graph of f in the xy-plane?

Answer:

638. AM-281 · Hard · SPR

For $x \geq 0$, the function h is defined by $h(x) = 12\left(\dfrac{1}{4}\right)^{x-2}$. What is the maximum value of h on this domain?

Answer:

639. AM-284 · Hard · SPR

The function f is defined by $f(x) = -(x-2)^2 + 7$. Let k be the maximum value of f. The table gives values of the function g.

x	4	5	6	7	8
$g(x)$	3	11	-2	9	14

What is the value of $g(k)$?

Answer:

640. AM-285 · Hard · SPR

The function f is defined by $f(x) = x^2 - 4x$. For what value of a is the average rate of change of f from $x = a$ to $x = a + 2$ equal to 6?

Answer:

641. AM-286 · Hard · SPR

The average rate of change of $h(x) = x^3$ from $x = 1$ to $x = k$, where $k > 1$, is 13. What is the value of k?

Answer:

642. AM-288 · Hard

The graph of $y = f(x)$ in the xy-plane has x-intercepts at $x = -2$ and $x = 5$. The function g is defined by $g(x) = f(2x)$. What are the x-intercepts of the graph of $y = g(x)$?

A) $x = -2$ and $x = \frac{5}{2}$ **B)** $x = -4$ and $x = 10$

C) $x = -1$ and $x = \frac{5}{2}$ **D)** $x = 0$ and $x = 3$

643. AM-289 · Hard · SPR

The functions f and g are defined for the values in the table.

x	1	2	3	4
$f(x)$	3	4	1	2
$g(x)$	4	1	2	3

What is the value of $g\left(f^{-1}(g(3))\right)$?

Answer:

644. AM-290 · Hard

The function f is odd, which means $f(-x) = -f(x)$ for every x. Which of the following functions must be even, meaning $h(-x) = h(x)$ for every x?

A) $[f(x)]^2$ **B)** $f(f(x))$

C) $f(x) + 1$ **D)** $f(x) + x$

645. AM-292 · Hard

Let $f(x) = \sqrt{x-1}$ and $g(x) = x^2 - 5$. Which describes all real values of x for which $f(g(x))$ is defined?

A) $x \geq 1$ **B)** $-\sqrt{6} \leq x \leq \sqrt{6}$

C) all real numbers **D)** $x \leq -\sqrt{6}$ or $x \geq \sqrt{6}$

646. AM-293 · Hard · SPR

The function f is defined by $f(x) = \frac{2x+1}{x-3}$ for $x \neq 3$. What is the value of $f^{-1}(5)$?

Answer:

647. AM-294 · Hard

The graph of $y = f(x)$ passes through the point $(4, -2)$. Which point must lie on the graph of $y = 3f(x-1) + 5$?

A) $(5, 11)$ **B)** $(5, -1)$

C) $(3, -1)$ **D)** $(3, 11)$

648. AM-298 · Hard · SPR

The graph of the quadratic function f has vertex $(4, -8)$ and passes through the point $(0, 8)$. The function g is defined by $g(x) = f(2x)$. What is the value of $g(1)$?

Answer:

649. AM-299 · Hard · SPR

The graph of $y = f(x)$ has x-intercepts at $x = 1$ and $x = 6$. The function g is defined by $g(x) = f(3 - x)$. What is the greatest x-intercept of the graph of $y = g(x)$?

Answer:

650. AM-300 · Hard · SPR

The function f is even, so $f(-x) = f(x)$ for every x. The function g is defined by $g(x) = f(x) + f(-x)$. If $f(3) = 7$ and $f(-5) = 2$, what is the value of $g(3) + g(5)$?

Answer:

651. AM-304 · Hard · SPR · No calculator needed

The function g is defined by $g(x) = \frac{x^2 - kx - 18}{x+3}$, where k is a constant. The graph of $y = g(x)$ in the xy-plane is a line with exactly one point removed. What is the y-coordinate of the removed point?

Answer:

652. AM-305 · Hard · No calculator needed

The function h is defined by $h(x) = \frac{x^2 - 4x - 12}{x^2 - 36}$. What is the sum of all values of x for which $h(x) = 0$?

A) 4 **B)** -8

C) 6 **D)** -2

653. AM-306 · Hard · No calculator needed

The function h is defined by $h(x) = x^2 - 9$ for $x < 2$ and $h(x) = -2x + 10$ for $x \geq 2$. What is the sum of all values of m for which $h(m) = 0$?

A) 0 **B)** 2

C) 5 **D)** 8

654. AM-308 · Hard · No calculator needed

The function f is defined by $f(x) = x^2 - 1$. How many distinct real solutions does the equation $f(f(x)) = 0$ have?

A) 1 **B)** 4

C) 2 **D)** 3

655. **AM-313** · Hard · No calculator needed

A kiosk's daily profit, in hundreds of dollars, is modeled by $P(x) = -2x^2 + 12x + 5$, where x is the price of an item in dollars. The model is used only for $4 \leq x \leq 7$. What is the greatest daily profit, in hundreds of dollars, predicted by the model on this interval?

A) 21 **B)** 23

C) -9 **D)** 5

656. **AM-315** · Hard · No calculator needed

When the polynomial $p(x)$ is divided by $x - 5$, the remainder is 8. What is the remainder when $p(2x - 1)$ is divided by $x - 3$?

A) 15 **B)** 8

C) 5 **D)** 3

657. **AM-316** · Hard · No calculator needed

For all real numbers x, $f(2x - 1) = 4x^2 + 3$. Which of the following defines $f(x)$?

A) $f(x) = x^2 + 4$ **B)** $f(x) = x^2 + 2x + 4$

C) $f(x) = 4x^2 + 3$ **D)** $f(x) = 4x^2 - 4x + 4$

658. **AM-319** · Hard · SPR · No calculator needed

The function f is defined by $f(x) = a\big((x-3)^2 - b\big)\big((x-3)^2 - c\big)$, where a, b and c are constants. If $f(1) = 40$ and $f(6) = -14$, what is the value of $f(5) + f(0)$?

Answer:

659. **AM-323** · Hard · No calculator needed

The function f is defined by $f(x) = 3x^2 - 5x + 2$. Which expression is equivalent to $f(a + 1) - f(a)$?

A) 0 **B)** $6a - 2$

C) $6a + 3$ **D)** $6a + 8$

660. **AM-328** · Hard · No calculator needed

The functions f and g are defined by $f(x) = 7 \cdot 4^x$ and $g(x) = f(x - 1.5)$. For all values of x, $g(x) = k \cdot f(x)$, where k is a constant. What is the value of k?

A) -6 **B)** $\frac{7}{8}$

C) $\frac{1}{8}$ **D)** 8

661. **AM-329** · Hard · No calculator needed

The function f is defined by $f(x) = 2(x - 1)(x - 7)$. For what value of k does the graph of $y = f(x) + k$ have exactly one x-intercept?

A) 9 **B)** 14

C) 18 **D)** -18

662. **AM-330** · Hard · No calculator needed

The function g is defined by $g(x) = 3^x$. If $g(a) = 4$, what is the value of $g(2a - 1)$?

A) $\frac{16}{3}$ **B)** 15

C) $\frac{8}{3}$ **D)** 7

Geometry and Trigonometry (274)

Area and volume (67)

663. GT-007 · Medium

Figure (described): a rectangle measuring 12 meters by 8 meters, with a semicircle of diameter 8 meters attached along one of the 8-meter sides, bulging outward

A school courtyard is shaped like a 12-meter by 8-meter rectangle with a semicircular reading nook attached to one of the 8-meter sides, as shown. The semicircle has a diameter of 8 meters. What is the total area of the courtyard, in square meters?

A) $96 + 4\pi$ **B)** $96 + 8\pi$

C) $96 + 16\pi$ **D)** 104π

664. GT-008 · Medium

A cylindrical water tank supplying a public fountain project in Jeddah has a radius of 15 meters and a volume of 4500π cubic meters. What is the height of the tank, in meters?

A) 10 **B)** 15

C) 20 **D)** 30

665. GT-026 · Medium

A rectangular storage box has length 8 cm, width 5 cm, and height 4 cm. What is the total surface area of the box, in square centimeters?

A) 92 **B)** 144

C) 160 **D)** 184

666. GT-032 · Medium

A cube has a total surface area of 150 square centimeters. What is the volume of the cube, in cubic centimeters?

A) 25 **B)** 75

C) 125 **D)** 150

667. GT-041 · Medium · SPR

A rectangular desalinated-water storage tank at a facility near Yanbu has a base measuring 8 feet by 5 feet. The tank is filled with water to a depth of 4.5 feet. Using a water density of 62.4 pounds per cubic foot, what is the weight of the water in the tank, in pounds?

Answer:

668. GT-058 · Medium · SPR

A cylindrical water tank near Riyadh has a base circumference of 44π meters and a height of 15 meters. What is the volume, in cubic meters, of the tank? (Express your answer in terms of π.)

Answer:

669. GT-060 · Medium · SPR

A design team is producing rectangular Saudi flag replica banners of two sizes. Rectangle A is similar to rectangle B, and the side lengths of rectangle A are 3 times the corresponding side lengths of rectangle B. If the area of rectangle B is 24 square centimeters, what is the area, in square centimeters, of rectangle A?

Answer:

670. GT-061 · Medium · SPR

For a school engineering fair project in Jeddah, a team builds a right circular cone-shaped sand mound model with a volume of 48π cubic centimeters and a height of 9 centimeters. What is the radius, in centimeters, of the base of the cone?

Answer:

671. GT-077 · Medium

A cylindrical water tank has a radius of 3 meters and a height of 8 meters. Which of the following is closest to the volume of the tank, in cubic meters? (Use $\pi \approx 3.14$.)

A) 150.7 **B)** 188.4

C) 226.1 **D)** 254.3

672. GT-078 · Medium

A trapezoid has parallel sides of length 14 cm and 22 cm, and a height of 9 cm. What is the area of the trapezoid, in cm^2?

A) 144 **B)** 153

C) 162 **D)** 171

673. GT-090 · Medium · SPR

For a particular square, the perimeter is numerically equal to the area. What is the perimeter of the square?

Answer:

674. GT-093 · Medium

A circular coaster has a diameter of 10 cm and a height (thickness) of 3 mm. If 8 coasters are stacked on top of each other, what is the volume of the stack?

A) 60π cm^3 **B)** 75π cm^3

C) 120π cm^3 **D)** 150π cm^3

675. GT-096 · Medium · SPR

If the surface area of a cube is 150 square centimeters, what is the volume of the cube, in cubic centimeters?

Answer:

676. GT-110 · Medium · SPR

Two similar rectangular prisms have volumes 972 and 36. An edge of the larger prism has length 12. What is the length of the corresponding edge of the smaller prism?

Answer:

677. GT-114 · Medium

The edges of cubes A and B are in the ratio $\frac{3}{88}$ (edge of A to edge of B). What is the ratio of the surface area of B to the surface area of A?

A) $\frac{9}{7{,}744}$ **B)** $\frac{3}{88}$

C) $\frac{88}{3}$ **D)** $\frac{7{,}744}{9}$

678. GT-201 · Medium · SPR

A right circular cylinder has volume 90π and height 10. The lateral surface area (excluding the bases) of the cylinder is $k\pi$. What is the value of k?

Answer:

679. GT-202 · Medium · SPR

A right circular cone has base radius 6 and slant height 10. The volume of the cone is $k\pi$. What is the value of k?

Answer:

680. GT-203 · Medium · SPR

A sphere has surface area 144π. The volume of the sphere is $k\pi$. What is the value of k?

Answer:

681. GT-205 · Medium · SPR

A rectangular box has a base that measures 6 by 8. The distance from one corner of the box to the opposite corner (the space diagonal) is 26. What is the height of the box?

Answer:

682. GT-208 · Medium · SPR

The space diagonal of a cube (the segment joining opposite vertices) has length $9\sqrt{3}$. What is the volume of the cube?

Answer:

683. GT-209 · Medium · SPR

A box has interior dimensions 10 cm by 12 cm by 16 cm. Cubes with edge length 2 cm are packed into the box with no gaps. How many cubes fit?

Answer:

684. GT-210 · Medium · SPR

A cone and a cylinder each have radius 5 and height 12. The cone is filled with water and the water is poured into the cylinder, repeatedly, until the cylinder is full. How many full cones of water does it take to fill the cylinder?

Answer:

685. GT-238 · Medium · No calculator needed

A triangle has two sides of lengths 10 and 12, and the angle between them measures $150°$. What is the area of the triangle?

A) $30\sqrt{3}$ **B)** 30

C) 15 **D)** 60

686. GT-248 · Medium · Calculator

The radius of a right circular cylinder is increased by 20% and its height is decreased by 25%. What is the effect on the volume of the cylinder?

A) It does not change. **B)** It increases by 8%.

C) It decreases by 5%. **D)** It increases by 19%.

687. GT-267 · Medium · No calculator needed

A parallelogram has sides of lengths 12 and 20. The cosine of one of its acute angles is 0.8. What is the area of the parallelogram?

A) 72 **B)** 144

C) 192 **D)** 240

688. GT-009 · Hard

A scale model of a cross-sectional segment of NEOM's "The Line" is geometrically similar to the actual structure, with a scale factor of 1 to 20 (model to actual). If the volume of the model is 3 cubic meters, what is the volume of the actual segment, in cubic meters?

A) 60 **B)** $1{,}200$

C) $8{,}000$ **D)** $24{,}000$

689. GT-010 · Hard

Figure (described): a cylindrical base topped by a hemispherical dome, both sharing the same radius

A mosque prayer hall is designed as a cylindrical base topped by a hemispherical dome of the same radius. The cylindrical base has radius 6 meters and height 10 meters, and the dome is a hemisphere with radius 6 meters. What is the total volume of the structure, in cubic meters, in terms of π?

A) 360π **B)** 432π

C) 504π **D)** 576π

690. GT-016 · Hard

Figure (described): a cylindrical grain silo topped by a cone, both sharing the same radius

A grain silo near Tabuk is designed as a cylindrical base topped by a conical roof of the same radius. The cylindrical base has radius 4 meters and height 12 meters, and the conical roof has radius 4 meters and height 3 meters. What is the total volume of the structure, in cubic meters, in terms of π?

A) 176π **B)** 192π

C) 208π **D)** 240π

691. GT-019 · Hard

A stone block used in a heritage restoration project in AlUla is a rectangular prism measuring 8 meters by 8 meters by 20 meters, with a cylindrical channel of radius 2 meters drilled completely through its 20-meter length. What is the volume of the remaining stone, in cubic meters, in terms of π?

A) $1280 - 80\pi$ **B)** $1280 - 160\pi$

C) $1280 - 20\pi$ **D)** $1200 - 80\pi$

692. GT-022 · Hard

A cone has base radius 6 centimeters and height 8 centimeters, apex pointing up. A plane parallel to the base cuts the cone at a height of 4 centimeters from the apex, removing the smaller cone on top. What is the volume of the removed smaller cone, in cubic centimeters, in terms of π?

A) 12π **B)** 24π

C) 48π **D)** 96π

693. GT-027 · Hard

A cylindrical water tank has a radius of 5 meters and a height of 12 meters. What is the total surface area of the tank, including both circular bases, in square meters, in terms of π?

A) 120π **B)** 150π

C) 170π **D)** 220π

694. GT-036 · Hard · SPR

In a figure, square $WXYZ$ has an area of 400 square units. Side AB of parallelogram $ABCD$ has the same length as a side of the square. Right triangle ABE, formed by dropping an altitude from vertex B to the extension of side AD, has hypotenuse AB and one leg $AE = 12$. If side AD of the parallelogram has length 33, what is the area of parallelogram $ABCD$?

Answer:

695. GT-037 · Hard · SPR

A rectangular vegetable garden at a NEOM community project originally has length s meters and width $0.4s$ meters. A surrounding path is widened, which decreases the garden's length by $0.1s$ and its width by $0.1s$. What is the percent decrease in the garden's area?

Answer:

696. GT-059 · Hard

A hemisphere is half of a sphere. If a hemisphere has a radius of 18 inches, which of the following is closest to the volume, in cubic inches, of the hemisphere?

A) $2{,}400$ **B)** $6{,}100$

C) $12{,}200$ **D)** $24{,}400$

697. GT-079 · Hard

A conical grain storage silo has a radius of 6 meters and a height of 10 meters. Which of the following is closest to the volume of the silo, in cubic meters? (Use $\pi \approx 3.14$.)

A) 282.6 **B)** 376.8

C) 452.2 **D)** 565.2

698. GT-080 · Hard

A spherical water tank has a radius of 5 meters. Which of the following is closest to the surface area of the tank, in square meters? (Use $\pi \approx 3.14$ and the formula $SA = 4\pi r^2$.)

A) 235.5 **B)** 261.7

C) 314.0 **D)** 392.5

699. GT-092 · Hard

Side AB of rectangle $ABCD$ measures 8 units. Side BC of the rectangle is shared with side BC of square $BCEF$. If the area of square $BCEF$ is 45 square units, what is the area of rectangle $ABCD$?

A) $12\sqrt{5}$ square units **B)** $24\sqrt{5}$ square units

C) $45 + 8\sqrt{5}$ square units **D)** 80 square units

700. GT-095 · Hard · SPR

A hexagon is formed by arranging 6 equilateral triangles, each of which has a vertex at the center of the hexagon. If the length of one side of a triangle is 6, what is the area, in square units, of the hexagon?

Answer:

701. GT-113 · Hard

Two similar rectangles have areas 630 and $2{,}520$. The perimeter of the smaller rectangle is 210, and the perimeter of the larger is n. What is the value of n?

A) $2{,}100$ **B)** $1{,}680$

C) 840 **D)** 420

702. GT-116 · Hard · SPR

A right square pyramid has a base area of 100 and a total surface area of $100 + 100\sqrt{5}$. What is the height of the pyramid?

Answer:

703. GT-119 · Hard · No calculator needed

Two similar rectangular crates have surface areas of 144 square feet and 400 square feet. The smaller crate has a volume of 108 cubic feet. What is the volume, in cubic feet, of the larger crate?

A) 1500 **B)** 180

C) 300 **D)** 500

704. GT-120 · Hard · SPR · No calculator needed

Two similar solid blocks have surface areas of 208 square centimeters and 325 square centimeters. The larger block has a volume of 375 cubic centimeters. What is the volume, in cubic centimeters, of the smaller block?

Answer:

705. GT-130 · Hard · Calculator

A closed right circular cylinder has a volume of 48π cubic centimeters and a total surface area of 56π square centimeters. The height of the cylinder is greater than its diameter. What is the height, in centimeters, of the cylinder?

A) 12 **B)** 2

C) 3 **D)** 4

706. GT-132 · Hard · No calculator needed

Sphere A has a great circle with circumference 8π centimeters. The surface area of sphere B is 80π square centimeters greater than the surface area of sphere A. What is the ratio of the volume of sphere B to the volume of sphere A?

A) $\frac{9}{4}$ **B)** $\frac{27}{8}$

C) $\frac{5}{4}$ **D)** $\frac{3}{2}$

707. GT-134 · Hard · Calculator

A right pyramid has a square base. The diagonal of the base is $10\sqrt{2}$ centimeters and the volume of the pyramid is 400 cubic centimeters. What is the lateral surface area, in square centimeters, of the pyramid (the four triangular faces, not the base)?

A) 260 **B)** 240

C) 130 **D)** 360

708. GT-138 · Hard · SPR · Calculator

A metal pipe is a hollow cylinder of length 15 centimeters and outer radius 10 centimeters. The wall has a uniform thickness of t centimeters, where $0 < t < 10$. The volume of metal in the pipe is 765π cubic centimeters. What is the value of t?

Answer:

709. GT-139 · Hard · No calculator needed

A right pyramid has a rectangular base. Its height is 12 centimeters, and the slant heights of its triangular faces are 13 centimeters for one pair of opposite faces and 20 centimeters for the other pair. What is the total surface area, in square centimeters, of the pyramid, including the base?

A) 866 **B)** 936

C) 1160 **D)** 616

710. GT-142 · Hard · SPR · Calculator

A raised garden bed is filled with soil. A vertical cross-section of the soil, perpendicular to the 12-foot length of the bed, is a trapezoid: the bed is 8 feet wide, and the soil is 1.5 feet deep along one side and 3.5 feet deep along the opposite side, with the surface sloping uniformly between them. Soil is sold at $\$45$ per cubic yard and billed for the exact volume (1 yard $= 3$ feet). What is the cost, in dollars, of the soil needed?

Answer:

711. GT-145 · Hard · Calculator

Cylinder P has radius 2 centimeters and height 6 centimeters. Cylinder Q is similar to cylinder P and has a total surface area of 200π square centimeters. What is the volume, in cubic centimeters, of cylinder Q?

A) 375π **B)** 600π

C) 60π **D)** 150π

712. GT-149 · Hard · SPR · Calculator

Rectangular box P has dimensions 3 inches by 4 inches by 6 inches. Box Q is similar to box P and has a volume of 1125 cubic inches. What is the total surface area, in square inches, of box Q?

Answer:

713. GT-153 · Hard · Calculator

A brass bushing is a solid right circular cylinder with diameter 60 millimeters and height 40 millimeters. A cylindrical hole of diameter 20 millimeters is drilled into one flat face, along the axis, to a depth of 25 millimeters. The density of brass is 7.8 grams per cubic centimeter. Which of the following is closest to the mass of the bushing, in grams? (1 cm $= 10$ mm)

A) 821 **B)** 882

C) $82{,}090$ **D)** 784

714. GT-158 · Hard · SPR · No calculator needed

A sphere is inscribed in a right circular cone whose base has radius 5 and whose height is 12, so that the sphere is tangent to the base and to the lateral surface of the cone. What fraction of the volume of the cone is occupied by the sphere? (Enter your answer as a fraction.)

Answer:

715. GT-163 · Hard · Calculator

A museum has a small solid model of a sculpture and a full-size copy; the two are similar solids. The sealant needed to coat the whole surface is proportional to surface area. The model needs 1.6 liters of sealant and the full-size copy needs 14.4 liters. The model has a volume of 2.5 cubic feet. What is the volume, in cubic feet, of the full-size copy?

A) 7.5 **B)** 22.5

C) 67.5 **D)** 202.5

716. GT-165 · Hard · SPR · No calculator needed

Two aquariums are similar in shape. The smaller aquarium holds 250 liters and has 400 square decimeters of glass in total. The larger aquarium holds 432 liters. How many square decimeters of glass does the larger aquarium have in total? (Note that 1 liter $= 1$ cubic decimeter.)

Answer:

717. GT-171 · Hard · No calculator needed

A roof ornament is a solid pyramid whose base is a square of area 100 square inches. The apex is directly above the center of the base, and the volume of the pyramid is 400 cubic inches. What is the combined area, in square inches, of the four triangular lateral faces?

A) 240 **B)** 130

C) 360 **D)** 260

718. GT-204 · Hard · SPR

Two similar rectangular prisms have volumes in the ratio $8 : 27$. The smaller prism has a surface area of 40. What is the surface area of the larger prism?

Answer:

719. GT-206 · Hard · SPR

A rectangular tank has a base that is 40 centimeters by 30 centimeters. A solid cube with edge 12 centimeters is placed in the tank and sinks to the bottom, fully under water. By how many centimeters does the water level rise?

Answer:

720. GT-207 · Hard · SPR

A square pyramid has a base with side length 12 and a height of 8. What is the lateral surface area of the pyramid (the four triangular faces, excluding the base)?

Answer:

721. GT-211 · Hard · SPR

The radius of a sphere is increased by 20%. By what percent does the volume of the sphere increase? (Enter as a decimal.)

Answer:

722. GT-212 · Hard · SPR

A solid is formed by placing a hemisphere of radius 3 on top of a right circular cylinder of radius 3 and height 7. The volume of the solid is $k\pi$. What is the value of k?

Answer:

723. GT-239 · Hard · SPR · No calculator needed

A rhombus has side length 13 and area 120. What is the sum of the lengths of its diagonals?

Answer:

724. GT-240 · Hard · No calculator needed

A right circular cone stands with its vertex pointing up. Its base radius is 6 and its height is 12. Water fills the cone to a depth of 6, measured from the base. What is the volume of the water?

A) 18π **B)** 72π

C) 108π **D)** 126π

725. GT-241 · Hard · SPR · No calculator needed

A sector of a circle has radius 10 and central angle $216°$. The sector is rolled into a right circular cone so that its two straight edges meet. The volume of the cone is $k\pi$. What is the value of k?

Answer:

726. GT-244 · Hard · No calculator needed

A square has one side on the diameter of a semicircle of radius 10, and its other two vertices lie on the arc. What is the area of the square?

A) 100 **B)** 50

C) 80 **D)** 200

727. GT-249 · Hard · No calculator needed

Sphere Q has volume 36π. Sphere P has a radius 7 times the radius of sphere Q. The spheres are externally tangent at point T. Segment $\overline{MT}$ is a diameter of sphere P and segment $\overline{TN}$ is a diameter of sphere Q, with M, T and N on one line. What is the length of $\overline{MN}$?

A) 66 **B)** 45

C) 24 **D)** 48

728. GT-252 · Hard · SPR · No calculator needed

Two identical right rectangular prisms each have a square base with side length s and a height of 35. Each prism has surface area K. The prisms are glued together base to base, and the new solid has surface area $\frac{15}{8}K$. What is the value of s?

Answer:

729. GT-253 · Hard · Calculator

Square pyramids X and Y have the same base, a square with side length 10. Pyramid X has height 12 and pyramid Y has height 24. The total surface area of Y is approximately what percent greater than the total surface area of X?

A) 89% **B)** 100%

C) 64% **D)** 50%

Lines, angles, and triangles (58)

730. GT-011 · Medium

Figure (described): line ℓ parallel to line m, both cut by transversal t; the angle formed at the intersection with ℓ measures $(3x+10)°$, and the corresponding angle formed at the intersection with m measures $(5x-30)°$

Line ℓ is parallel to line m and both are intersected by transversal t. One angle formed at the intersection with ℓ measures $(3x+10)°$, and its corresponding angle at the intersection with m measures $(5x-30)°$. What is the value of x?

A) 10 **B)** 15

C) 20 **D)** 40

731. GT-020 · Medium

In triangle ABC, the measure of angle A is twice the measure of angle B, and the measure of angle C is $30°$ more than the measure of angle B. What is the measure of angle A, in degrees?

A) 37.5 **B)** 67.5

C) 75 **D)** 82.5

732. GT-024 · Medium

A regular polygon has exterior angles that each measure $24°$. How many sides does the polygon have?

A) 12 **B)** 15

C) 18 **D)** 24

733. GT-033 · Medium

In a figure, angles b and d together form a straight line, with $d = 112°$. Angles a, b, and c are the three interior angles of a triangle, where $a = 3c$. What is the value of $a + b$?

A) 140 **B)** 146

C) 152 **D)** 158

734. GT-034 · Medium · SPR

What is the positive difference, in degrees, between the measure of an exterior angle of a regular octagon (8 sides) and the measure of an exterior angle of a regular decagon (10 sides)?

Answer:

735. GT-048 · Medium · SPR

At a certain time of day in Jeddah, a 6-meter flagpole casts a shadow that is 4 meters long. At the same time, a nearby minaret casts a shadow that is 30 meters long. What is the height, in meters, of the minaret?

Answer:

736. GT-049 · Medium

Triangle DEF is similar to triangle GHI, where D, E, and F correspond to G, H, and I, respectively. The measure of angle D is $48°$, and the measure of angle E is $63°$. What is the measure of angle H?

A) $48°$ **B)** $63°$

C) $69°$ **D)** $117°$

737. GT-062 · Medium

In a triangle, the measures of the three interior angles are x, $2x$, and $(3x - 30)$ degrees. What is the value of x?

A) 30 **B)** 32

C) 35 **D)** 40

738. GT-063 · Medium

Line ℓ is parallel to line m, and a transversal crosses both lines. One angle formed at line ℓ measures $(3x + 15)°$, and the corresponding angle formed at line m measures $(5x - 25)°$. What is the value of x?

A) 15 **B)** 18

C) 20 **D)** 25

739. GT-081 · Medium

Two lines intersect, forming four angles. One angle measures $(5x + 10)°$ and the angle vertical to it measures $(3x + 30)°$. What is the value of x?

A) 8 **B)** 10

C) 12 **D)** 15

740. GT-082 · Medium

A triangle has side lengths of 7, 10, and x, where x is an integer. What is the greatest possible value of x?

A) 15 **B)** 16

C) 17 **D)** 18

741. GT-094 · Medium

Triangles ABC and DEF are similar. Side AB of triangle ABC corresponds to side DE of triangle DEF. If the length of $\overline{AB}$ is 12 units and the length of $\overline{DE}$ is 8 units, then the area of triangle DEF is equal to

A) $\frac{2}{3}$ (area of triangle ABC)

B) $\frac{4}{9}$ (area of triangle ABC)

C) $\frac{9}{4}$ (area of triangle ABC)

D) $\frac{16}{9}$ (area of triangle ABC)

742. GT-213 · Medium · SPR

Two parallel lines are cut by a transversal. A pair of alternate interior angles measure $(3x + 10)°$ and $(5x - 30)°$. What is the measure, in degrees, of each of these angles?

Answer:

743. GT-214 · Medium · SPR

An exterior angle of a triangle measures $(7x - 4)°$. The two remote interior angles measure $(2x + 15)°$ and $(3x + 9)°$. What is the measure, in degrees, of the exterior angle?

Answer:

744. GT-215 · Medium · SPR

In an isosceles triangle, the two base angles each measure $(x+20)°$ and the vertex angle measures $(4x-10)°$. What is the measure, in degrees, of the vertex angle?

Answer:

745. GT-216 · Medium · SPR

In triangle ABC, point D lies on $\overline{AB}$ and point E lies on $\overline{AC}$ such that $\overline{DE}$ is parallel to $\overline{BC}$. If $AD = 6$, $DB = 4$ and $AE = 9$, what is the length of $\overline{EC}$?

Answer:

746. GT-218 · Medium · SPR

Each interior angle of a regular polygon measures $140°$. How many sides does the polygon have?

Answer:

747. GT-219 · Medium

A trapezoid has parallel sides of length 10 and 24. A segment joins the midpoints of the two non-parallel sides. What is the length of this segment?

A) 34 **B)** 14

C) 7 **D)** 17

748. GT-221 · Medium · SPR

In the xy-plane, a triangle has vertices $(0, 0)$, $(8, 0)$ and $(3, k)$, where $k > 0$. The area of the triangle is 20. What is the value of k?

Answer:

749. GT-228 · Medium

Two similar triangles have areas 18 and 50. The altitude to the longest side of the larger triangle is 15. What is the corresponding altitude of the smaller triangle?

A) 10.8 **B)** 5.4

C) 12.5 **D)** 9

750. GT-274 · Medium · No calculator needed

In triangle ABC, $AB = 2x + 1$, $BC = x + 6$ and $CA = 3x - 4$. The perimeter of the triangle is 45. Which angle of the triangle has the greatest measure?

A) Angle A
B) It cannot be determined from the information given.
C) Angle C
D) Angle B

751. GT-012 · Hard · SPR

Figure (described): triangle PQR with point S on side PQ and point T on side PR, and segment ST drawn parallel to side QR

In triangle PQR, segment ST is drawn parallel to side QR, where S lies on PQ and T lies on PR. If $PS = x$, $SQ = x + 4$, $PT = 15$, and $TR = 25$, what is the value of x?

Answer:

752. GT-013 · Hard · SPR

Figure (described): triangle XYZ with an exterior angle drawn at vertex X by extending one of the triangle's sides

In triangle XYZ, the exterior angle at vertex X measures $(5x - 10)°$. This exterior angle equals the sum of the two non-adjacent (remote) interior angles, which measure $(2x + 15)°$ at vertex Y and $(x + 25)°$ at vertex Z. What is the measure of angle Y, in degrees?

Answer:

753. GT-031 · Hard

Figure (described): quadrilateral $ABCD$ formed by two right triangles, ABC and ACD, sharing diagonal AC; triangle ABC has a right angle at B with $AB = 9$ and $BC = 12$; triangle ACD has a right angle at A with $AD = 8$

In the figure, quadrilateral $ABCD$ is formed by two right triangles, ABC and ACD, sharing diagonal AC. Triangle ABC has a right angle at B with $AB = 9$ and $BC = 12$. Triangle ACD has a right angle at A with $AD = 8$. What is the area of quadrilateral $ABCD$?

A) 90 **B)** 96

C) 108 **D)** 114

754. GT-035 · Hard · SPR

In a figure, two isosceles triangles share a common vertex. In the top triangle, the vertex angle is c, and each of its two equal base angles measures $40°$. In the bottom triangle, the vertex angle b equals half of c, and angle a is supplementary to one of the bottom triangle's two equal base angles. What is the value of a, in degrees?

Answer:

755. GT-050 · Hard · SPR

In triangle ABC, the exterior angle at vertex C measures $130°$. If the measure of angle A is $55°$, what is the measure of angle B?

Answer:

756. GT-051 · Hard

Lines l and m are parallel and are cut by transversal t. Which of the following pairs of measures could NOT represent the measures of two same-side interior angles formed by l, m, and t?

A) $100°$ and $80°$
B) $95°$ and $85°$
C) $112°$ and $68°$
D) $95°$ and $84°$

757. GT-064 · Hard · SPR

In triangle ABC, the exterior angle at vertex C measures $(4x + 10)°$. This exterior angle equals the sum of the two non-adjacent interior angles, which measure $(2x + 5)°$ and $(x + 25)°$. What is the value of x?

Answer:

758. GT-065 · Hard

A pole that is 2.4 meters tall casts a shadow 3.2 meters long. At the same time, a nearby communications tower casts a shadow 52 meters long. Using similar triangles, which of the following is closest to the height, in meters, of the tower?

A) 34.7 **B)** 36.4

C) 39.0 **D)** 41.6

759. GT-083 · Hard · SPR

In isosceles triangle ABC, $AB = AC$, and the measure of angle A is $30°$ more than the measure of angle B. What is the measure, in degrees, of angle C?

Answer:

760. GT-084 · Hard

In triangle ABC, points M and N are the midpoints of sides AB and AC, respectively. If $MN = 3x - 4$ and $BC = 5x + 2$, what is the length of BC?

A) 42 **B)** 47

C) 52 **D)** 56

761. GT-089 · Hard

In $\triangle XYZ$, the side lengths are a, b, and c, where $4 < a < 6$ and $9 < b < 11$. Which of the following represents all possible values of c?

A) $3 < c < 15$ **B)** $3 < c < 17$

C) $5 < c < 15$ **D)** $5 < c < 17$

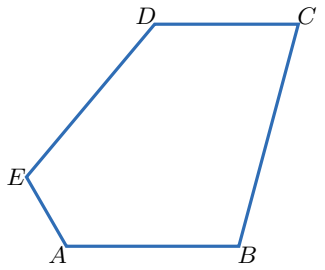
762. GT-098 · Hard · SPR

In triangle ABC, sides AC and AB are congruent. Points D, C, and B are collinear, with C between D and B. If the measure of angle DCA is $130°$, what is the measure, in degrees, of angle A?

Answer:

763. GT-124 · Hard · No calculator needed

In convex pentagon $ABCDE$ shown (not drawn to scale), side $\overline{AB}$ is parallel to side $\overline{DC}$. The measures of four interior angles are $\angle A = (6x)°$, $\angle B = (5x + 5)°$, $\angle C = (3x + 15)°$ and $\angle D = (7x - 10)°$. What is the measure, in degrees, of $\angle E$?

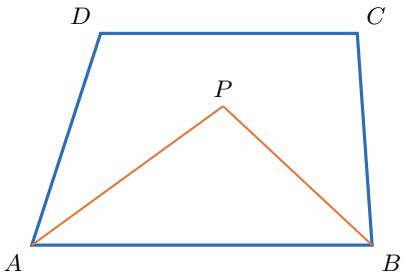


A) 110 **B)** 70

C) 105 **D)** 130

764. GT-126 · Hard · No calculator needed

In trapezoid $ABCD$ shown (not drawn to scale), $\overline{AB} \parallel \overline{DC}$, $\angle D = 108°$, and $\angle C = 94°$. The bisector of $\angle DAB$ and the bisector of $\angle ABC$ intersect at point P inside the trapezoid. What is the measure, in degrees, of the obtuse angle formed at P by these two bisectors?



A) 101 **B)** 79

C) 22 **D)** 108

765. GT-129 · Hard · No calculator needed

Two lines intersect at point O, and the smallest of the four angles they form measures $50°$. A ray from O bisects each of the four angles. Together with the two lines, this gives eight rays from O, which divide the full turn into eight small angles. A *span* is an angle formed by 1, 2 or 3 consecutive small angles. Which of the following is NOT the measure of any span?

A) $130°$ **B)** $155°$

C) $140°$ **D)** $115°$

766. GT-141 · Hard · No calculator needed

In equilateral triangle ABC, point D lies on side $\overline{BC}$ such that $BD = \frac{1}{4}BC$. What is the value of $\tan \angle DAC$?

A) $\frac{5\sqrt{3}}{9}$ **B)** $\frac{3\sqrt{3}}{7}$

C) $\frac{3\sqrt{3}}{5}$ **D)** $\frac{\sqrt{3}}{7}$

767. GT-147 · Hard · No calculator needed

In triangle ABC, point D lies on $\overline{AB}$ and point E lies on $\overline{AC}$ so that $\overline{DE}$ is parallel to $\overline{BC}$. The measure of $\angle ADE$ is $(4x - 10)°$, the measure of $\angle ABC$ is $(2x + 30)°$, and the measure of $\angle ACB$ is $(x + 25)°$. What is the measure, in degrees, of $\angle BAC$?

A) 115 **B)** 45

C) 65 **D)** 70

768. GT-152 · Hard · No calculator needed

Points A, B, C and D are consecutive vertices of a regular polygon with n sides, and $\overline{AC}$ is a diagonal. The measure of $\angle BAC$ is $(x + 3)°$ and the measure of $\angle ACD$ is $(10x+15)°$, where x is a constant. What is the value of n?

A) 24 **B)** 8

C) 12 **D)** 15

769. GT-160 · Hard · No calculator needed

A convex polygon has n sides. Its interior angles, listed from smallest to largest, form an arithmetic sequence with smallest angle $120°$ and common difference $5°$. How many sides does the polygon have?

A) 7 **B)** 9

C) 16 **D)** 18

770. GT-161 · Hard · No calculator needed

Two lines intersect and form four angles. The acute angles each measure $(2k + 4)°$, where k is a constant. Each obtuse angle measures 4 degrees more than 3 times the measure of an acute angle. Which of the following CANNOT be the sum, in degrees, of the measures of two of the four angles?

A) 272 **B)** 88

C) 180 **D)** 224

771. GT-162 · Hard · No calculator needed

In triangle ABC, the measure of angle A is $(3x + 11)°$, the measure of angle B is $(2x + 4)°$, and the measure of angle C is $30°$. What is the value of

$$(\sin A \cos B + \cos A \sin B)(\cos A \cos B - \sin A \sin B)\,?$$

A) $-\dfrac{1}{4}$ **B)** $\dfrac{1}{4}$

C) $-\dfrac{\sqrt{3}}{4}$ **D)** $\dfrac{\sqrt{3}}{4}$

772. GT-217 · Hard · SPR

In triangle ABC, $AB = 12$ and $AC = 18$. The bisector of angle A meets $\overline{BC}$ at D, and $BC = 20$. What is the length of the longer of the two segments BD and DC?

Answer:

773. GT-220 · Hard · SPR

Two sides of a triangle have lengths 7 and 12. The third side has an integer length. How many different integer lengths are possible for the third side?

Answer:

774. GT-222 · Hard · SPR

In triangle ABC, $AB = AC$ and angle A measures $40°$. Point D lies on $\overline{AC}$ such that $BD = BC$. What is the measure, in degrees, of angle ABD?

Answer:

775. GT-226 · Hard · SPR

Triangle ABC is similar to triangle DEF. The area of triangle ABC is 36 and the area of triangle DEF is 100. The perimeter of triangle ABC is 24. What is the perimeter of triangle DEF?

Answer:

776. GT-227 · Hard · SPR

In triangle ABC, point D lies on $\overline{AB}$ and point E lies on $\overline{AC}$ so that $\overline{DE}$ is parallel to $\overline{BC}$. If $AD : DB = 2 : 3$ and the area of triangle ADE is 20, what is the area of quadrilateral $DBCE$?

Answer:

777. GT-229 · Hard · SPR

In triangle ABC, point D is on $\overline{AB}$ and point E is on $\overline{AC}$ with $\overline{DE} \parallel \overline{BC}$. The segment $\overline{DE}$ divides the triangle into two regions of equal area. If $AB = 20$, what is the length of $\overline{AD}$, to the nearest tenth?

Answer:

778. GT-230 · Hard · SPR

A right triangle has legs of length 6 and 8. A larger, similar right triangle has area 150. What is the length of the hypotenuse of the larger triangle?

Answer:

779. GT-231 · Hard · SPR

A square pyramid has a base with area 144. A plane parallel to the base cuts the pyramid at a point one-third of the way from the apex to the base, forming a smaller square cross-section. What is the area of the cross-section?

Answer:

780. GT-236 · Hard · No calculator needed

In triangle ABC, point D lies on $\overline{AB}$ and point E lies on $\overline{AC}$ so that the measure of angle AED equals the measure of angle ABC. If $AD = 4$, $DB = 11$ and $AE = 6$, what is the length of $\overline{EC}$?

A) 10 **B)** 16.5

C) 22.5 **D)** 4

781. GT-243 · Hard · No calculator needed

In triangles PQR and XYZ, $PQ = XY = 9$ and $m\angle R = m\angle Z = 35°$, where $\overline{PQ}$ is opposite angle R and $\overline{XY}$ is opposite angle Z. Which additional fact is enough to prove that triangle PQR is congruent to triangle XYZ?

A) $PR = XZ$ **B)** $QR = YZ = 14$

C) $m\angle Q = m\angle Y$ **D)** $PR = QR$

782. GT-246 · Hard · No calculator needed

Triangle ABC has area 60. Point D lies on $\overline{BC}$ with $BD : DC = 2 : 3$, and point E lies on $\overline{AD}$ with $AE : ED = 3 : 1$. What is the area of triangle BED?

A) 15 **B)** 18

C) 10 **D)** 6

783. GT-254 · Hard · No calculator needed

In triangle ABC, the measure of angle A is $18°$. Which additional fact is NOT sufficient to determine the measure of angle B?

A) $m\angle B = m\angle C$ **B)** $m\angle C = 3\,m\angle B$

C) Triangle ABC is isosceles. **D)** $m\angle C = 90°$

784. GT-256 · Hard · No calculator needed

Parallel lines m and n are cut by a transversal. Angles b and c are same-side interior angles, and angle a is one of the other angles formed. If $200 < a + b + c < 236$, where all measures are in degrees, which of the following could be the measure of angle a, in degrees?

A) 56 **B)** 18

C) 64 **D)** 40

785. GT-270 · Hard · SPR · No calculator needed

A rectangle has side lengths 6 and $6\sqrt{3}$. What is the measure, in degrees, of the acute angle formed by its diagonals?

Answer:

786. GT-272 · Hard · No calculator needed

Parallel lines ℓ and m are horizontal, with ℓ above m. Point P lies between the lines. Point A on ℓ and point B on m are both to the right of P. Segment $\overline{PA}$ makes a $38°$ angle with line ℓ, and segment $\overline{PB}$ makes a $47°$ angle with line m, both angles opening toward P. What is the measure, in degrees, of angle APB?

A) 42.5 **B)** 95

C) 85 **D)** 9

787. GT-273 · Hard · No calculator needed

In triangle ABC, the measure of angle A is $52°$. The bisectors of angles B and C intersect at point I. What is the measure, in degrees, of angle BIC?

A) 116 **B)** 128

C) 64 **D)** 104

Right triangles and trigonometry (70)

788. GT-001 · Medium

Figure (described): right triangle ABC with the right angle at vertex C

In right triangle ABC, the right angle is at C. If $\sin(A) = \dfrac{3}{5}$, what is the value of $\cos(B)$?

A) $\dfrac{3}{4}$ **B)** $\dfrac{3}{5}$

C) $\dfrac{4}{5}$ **D)** $\dfrac{5}{3}$

789. GT-014 · Medium

Figure (described): right triangle DEF with the right angle at vertex F

In right triangle DEF, the right angle is at F. If $\tan(D) = \dfrac{8}{15}$, what is the value of $\tan(E)$?

A) $\dfrac{8}{15}$ **B)** $\dfrac{15}{8}$

C) $\dfrac{8}{17}$ **D)** $\dfrac{17}{15}$

790. GT-023 · Medium

If θ is an acute angle and $\sin(\theta) = \dfrac{5}{13}$, what is the value of $\tan(\theta)$?

A) $\dfrac{5}{12}$ **B)** $\dfrac{12}{5}$

C) $\dfrac{5}{13}$ **D)** $\dfrac{12}{13}$

791. GT-025 · Medium

In a $30°$-$60°$-$90°$ right triangle, the side opposite the $30°$ angle has length 7. What is the length of the hypotenuse?

A) 7 **B)** $7\sqrt{2}$

C) $7\sqrt{3}$ **D)** 14

792. GT-030 · Medium

What is the value of $\cos(60°) + \sin(30°)$?

A) $\frac{1}{2}$ **B)** $\frac{\sqrt{3}}{2}$

C) 1 **D)** $\sqrt{3}$

793. GT-043 · Medium · SPR

If $\sin(2x)° = \cos(x + 15)°$, what is the value of x?

Answer:

794. GT-044 · Medium

In right triangle ABC, the right angle is at C. If $AC = 16$ and $BC = 30$, what is $\tan A$?

A) $\frac{8}{15}$ **B)** $\frac{15}{8}$

C) $\frac{16}{34}$ **D)** $\frac{34}{16}$

795. GT-046 · Medium

In triangle ABC, angle $C = 90°$, angle $B = 30°$, and $BC = 18$. What is the length of AC?

A) 9 **B)** $6\sqrt{3}$

C) $9\sqrt{3}$ **D)** $18\sqrt{3}$

796. GT-066 · Medium

In right triangle ABC, angle C is the right angle, $AC = 9$, and $BC = 12$. What is $\sin(A)$?

A) $\frac{3}{5}$ **B)** $\frac{4}{5}$

C) $\frac{3}{4}$ **D)** $\frac{4}{3}$

797. GT-069 · Medium

In right triangle ABC, angle C is the right angle, $AB = 13$ (the hypotenuse), and $BC = 5$. What is $\cos(A)$?

A) $\frac{5}{13}$ **B)** $\frac{12}{13}$

C) $\frac{5}{12}$ **D)** $\frac{12}{5}$

798. GT-088 · Medium

In $\triangle PQR$, $PQ = 8$, $QR = 10$, and $m\angle Q = 60°$. What is the area of $\triangle PQR$?

A) 20 **B)** $20\sqrt{2}$

C) $20\sqrt{3}$ **D)** $40\sqrt{3}$

799. GT-097 · Medium · SPR

What is the distance between the points $(-3, -9)$ and $(9, -4)$?

Answer:

800. GT-102 · Medium

In triangle ABC, angle B is a right angle and the measure of angle A is twice the measure of angle C. If $AB = 5$, what is the length of BC?

A) 5 **B)** $5\sqrt{2}$

C) $5\sqrt{3}$ **D)** 10

801. GT-107 · Medium

In right triangle ABC, the right angle is at B, and BD is the altitude drawn to hypotenuse AC (with D on AC). If $AD = 3$ and $DC = 12$, what is the length of BD?

A) 4 **B)** 6

C) 7.5 **D)** 9

802. GT-173 · Medium · SPR

$$\sin(x°) = \cos((3x - 10)°)$$

In the equation above, x is between 0 and 30. What is the value of x?

Answer:

803. GT-174 · Medium · SPR

In right triangle ABC, the right angle is at C, $\tan A = \frac{5}{12}$, and $AB = 39$. What is the perimeter of triangle ABC?

Answer:

804. GT-176 · Medium

An isosceles right triangle has a hypotenuse of length 14. What is the perimeter of the triangle?

A) 28 **B)** $14 + 7\sqrt{2}$

C) $14 + 14\sqrt{2}$ **D)** $14 + 7\sqrt{3}$

805. GT-177 · Medium · SPR

The height of an equilateral triangle is $6\sqrt{3}$. What is the perimeter of the triangle?

Answer:

806. GT-179 · Medium · SPR

A 13-foot ladder leans against a vertical wall. Its base is 5 feet from the wall. The base is then pulled 2 feet farther from the wall, so the ladder is 7 feet from the wall. To the nearest tenth of a foot, how high up the wall does the ladder reach now?

Answer:

807. GT-181 · Medium

What is the value of $\cos\dfrac{5\pi}{3}$?

A) $-\dfrac{\sqrt{3}}{2}$ **B)** $-\dfrac{1}{2}$

C) $\dfrac{\sqrt{3}}{2}$ **D)** $\dfrac{1}{2}$

808. GT-182 · Medium · SPR

From a point on level ground 40 meters from the base of a tower, the angle of elevation to the top of the tower is $62°$. To the nearest meter, what is the height of the tower?

Answer:

809. GT-184 · Medium · SPR

In a right triangle, the hypotenuse is 50 and the cosine of an acute angle θ is $\dfrac{7}{25}$. What is the area of the triangle?

Answer:

810. GT-186 · Medium · SPR

The diagonal of a rectangle is 26 and the ratio of its two side lengths is $5:12$. What is the area of the rectangle?

Answer:

811. GT-002 · Hard · SPR

Figure (described): a vertical tower of height h meters; two observation points lie on level ground along the same line as the base of the tower, with the farther point 50 meters beyond the nearer point

Engineers surveying a proposed tower near King Abdullah Financial District measure the angle of elevation to the top of the tower from a point on the ground to be $30°$. Standing 50 meters closer to the base of the tower, along the same straight line, the angle of elevation is $60°$. What is the height of the tower, in meters, to the nearest tenth?

Answer:

812. GT-003 · Hard

Figure (described): triangle with two sides of length 7 km and 8 km meeting at a vertex, with an included angle of $120°$ between them

A triangular plot of land has two sides of length 7 kilometers and 8 kilometers, with an included angle of $120°$ between them. What is the length, in kilometers, of the third side of the plot?

A) 11 **B)** 12

C) 13 **D)** 15

813. GT-017 · Hard

Figure (described): right triangle JKL with the right angle at vertex L

In right triangle JKL, the right angle is at L. If $\cos(J)=\dfrac{7}{25}$, what is the value of $\sin(J)$?

A) $\dfrac{7}{25}$ **B)** $\dfrac{24}{25}$

C) $\dfrac{25}{7}$ **D)** $\dfrac{25}{24}$

814. GT-042 · Hard · SPR

In a figure, right triangle ACD has a right angle at D, and right triangle BCD (with B on segment AD) has a right angle at B; both triangles share angle C. If $\cos C=0.75$ and $AC=48$, what is the length of BC?

Answer:

815. GT-045 · Hard · SPR

In right triangle ABC, the right angle is at C, and $\cos A=\dfrac{3}{5}$. If $AC=36$, what is the length of AB?

Answer:

816. GT-047 · Hard · SPR

An isosceles right triangle has a perimeter of $40+20\sqrt{2}$. What is the length of the hypotenuse of the triangle?

Answer:

817. GT-067 · Hard

In right triangle DEF, angle F is the right angle, angle D measures $35°$, and side DF (adjacent to angle D) measures 18. Which of the following is closest to the length of side EF (opposite angle D)?

A) 10.3 **B)** 12.6

C) 14.9 **D)** 17.1

818. GT-068 · Hard

A triangular plot of land has two sides measuring 80 meters and 95 meters, with an included angle of $62°$ between them. Which of the following is closest to the length, in meters, of the third side?

A) 83.4 **B)** 87.9

C) 91.0 **D)** 95.7

819. GT-070 · Hard

From a point on the ground 40 meters from the base of a communications tower, the angle of elevation to the top of the tower is $58°$. Which of the following is closest to the height, in meters, of the tower?

A) 33.9 **B)** 47.1

C) 64.0 **D)** 75.5

820. GT-071 · Hard

In triangle DEF, angle $D = 40°$, angle $E = 65°$, and side $DE = 22$. Which of the following is closest to the length of side EF (opposite angle D)?

A) 12.6 **B)** 14.6

C) 16.9 **D)** 19.4

821. GT-072 · Hard

In right triangle PQR, angle R is the right angle. If $\sin(P) = \frac{7}{25}$, what is the value of $\cos(Q)$?

A) $\frac{7}{25}$ **B)** $\frac{24}{25}$

C) $\frac{25}{7}$ **D)** $\frac{25}{24}$

822. GT-087 · Hard · SPR

Layla walked from Noor's house 4 miles due north, then 3 miles due east, while Noor walked 2 miles due south and 5 miles due west. If Noor were to walk directly to Layla's location, how many hours would it take her, if Noor walks at a rate of 5 miles per hour?

Answer:

823. GT-100 · Hard · SPR

The legs of a right triangle have lengths $x+1$ and $x+8$, and the hypotenuse has length $x+9$. What is the area of the triangle?

Answer:

824. GT-101 · Hard

A $30°$-$60°$-$90°$ triangle has a perimeter of $18 + 6\sqrt{3}$. What is the length of its hypotenuse?

A) 6 **B)** $6\sqrt{3}$

C) 12 **D)** $12\sqrt{3}$

825. GT-112 · Hard · SPR

In right triangle PQR, the right angle is at Q, $PQ = 18$, and $PR = 30$. Point S lies on $\overline{PR}$ such that $\overline{QS} \perp \overline{PR}$. What is the value of $\frac{QR}{QS}$?

Answer:

826. GT-118 · Hard · SPR · No calculator needed

In triangle ABC, the angle at C is a right angle. Point D lies on $\overline{AB}$ so that $\overline{CD} \perp \overline{AB}$. The area of triangle ACD is 48 and the area of triangle CDB is 108. What is the length of $\overline{AB}$?

Answer:

827. GT-121 · Hard · No calculator needed

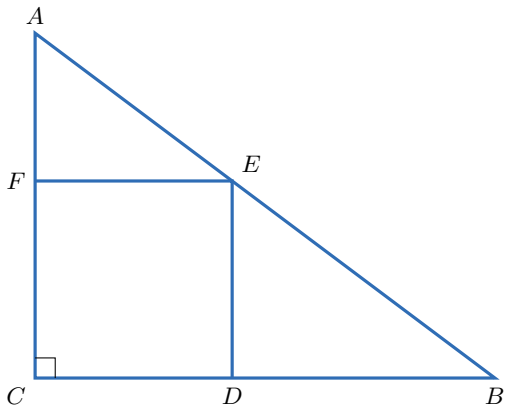
In triangle PQR, the angle at Q is a right angle. Point S lies on side $\overline{PR}$ so that $\overline{QS} \perp \overline{PR}$. If $\tan\angle QRP = \frac{4}{3}$ and $PS = 16$, what is the length of $\overline{SR}$?

A) 25 **B)** 9

C) 12 **D)** 15

828. GT-125 · Hard · No calculator needed

In right triangle ABC, $\angle C = 90°$, $AB = 20$, and the area of the triangle is 96. Square $CDEF$ is inscribed in the triangle, with D on $\overline{BC}$, F on $\overline{AC}$, and E on $\overline{AB}$. What is the side length of the square?



A) $\frac{24}{7}$ **B)** $\frac{48}{5}$

C) $\frac{96}{7}$ **D)** $\frac{48}{7}$

829. GT-128 · Hard · SPR · No calculator needed

In right triangle PQR, $\angle R = 90°$ and $\sin P + \sin Q = \frac{\sqrt{6}}{2}$. What is the value of $\tan P + \tan Q$?

Answer:

830. GT-133 · Hard · Calculator

In right triangle ABC, the right angle is at C and $AB = 10$. Point D lies on $\overline{AB}$ with $\overline{CD} \perp \overline{AB}$ and $CD = 4$. If $AD > DB$, what is the value of $\cos A$?

A) $\frac{\sqrt{5}}{5}$ **B)** $\frac{4}{5}$

C) $\frac{1}{2}$ **D)** $\frac{2\sqrt{5}}{5}$

831. GT-136 · Hard · Calculator

In triangle ABC, angle C is a right angle and angle A measures $30°$. The circle inscribed in the triangle has radius 5. What is the length of $\overline{AC}$?

A) $10 + 10\sqrt{3}$ **B)** $10\sqrt{3} - 10$

C) $5 + 5\sqrt{3}$ **D)** $15 + 5\sqrt{3}$

832. GT-143 · Hard · No calculator needed

In right triangle ABC, the right angle is at C and $\tan A = \frac{5}{12}$. The area of the triangle is 120. Point D lies on $\overline{AB}$ such that $\overline{CD} \perp \overline{AB}$. What is the length of $\overline{AD}$?

A) $\frac{50}{13}$ **B)** $\frac{288}{13}$

C) $\frac{120}{13}$ **D)** 24

833. GT-151 · Hard · SPR · Calculator

For $0 < x < 45$, the equation

$$\sin((5x - 20)^\circ) = \cos((x + 14)^\circ)$$

has more than one solution. What is the sum of all solutions x in this interval?

Answer:

834. GT-154 · Hard · No calculator needed

In right triangle ABC, the angle at C is a right angle. Point D lies on $\overline{AB}$ so that $\overline{CD} \perp \overline{AB}$. The length of $\overline{CD}$ is 6, and DB is 5 units longer than AD. What is the value of $\sin B$?

A) $\frac{6}{13}$ **B)** $\frac{2\sqrt{13}}{13}$

C) $\frac{3\sqrt{13}}{13}$ **D)** $\frac{4}{13}$

835. GT-168 · Hard · SPR · No calculator needed

In triangle PQR, the measure of angle P is 30° and $PR = 14$. The length $QR = a$ is a positive integer with $a \leq 20$. For how many values of a does the given information determine exactly one triangle, up to congruence?

Answer:

836. GT-175 · Hard · SPR

In right triangle PQR, the right angle is at Q, $PQ = 9$ and $QR = 12$. The altitude from Q to hypotenuse $\overline{PR}$ has length h. What is the value of h? (Enter as a decimal.)

Answer:

837. GT-178 · Hard

In a right triangle, angle θ is acute and $\sin\theta = 0.6$. What is the value of $\cos(90^\circ - \theta) + \tan\theta$?

A) 1.35 **B)** 1.6

C) 0.75 **D)** 1.2

838. GT-180 · Hard · SPR

In right triangle PQR, the right angle is at Q, $PQ = 8$ and $QR = 15$. Point S is on $\overline{PR}$ such that $\overline{QS}$ is perpendicular to $\overline{PR}$. What is the length of $\overline{QS}$? (Enter as a fraction.)

Answer:

839. GT-183 · Hard · SPR

In a right triangle, the sine of one acute angle is $\frac{3}{5}$ and the area of the triangle is 24. What is the length of the hypotenuse?

Answer:

840. GT-185 · Hard · SPR

The legs of a right triangle have lengths x and $x+7$, and the hypotenuse is 13. What is the area of the triangle?

Answer:

841. GT-234 · Hard · No calculator needed

For $0 < t < \frac{\pi}{2}$, $\sin t = 0.6$. What is the value of $\cos(\pi + t) + \sin(\pi - t)$?

A) 0.2 **B)** -1.4

C) -0.2 **D)** 1.4

842. GT-237 · Hard · No calculator needed

In right triangle ABC, angle C is the right angle, $AC = 16$ and $BC = 12$. Point E lies on $\overline{AC}$ with $AE = 10$, and point F lies on $\overline{AB}$ so that $\overline{EF} \perp \overline{AB}$. What is the length of $\overline{EF}$?

A) 8 **B)** 6

C) 7.5 **D)** 12.5

843. GT-242 · Hard · No calculator needed

Line ℓ passes through $(0, 0)$ and $(a, 12)$, where $a > 0$. The acute angle between ℓ and the positive y-axis measures 60°. What is the value of a?

A) $12\sqrt{3}$ **B)** $4\sqrt{3}$

C) 24 **D)** 6

844. GT-250 · Hard · SPR · Desmos

In the equation $\sin\left(\frac{39\pi}{p}\right) = \frac{\sqrt{3}}{2}$, p is a constant and $p > 60$. What is the value of p?

Answer:

845. GT-257 · Hard · No calculator needed

In triangle JKL, the measure of angle J is $(ab)^\circ$ and the measure of angle K is $(cd)^\circ$, where $ab + cd = 90$. If $\sin J = 0.6$, what is the value of $\tan K$?

A) $\frac{3}{4}$
B) $\frac{4}{3}$
C) $\frac{4}{5}$
D) The value cannot be determined from the information given.

846. GT-258 · Hard · No calculator needed

Right triangles PQR and XYZ are similar, with P, Q and R corresponding to X, Y and Z, and right angles at Q and Y. If $\tan P = \frac{8}{15}$ and $XZ = 51$, what is the length of $\overline{XY}$?

A) 27.2
B) 45
C) 24
D) 95.625

847. GT-259 · Hard · No calculator needed

In triangle ABC, $AB = AC = 17$ and $BC = 16$. What is the value of $\tan B$?

A) $\frac{8}{15}$
B) $\frac{15}{8}$
C) $\frac{15}{17}$
D) $\frac{16}{17}$

848. GT-260 · Hard · No calculator needed

A rectangular box has a base that measures 9 by 12 and a height of 20. Let θ be the angle between a diagonal of the box (from a bottom corner to the opposite top corner) and the base. What is the value of $\tan\theta$?

A) $\frac{20}{9}$
B) $\frac{5}{3}$
C) $\frac{3}{4}$
D) $\frac{4}{3}$

849. GT-261 · Hard · No calculator needed

In right triangle ABC, angle C is the right angle and $\cos A = k$. Which expression is equal to $\sin^2 A + \sin^2 B$?

A) 1
B) The value cannot be determined from the information given.
C) $2 - 2k^2$
D) $2k^2$

850. GT-262 · Hard · No calculator needed

Right triangle ABC has its right angle at C, $AC = 9$ and $BC = 12$. Each leg is lengthened by 3 and the angle at C stays a right angle. What is $\tan A$ in the new triangle?

A) $\frac{4}{5}$
B) $\frac{4}{3}$
C) $\frac{5}{4}$
D) $\frac{5}{3}$

851. GT-263 · Hard · No calculator needed

A straight railway track up a hill has a constant grade of 75%, meaning it rises 75 meters for every 100 meters of horizontal distance. The track is 300 meters long, measured along the slope. What is the vertical rise of the track, in meters?

A) 400
B) 240
C) 180
D) 225

852. GT-264 · Hard · No calculator needed

A drone flies 12 kilometers in a direction 60° north of east, then flies 12 kilometers due west. How far, in kilometers, is the drone from its starting point?

A) 12
B) 24
C) $12\sqrt{2}$
D) $6\sqrt{3}$

853. GT-265 · Hard · No calculator needed

In triangle ABC, the measure of angle B is 45° and the measure of angle C is 30°. The altitude from A to $\overline{BC}$ has length 6. What is the length of $\overline{BC}$?

A) 12
B) $6 + 2\sqrt{3}$
C) $12 + 6\sqrt{2}$
D) $6 + 6\sqrt{3}$

854. GT-266 · Hard · No calculator needed

In the xy-plane, the points $A(0,0)$, $B(6,3)$ and $C(4,7)$ are the vertices of a triangle. What is the value of $\tan(\angle CAB)$?

A) $\frac{1}{2}$
B) $\frac{2}{3}$
C) $\frac{3}{2}$
D) $\frac{7}{4}$

855. GT-268 · Hard · No calculator needed

From the top of a cliff 45 meters high, the angles of depression to two boats on opposite sides of the cliff are 30° and 45°. The boats and the base of the cliff are on a straight line. What is the distance, in meters, between the boats?

A) $45\sqrt{3} - 45$
B) $45 + 15\sqrt{3}$
C) $45\sqrt{3}$
D) $45 + 45\sqrt{3}$

856. GT-269 · Hard · No calculator needed

In a right triangle, the measure of acute angle A is between 45° and 90°. Which of the following is true?

A) $\cos A < \sin A < \tan A$
B) $\tan A < \sin A < \cos A$
C) $\sin A < \cos A < \tan A$
D) $\cos A < \tan A < \sin A$

857. GT-271 · Hard · No calculator needed

In the xy-plane, line ℓ has slope $\frac{3}{4}$ and line k is perpendicular to line ℓ. Line k makes an angle θ with the positive x-axis, measured counterclockwise, where $0^\circ < \theta < 180^\circ$. What is the value of $\cos\theta$?

A) $-\frac{4}{5}$
B) $\frac{3}{5}$
C) $-\frac{3}{5}$
D) $\frac{4}{5}$

Circles (79)

858. GT-004 · Medium

A circle in the xy-plane is defined by the equation $x^2 + y^2 - 6x + 8y = 0$. What is the radius of the circle?

A) 3 **B)** 4

C) 5 **D)** 25

859. GT-015 · Medium

A circle in the xy-plane is defined by the equation $x^2 + y^2 + 10x - 4y + 13 = 0$. What is the radius of the circle?

A) 4 **B)** 8

C) 13 **D)** 16

860. GT-028 · Medium

In a circle with radius 10, a chord subtends a central angle of $60°$. What is the length of the chord?

A) 5 **B)** $5\sqrt{3}$

C) 10 **D)** $10\sqrt{3}$

861. GT-029 · Medium

Figure (described): circle with center O; line ℓ is tangent to the circle at point P; point Q lies on line ℓ

In the figure, line ℓ is tangent to the circle with center O at point P. If $OP = 9$ and $OQ = 15$ for a point Q on line ℓ, what is the length of PQ?

A) 6 **B)** 12

C) $\sqrt{306}$ **D)** 18

862. GT-052 · Medium · SPR

An angle has a measure of $\dfrac{7\pi}{12}$ radians. What is the measure of the angle, in degrees?

Answer:

863. GT-053 · Medium · SPR

The number of radians in a 540-degree angle can be written as $a\pi$, where a is a constant. What is the value of a?

Answer:

864. GT-054 · Medium

Points X, Y, and Z lie on a circular fountain in a Jeddah park, as shown below. On this circle, minor arc XY has a length of 52 meters, and major arc XZY has a length of 260 meters. What is the circumference, in meters, of the circle?

A) 52 **B)** 208

C) 260 **D)** 312

865. GT-056 · Medium

What is the center of the circle in the xy-plane defined by the equation $x^2 + y^2 - 14x + 22y + 10 = 0$?

A) $(7, -11)$ **B)** $(7, 11)$

C) $(-7, 11)$ **D)** $(-7, -11)$

866. GT-073 · Medium

A circle in the xy-plane has equation $(x+2)^2+(y-5)^2 = 49$. What is the radius of the circle?

A) 7 **B)** 14

C) 24.5 **D)** 49

867. GT-074 · Medium

A circle has a radius of 9 cm. What is the length, in cm, of an arc that subtends a central angle of $80°$? (Use $\pi \approx 3.14$.)

A) 9.4 **B)** 12.6

C) 15.7 **D)** 18.8

868. GT-085 · Medium

The equation of a circle in the xy-plane is $x\,(x - 8) + y\,(y + 6) = 39$. What is the radius of the circle?

A) 6 **B)** 7

C) 8 **D)** 9

869. GT-091 · Medium

A circle has an area of 144π ft^2. What is the circumference of the circle?

A) 12 ft **B)** 12π ft

C) 24π ft **D)** 48π ft

870. GT-104 · Medium

A circle has radius $2x - 3$ and circumference $(3x + 4)\pi$. What is the area of the circle?

A) 17π **B)** 34π

C) 196π **D)** 289π

871. GT-108 · Medium · SPR

Chords $\overline{AC}$ and $\overline{BE}$ of a circle intersect at point D inside the circle. If $AD = 4$, $DC = 15$, and $BD = 6$, what is the length of $\overline{BE}$?

Answer:

872. GT-109 · Medium

Segment AC is a diameter of a circle, and point B lies on the circle. If $AC = 26$ and $AB = 10$, what is the length of BC?

A) 16 **B)** 24

C) $\sqrt{776}$ **D)** 26

873. GT-117 · Medium

The equation $x^2 - 18x + y^2 - 6y = -38$ represents a circle in the xy-plane. What is the radius of the circle?

A) 4 **B)** 6

C) $\sqrt{47}$ **D)** $\sqrt{52}$

874. GT-187 · Medium

A circle has radius 15. An arc of the circle subtends a central angle of $72°$. What is the length of the arc?

A) 12π **B)** 6π

C) 3π **D)** 30π

875. GT-188 · Medium · SPR

A sector of a circle with radius 6 has area 12π. What is the measure, in degrees, of the central angle of the sector?

Answer:

876. GT-192 · Medium · SPR

A chord of a circle with radius 13 is 5 units from the center of the circle. What is the length of the chord?

Answer:

877. GT-193 · Medium · SPR

Points A and B are the endpoints of a diameter of a circle, and point C is on the circle. The measure of angle CAB is $28°$. What is the measure, in degrees, of angle ABC?

Answer:

878. GT-195 · Medium

In the xy-plane, a circle has center $(-3, 5)$ and is tangent to the x-axis. Which of the following is an equation of the circle?

A) $(x+3)^2+(y-5)^2=25$ **B)** $(x+3)^2+(y-5)^2=9$

C) $(x+3)^2+(y-5)^2=5$ **D)** $(x-3)^2+(y+5)^2=25$

879. GT-196 · Medium · SPR

In the xy-plane, the line $y = 6$ intersects the circle $x^2 + y^2 = 100$ at two points. What is the distance between the two points?

Answer:

880. GT-197 · Medium · SPR

An arc of a circle has length 10π centimeters and subtends a central angle of $\dfrac{2\pi}{3}$ radians. What is the radius of the circle, in centimeters?

Answer:

881. GT-198 · Medium · SPR

Two concentric circles have the same center. The outer circle has radius 7, and the area of the region between the two circles is 48π. What is the radius of the inner circle?

Answer:

882. GT-199 · Medium

A sector of a circle has radius 12 and a central angle of $60°$. What is the perimeter of the sector?

A) $24 + 24\pi$ **B)** $24 + 4\pi$

C) $12 + 4\pi$ **D)** 4π

883. GT-245 · Medium · No calculator needed

In the xy-plane, the endpoints of a diameter of a circle are $(-2, 3)$ and $(6, -1)$. Which equation represents the circle?

A) $x^2+y^2-4x-2y-15=0$ **B)** $x^2+y^2-4x-2y-20=0$

C) $x^2+y^2-4x-2y-75=0$ **D)** $x^2+y^2+4x+2y-15=0$

884. GT-005 · Hard · SPR

In the xy-plane, the graph of $x^2 + y^2 + 4x - 10y + k = 0$ is a circle with radius 4. What is the value of k?

Answer:

885. GT-006 · Hard

Figure (described): a circular plaza with center O and radius 45 meters; a decorative walking path traces an arc along the circle's edge, subtending a central angle at O

As part of a Vision 2030 public-space project in Riyadh, a circular plaza has a radius of 45 meters. A decorative walking path forms an arc along the plaza's boundary that subtends a central angle of $\dfrac{5\pi}{6}$ radians. What is the length of the arc, in meters?

A) $\dfrac{75\pi}{2}$ **B)** 45π

C) 150π **D)** $\dfrac{25\pi}{2}$

886. GT-018 · Hard · SPR

A circle in the xy-plane has the equation $x^2 + y^2 - 8x + 2y - 8 = 0$. What is the diameter of the circle?

Answer:

887. GT-021 · Hard · SPR

Figure (described): circle with center O; chord AB subtends a central angle of $100°$ at O; point C lies on the major arc

In the circle above, chord AB subtends a central angle of $100°$ at center O. Point C lies on the major arc. What is the measure of angle ACB, in degrees?

Answer:

888. GT-038 · Hard · SPR

A circle has center A and radius $20\sqrt{3}$. Segment CD is tangent to the circle at point D, so triangle ACD has a right angle at D. Point B lies on the circle along ray AC, and minor arc BD has length $\frac{10\sqrt{3}}{3}\pi$. What is the length of CD?

Answer:

889. GT-039 · Hard

In a figure, PA and PB are tangent to a circle centered at O, at points A and B respectively, so $OA \perp PA$ and $OB \perp PB$. If $\angle APB = 100°$ and the circle has a circumference of 36π, what is the length of minor arc AB?

A) 6π **B)** 8π

C) 9π **D)** 10π

890. GT-040 · Hard · SPR

A circle has equation $(x+4)^2+(y-b)^2=100$, where b is a constant. The points $(4,13)$ and $(-10,-1)$ both lie on the circle. What is the value of b?

Answer:

891. GT-055 · Hard

In the xy-plane, the graph of $2x^2-8x+2y^2+12y=24$ is a circle. What is the radius of the circle?

A) 5 **B)** 6.5

C) $\sqrt{40}$ **D)** $\sqrt{50}$

892. GT-057 · Hard · SPR

A circular roundabout on the Jeddah Corniche has its center at $(2,-5)$ in a coordinate model of the area, where distances are measured in hundreds of meters. Road k is tangent to the roundabout at the point $(9,-1)$. What is the slope of road k?

Answer:

893. GT-075 · Hard

A circle has a radius of 12 meters. What is the area, in square meters, of a sector with a central angle of $150°$? (Use $\pi \approx 3.14$.)

A) 150.8 **B)** 172.3

C) 188.4 **D)** 205.6

894. GT-076 · Hard

A circle in the xy-plane is defined by the equation $x^2+y^2-6x+10y+18=0$. What is the center of the circle?

A) $(3,-5)$ **B)** $(-3,5)$

C) $(6,-10)$ **D)** $(3,5)$

895. GT-086 · Hard

A circle in the xy-plane has center $(0,3)$ and passes through the point $(5,15)$. Which of the following is an equation of the circle?

A) $x^2+(y-3)^2=144$
B) $x^2+(y-3)^2=169$
C) $(x-5)^2+(y-3)^2=169$
D) $x^2+(y-3)^2=13$

896. GT-099 · Hard · SPR

O is the center of a circle. Chords $\overline{AC}$ and $\overline{BD}$ are diameters that intersect at O (so A, O, C are collinear, and B, O, D are collinear). OB and OC are radii of the circle. If the measure of angle OBC is $64°$, what is the measure, in degrees, of arc AD?

Answer:

897. GT-103 · Hard · SPR

A sector of a circle with radius 9 has an area of 18π. The central angle of the sector measures $k\pi$ radians. What is the value of k?

Answer:

898. GT-105 · Hard · SPR

The equation $x^2+y^2-8x+10y=8$ defines a circle in the xy-plane. What is the radius of the circle?

Answer:

899. GT-106 · Hard · SPR

Points A, B, and C lie on a circle, and $\angle ABC$ is a right angle. Point D lies on segment AC such that $BD \perp AC$, with $BD = 6$. The diameter of the circle is 13 and $AB < BC$. What is the value of $\frac{CD}{AD}$?

Answer:

900. GT-111 · Hard · SPR

In a circle with center O, segment AC is a diameter and $AC = 426$. Point D lies outside the circle with $OD = AC$, and $\overline{CD}$ is tangent to the circle at C. Segment OD intersects the circle at E. The length of minor arc AE is $b\pi$. What is the value of b?

Answer:

901. GT-115 · Hard · SPR

Circle A in the xy-plane has equation $x^2+y^2-8x+6y-8n=0$, where n is a constant. Circle B has the same center as circle A, has twice the diameter of circle A, and passes through $(10,-3)$. What is the value of n?

Answer:

902. GT-122 · Hard · No calculator needed

In the xy-plane, a circle passes through the points $(-1,2)$, $(7,2)$ and $(6,3)$. Which of the following is an equation of the circle?

A) $(x-3)^2+(y+1)^2=25$ **B)** $(x-3)^2+(y-2)^2=16$

C) $(x-3)^2+(y+1)^2=5$ **D)** $(x-3)^2+(y-1)^2=17$

903. GT-123 · Hard · Desmos

In the xy-plane, the line $x+y=4$ intersects the circle $x^2+y^2-4x+2y=24$ at two points. What is the distance between these two points?

A) 7 **B)** $7\sqrt{2}$

C) 14 **D)** $\dfrac{7\sqrt{2}}{2}$

904. GT-127 · Hard · No calculator needed

In the xy-plane, the line $y=k(x-1)-2$, where k is a positive constant, is tangent to the parabola $y=x^2+2x+4$. What is the value of k?

A) 8 **B)** 10

C) -2 **D)** 4

905. GT-131 · Hard · SPR · No calculator needed

$$x^2+y^2-6x+8y=k$$

In the xy-plane, the graph of the equation above, where k is a constant, is a circle that is tangent to the line $4x-3y=-11$. What is the value of k?

Answer:

906. GT-135 · Hard · No calculator needed

$$(x-a-2)^2+(y-2a+1)^2=25$$

In the xy-plane, the circle above is defined for each real number a. Which of the following points lies on the circle for every real value of a?

A) $(a-1,\ 2a+2)$ **B)** $(a+5,\ 2a+3)$

C) $(a+5,\ 2a+4)$ **D)** $(2a+5,\ 2a+3)$

907. GT-137 · Hard · Calculator

$$(x-1)^2+(y+3)^2=40$$

$$x^2+y^2-4kx+2ky=9$$

In the xy-plane, the two equations above define circles. The center of the second circle lies on the first circle. If k is a positive constant, what is the value of k?

A) $-1+\sqrt{7}$ **B)** $2+\sqrt{7}$

C) $1+2\sqrt{7}$ **D)** $1+\sqrt{7}$

908. GT-140 · Hard · SPR · No calculator needed

In the xy-plane, a circle has equation $x^2+y^2+Dx+Ey+F=0$, where D, E, and F are constants. The circle passes through the points $(2,7)$ and $(10,7)$, has radius 5, and has its center below the line $y=7$. What is the value of $E+F$?

Answer:

909. GT-144 · Hard · No calculator needed

$$x^2+y^2-4x+6y-12=0$$

In the xy-plane, the graph of the equation above is a circle. Line ℓ has slope $\dfrac{3}{4}$ and is tangent to the circle at a point with a positive y-coordinate. What is the y-intercept of ℓ?

A) 1 **B)** $\dfrac{7}{4}$

C) $-\dfrac{1}{3}$ **D)** $-\dfrac{43}{4}$

910. GT-146 · Hard · SPR · No calculator needed

In the xy-plane, the circle with equation $(x-7)^2+(y+2)^2=9$ has two distinct tangent lines that pass through the point $(2,-2)$. What is the product of the slopes of these two lines?

Answer:

911. GT-148 · Hard · SPR · Calculator

In the xy-plane, a circle lies in the first quadrant and is tangent to both the x-axis and the y-axis. The line $3x + 4y = 20$ intersects the circle at two points 8 units apart. What is the sum of all possible values of the radius of the circle?

Answer:

912. GT-150 · Hard · No calculator needed

In the xy-plane, the center of a circle lies on the y-axis. The circle passes through the points $(6, 1)$ and $(-2, 5)$. Which of the following is an equation of the circle?

A) $x^2 + (y+1)^2 = 44$ **B)** $x^2 + (y+1)^2 = 40$

C) $x^2 + (y-1)^2 = 40$ **D)** $x^2 + (y+1)^2 = 20$

913. GT-155 · Hard · No calculator needed

Triangle ABC is inscribed in a circle with circumference 52π. Side AC has length 52 and side AB has length 48. What is the ratio of the length of the shortest side of the triangle to the radius of the circle?

A) $\frac{10}{13}$ **B)** $\frac{24}{13}$

C) $\frac{5}{12}$ **D)** $\frac{5}{13}$

914. GT-156 · Hard · SPR · Calculator

In the xy-plane, the line $y = mx + 10$, where m is a constant, intersects the circle $x^2 + y^2 = 12$ at exactly one point. What is the value of m^2?

Answer:

915. GT-157 · Hard · No calculator needed

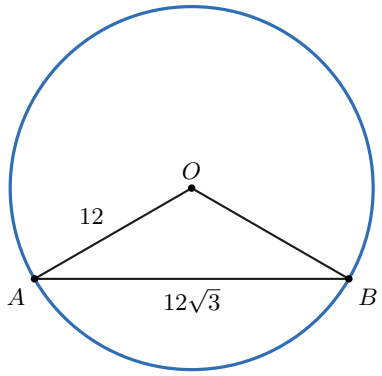
In the xy-plane, the circle $x^2 + y^2 + 8x - 2y - 8 = 0$ and the line $5x + 12y = 44$ intersect at two points, P and Q. What is the length of $\overline{PQ}$?

A) 6 **B)** $2\sqrt{21}$

C) 8 **D)** 3

916. GT-159 · Hard · No calculator needed

In the circle shown, O is the center, the radius is 12, and chord $\overline{AB}$ has length $12\sqrt{3}$. What is the area of the region bounded by chord $\overline{AB}$ and the minor arc AB?



A) $24\pi - 36\sqrt{3}$ **B)** $48\pi - 36\sqrt{3}$

C) 48π **D)** $96\pi + 36\sqrt{3}$

917. GT-164 · Hard · No calculator needed

A circular clock face has radius 15 centimeters. A shaded sector of the clock face has area 45π square centimeters. What is the perimeter, in centimeters, of the shaded sector (the arc together with the two bounding radii)?

A) $30 + 6\pi$ **B)** $15 + 6\pi$

C) $30 + 3\pi$ **D)** $30 + 45\pi$

918. GT-166 · Hard · No calculator needed

Segment $\overline{AB}$ is a diameter of a circle and has length 20. Point C lies on the circle, and point D lies on $\overline{AB}$ with $\overline{CD} \perp \overline{AB}$ and $CD = 8$. If $AD < DB$, what is the value of $\frac{AC}{CB}$?

A) $\frac{1}{4}$ **B)** $\frac{2}{5}$

C) $\frac{1}{2}$ **D)** $\frac{4}{5}$

919. GT-167 · Hard · Calculator

A circle has area 48π. A square and an equilateral triangle are each inscribed in the circle (all vertices on the circle). By how much does the area of the square exceed the area of the triangle?

A) $96 - 27\sqrt{3}$ **B)** $96 - 36\sqrt{3}$

C) $96 - 12\sqrt{3}$ **D)** $192 - 36\sqrt{3}$

920. GT-169 · Hard · Desmos

In the xy-plane, the circle $x^2 + y^2 + 8x - 2y = 8$ and the line $5x - 12y = k$, where k is a constant, intersect at exactly one point. What is the greatest possible value of k?

A) 33 **B)** -97

C) 65 **D)** 32

921. GT-170 · Hard · No calculator needed

In a circle with center O, chord $\overline{AB}$ has $\angle AOB = 120°$ and lies at a perpendicular distance of 3 from O. The lines tangent to the circle at A and at B intersect at point P. What is the length of $\overline{PA}$?

A) 6 **B)** $6\sqrt{3}$

C) 12 **D)** $3\sqrt{3}$

922. GT-172 · Hard · SPR · No calculator needed

In the xy-plane, a circle is tangent to both the x-axis and the y-axis, has its center in the second quadrant, and passes through the point $(-2, 9)$. The diameter of the circle is less than 15. The circle has equation $x^2 + y^2 + Dx + Ey + F = 0$, where D, E and F are constants. What is the value of F?

Answer:

923. GT-189 · Hard · SPR

$$x^2 + y^2 + 10x - 24y = 0$$

In the xy-plane, the graph of the equation above is a circle. What is the diameter of the circle?

Answer:

924. GT-190 · Hard · SPR

In the xy-plane, a circle has its center on the x-axis and passes through the points $(0, 4)$ and $(8, 4)$. What is the square of the radius of the circle?

Answer:

925. GT-191 · Hard

In the xy-plane, the circle $(x-2)^2+(y+1)^2 = 25$ passes through the point $(6, 2)$. What is the slope of the line tangent to the circle at $(6, 2)$?

A) $-\frac{3}{4}$ **B)** $\frac{4}{3}$

C) $-\frac{4}{3}$ **D)** $\frac{3}{4}$

926. GT-194 · Hard · SPR

Point P is 10 units from the center O of a circle with radius 6. Segments $\overline{PA}$ and $\overline{PB}$ are tangent to the circle at A and B. What is the area of quadrilateral $PAOB$?

Answer:

927. GT-200 · Hard · SPR

A circle has area 50π. A square is inscribed in the circle so that all four vertices lie on the circle. What is the area of the square?

Answer:

928. GT-223 · Hard

Circle C_1 has equation $x^2 + y^2 - 6x + 4y + 4 = 0$. Circle C_2 has equation $x^2 + y^2 + 2x - 2y + k = 0$, where k is a constant. If the two circles are externally tangent, what is the value of k?

A) -62 **B)** -7

C) -2 **D)** 1

929. GT-224 · Hard

In the xy-plane, circle A has equation $(x - 2)^2 + (y + 1)^2 = 9$. Circle B is the image of circle A after a translation 3 units left and 4 units up, followed by a dilation about its own centre that doubles the radius. Which is an equation of circle B?

A) $(x+1)^2+(y-3)^2 = 36$ **B)** $(x+1)^2+(y-3)^2 = 18$

C) $(x-5)^2+(y+5)^2 = 36$ **D)** $(x+1)^2+(y-3)^2 = 6$

930. GT-225 · Hard · SPR

An equilateral triangle with perimeter 36 is inscribed in a circle. What is the area of the circle divided by π?

Answer:

931. GT-232 · Hard · No calculator needed

In the xy-plane, a circle has equation $(x-4)^2+(y+2)^2 = 50$. Which point lies in the interior of the circle?

A) $(10, 2)$ **B)** $(9, 3)$

C) $(-1, -6)$ **D)** $(-3, 0)$

932. GT-233 · Hard · SPR · No calculator needed

The graph of $x^2 + y^2 + 10x - 8y + k = 0$ in the xy-plane is a circle, where k is a constant. What is the greatest possible integer value of k?

Answer:

933. GT-235 · Hard · No calculator needed

Points A, B and C lie on a circle with radius 12, and C lies on the major arc AB. The measure of angle ACB is $\frac{2\pi}{9}$ radians. What is the length of minor arc AB?

A) $\frac{16\pi}{3}$ **B)** $\frac{8\pi}{3}$

C) $\frac{64\pi}{3}$ **D)** $\frac{56\pi}{3}$

934. GT-247 · Hard · No calculator needed

In a circle of radius 13, two parallel chords have lengths 24 and 10. Which of the following could be the distance between the two chords?

A) 12 **B)** 17

C) 26 **D)** 5

935. GT-251 · Hard · No calculator needed

A rectangle is inscribed in a circle with area 676π. The length of the rectangle's diagonal is twice the length of its shorter side. What is the perimeter of the rectangle?

A) $52 + 52\sqrt{3}$ **B)** 104

C) $26 + 26\sqrt{3}$ **D)** $104 + 52\sqrt{3}$

936. GT-255 · Hard · Calculator

Model Earth as a sphere with radius $6{,}371$ kilometers. An observer's eye is 1 kilometer above the surface. The line of sight from the eye to the horizon is tangent to Earth. Which is closest to the distance, in kilometers, from the eye to the point where the line of sight touches Earth?

A) 112.9 **B)** 1.0

C) $6{,}371.0$ **D)** 79.8

Problem-Solving and Data Analysis (210)

Ratios, rates, proportional relationships, and units (43)

937. PS-004 · Medium

A car consumes fuel at a rate of 6.5 liters per 100 kilometers. At this rate, how many liters of fuel are needed to travel 340 kilometers?

A) 20.4 **B)** 21.0

C) 22.1 **D)** 23.9

938. PS-017 · Medium

A recipe for a traditional Saudi dish requires flour, sugar, and dates in the ratio $5 : 2 : 3$ by weight. If a bakery uses 18 kilograms of dates to prepare a large batch, how many kilograms of flour are used?

A) 12 **B)** 24

C) 30 **D)** 36

939. PS-054 · Medium

A car travels at a constant speed of 90 kilometers per hour. What is this speed in meters per second?

A) 15 **B)** 25

C) 32.4 **D)** 324

940. PS-055 · Medium

A recipe that serves 8 people uses 3 cups of orange juice and 5 cups of water. If the same ratio of ingredients is used to make a batch that serves 24 people, how many cups of orange juice are needed?

A) 6 **B)** 7.5

C) 9 **D)** 15

941. PS-056 · Medium · SPR

A factory produces cardboard boxes at a constant rate of 480 boxes every 8 hours. At this rate, how many boxes does the factory produce in 5.5 hours?

Answer:

942. PS-057 · Medium

An architect built a scale model of a tower planned for the King Abdullah Financial District, where every 1 centimeter on the model represents 2.5 meters of the actual tower. If the model is 18 centimeters tall, what is the height, in meters, of the actual tower?

A) 20 **B)** 36

C) 45 **D)** 50

943. PS-058 · Medium

A saline solution is mixed so that the ratio of salt to water is 2 to 15. If a container holds 51 liters of the solution in total, how many liters of salt are in the container?

A) 3 **B)** 6

C) 9 **D)** 12

944. PS-072 · Medium

A gift box contains dates that are either Ajwa or Sukkari, in a ratio of 7 to 5. When 8 Ajwa dates and 8 Sukkari dates are removed from the box, the ratio of Ajwa to Sukkari dates becomes 3 to 2. How many Sukkari dates were originally in the box?

A) 24 **B)** 32

C) 40 **D)** 48

945. PS-087 · Medium · SPR

A rectangular banner is 2.4 meters by 1.5 meters. Printing costs $\$0.05$ per square centimeter. What is the cost, in dollars, to print the banner? (1 meter $= 100$ centimeters.)

Answer:

946. PS-089 · Medium

A car travels at a constant speed of 90 kilometers per hour. What is the car's speed, in meters per second?

A) 1.5 **B)** 25

C) 90 **D)** 324

947. PS-090 · Medium · SPR

A rectangular mat is 1.2 meters long and 0.5 meter wide. What is the area of the mat, in square centimeters? (1 meter $= 100$ centimeters.)

Answer:

948. PS-173 · Medium · SPR · No calculator needed

The ratio of grade 9 students to grade 10 students in a club is $3 : 5$, and the ratio of grade 10 students to grade 11 students is $4 : 7$. The club has 201 students in grades 9, 10 and 11. How many more grade 11 students than grade 9 students are in the club?

Answer:

949. PS-187 · Medium · Calculator

A runner's pace is 6 minutes 15 seconds per kilometer. What is the runner's speed, in kilometers per hour?

A) 9.6 **B)** 6.25

C) 9.76 **D)** 0.16

950. PS-003 · Hard · SPR

For a construction project at NEOM, a concrete mixture is made from cement, sand, and gravel in the ratio $1:2:3$ by weight. A batching plant produces gravel at a constant rate of 6 metric tons per hour. At this rate, how many hours will it take the plant to produce 90 metric tons of the mixture?

Answer:

951. PS-059 · Hard

Car A travels 240 kilometers in 3 hours at a constant speed, and Car B travels 350 kilometers in 4.5 hours at a constant speed. Which of the following is closest to how much faster, in kilometers per hour, Car A's speed is compared with Car B's speed?

A) 0.9 **B)** 2.2

C) 4.4 **D)** 10.0

952. PS-060 · Hard

A metal block has a mass of 540 grams and a volume of 60 cubic centimeters. What is the density of the block, in kilograms per cubic meter? (1 gram per cubic centimeter $= 1{,}000$ kilograms per cubic meter.)

A) 9 **B)** 90

C) 900 **D)** $9{,}000$

953. PS-061 · Hard · SPR

A desalination plant in the Eastern Province produces fresh water at a constant rate of $1{,}250{,}000$ liters per hour. At this rate, how many cubic meters of water does the plant produce in one day? (1 cubic meter $= 1{,}000$ liters.)

Answer:

954. PS-062 · Hard

The value of y varies directly with x. When $x = 9$, $y = 45$. What is the value of y when $x = 14$?

A) 56 **B)** 63

C) 70 **D)** 77

955. PS-063 · Hard

A bus rapid transit line in Riyadh carries passengers at a constant average rate. The line carries $15{,}000$ passengers over a 6-hour morning service window. At this same constant rate, which of the following is closest to the number of passengers carried during a 2.5-hour afternoon service window?

A) $3{,}750$ **B)** $5{,}000$

C) $6{,}250$ **D)** $7{,}500$

956. PS-073 · Hard · SPR

Two cars are racing at constant speeds around a circular track. Car A requires 12 seconds to travel once around the track, and Car B requires 20 seconds to travel once around the track. If Car A passes Car B at a certain moment, how many seconds will elapse before Car A next passes Car B?

Answer:

957. PS-088 · Hard · SPR

A pump fills a rectangular tank at a rate of 0.8 cubic meter per minute. The tank is 12 meters long, 5 meters wide and 2 meters deep. How many hours does it take to fill the tank completely? (Enter as a decimal.)

Answer:

958. PS-093 · Hard · Calculator

A test vehicle accelerates at a constant 22 feet per second squared. Which of the following is equal to this acceleration in miles per minute squared? (1 mile $= 5280$ feet)

A) $\frac{1}{4}$ **B)** $\frac{1}{240}$

C) 15 **D)** $79{,}200$

959. PS-098 · Hard · SPR · Calculator

A car's fuel economy is 40 miles per gallon. Use 1 mile $= 1.6$ kilometers and 1 gallon $= 3.8$ liters. How many liters of fuel does the car need to drive 304 kilometers?

Answer:

960. PS-101 · Hard · SPR · Calculator

A commuter train accelerates at a constant rate of 0.5 meters per second squared. What is this acceleration, in kilometers per hour squared? (1 kilometer $= 1000$ meters and 1 hour $= 3600$ seconds.)

Answer:

961. PS-104 · Hard · SPR · No calculator needed

A table has 4 legs, each with the same ground-contact area of A square centimeters. With a box on it, the table presses on the floor with a pressure of 12 newtons per square centimeter on each leg. A 120-newton weight is then added to the box, and the table is tilted so that only 3 legs touch the floor. The pressure on each of those 3 legs is now 24 newtons per square centimeter. (Pressure is force divided by contact area, and the load is shared equally by the legs touching the floor.) What is the value of A?

Answer:

962. PS-108 · Hard · SPR · No calculator needed

A box contains only red, green and blue flash drives, 140 in all. The ratio of red drives to green drives is 7 to 4. The ratio of blue drives to drives that are not blue is 3 to 11. How many more red drives than blue drives are in the box?

Answer:

963. PS-110 · Hard · Calculator

A site plan was drawn so that 1 centimeter on the plan represents 40 meters of ground. A copier then enlarged the plan to 125% of its size. On the enlarged copy, the parcel appears as a rectangle measuring 7.5 centimeters by 5 centimeters. What is the actual area of the parcel, in hectares? (1 hectare $= 10{,}000$ square meters.)

A) 384 **B)** 3.84

C) 6 **D)** 38.4

964. PS-112 · Hard · SPR · No calculator needed

A rectangular banner has width w and length $1.8w$, where $w > 0$. The banner is covered exactly, with no gaps or overlaps, by 4 identical square tiles. The side length of each tile is $k\,w\sqrt{5}$, where k is a positive constant. What is the value of k?

Answer:

965. PS-116 · Hard · No calculator needed

Study room A has x students and study room B has 72 students. One-third of the students in room A move to room B. Then one-fourth of the students who are now in room B move to room A. After both moves, the ratio of the number of students in room A to the number in room B is 5 to 7. How many students were in room A originally?

A) 72 **B)** 36

C) 45 **D)** 63

966. PS-117 · Hard · SPR · Calculator

A tram increases its speed at a constant rate from 54 kilometers per hour to 90 kilometers per hour in 20 seconds. What is the tram's acceleration, in meters per minute squared?

Answer:

967. PS-119 · Hard · Calculator

A car uses fuel at a rate of 6.0 liters per 100 kilometers on a highway and 10.0 liters per 100 kilometers in the city. On one trip the car travels 240 kilometers on the highway and 80 kilometers in the city. Using 1 mile $= 1.6$ kilometers and 1 gallon $= 3.8$ liters, which of the following is closest to the car's fuel efficiency for the whole trip, in miles per gallon?

A) 87 **B)** 6

C) 30 **D)** 34

968. PS-120 · Hard · No calculator needed

On a map, 1 inch represents $1{,}500$ feet. A rectangular parking lot appears on the map as 2.4 inches by 1.6 inches. Which of the following is the actual area of the lot, in square yards? (1 yard $= 3$ feet)

A) $8{,}640{,}000$ **B)** $320{,}000$

C) $960{,}000$ **D)** $2{,}880{,}000$

969. PS-123 · Hard · Calculator

A cyclist rides a 90-kilometer route. She covers the first 60% of the distance at an average speed of 27 kilometers per hour, rests for 30 minutes, and then covers the rest of the route at an average speed of 24 kilometers per hour. What was her average speed for the whole trip, including the rest, in meters per second?

A) 22.5 **B)** 6.25

C) 7.08 **D)** 7.14

970. PS-127 · Hard · SPR

The intensity of light from a small lamp is inversely proportional to the square of the distance from the lamp. A student moves 20% closer to the lamp. By what percent does the intensity at the student's position increase?

Answer:

971. PS-129 · Hard · SPR

In a steady state, the average number of customers in a shop equals the average arrival rate multiplied by the average time a customer spends in the shop. A shop averages 12 customers per hour arriving, and each stays 45 minutes on average. After a promotion the arrival rate rises by 50% and the average time spent falls by 20%. By what percent does the average number of customers in the shop increase?

Answer:

972. PS-130 · Hard · SPR

The rainwater collected by a flat roof is proportional to the roof's area. A square roof with side 3 m collects 18 litres in a shower. In the same shower, how many litres, to the nearest tenth, does a circular roof with radius 2 m collect?

Answer:

973. PS-131 · Hard · SPR

Gear A has 24 teeth and drives gear B, which has 8 teeth. Gear C is fixed on the same axle as gear B and has 36 teeth. Gear C drives gear D, which has 12 teeth. Gear A turns at 90 revolutions per minute. How many revolutions per minute does gear D turn?

Answer:

974. PS-166 · Hard · Calculator

District A has $120{,}000$ residents and a population density of $4{,}000$ residents per square kilometer. District B has $60{,}000$ residents and a population density of $1{,}500$ residents per square kilometer. Which is closest to the population density, in residents per square kilometer, of the two districts combined?

A) $5{,}500$ **B)** $3{,}167$

C) $2{,}571$ **D)** $2{,}750$

975. PS-188 · Hard · No calculator needed

A pipe can fill an empty tank in 6 hours, and a drain can empty the full tank in 9 hours. Starting with the tank empty, the pipe and the drain are both open for 4 hours, and then the drain is closed. How many more hours does it take the pipe to fill the tank?

A) 2 **B)** $\dfrac{26}{3}$

C) $\dfrac{14}{3}$ **D)** 14

976. PS-189 · Hard · No calculator needed

A crew of 12 workers can finish a job in 20 days. After the crew has worked for 5 days, 3 more workers join, all working at the same rate. How many days in total does the job take?

A) 16 **B)** 25

C) 17 **D)** 12

977. PS-190 · Hard · No calculator needed

A driver travels from one city to another at an average speed of 60 kilometers per hour. At what average speed, in kilometers per hour, must the driver return along the same road so that the average speed for the whole round trip is 80 kilometers per hour?

A) 120 **B)** 106.7

C) 160 **D)** 100

978. PS-191 · Hard · No calculator needed

A printer prints p pages every m seconds. Which expression gives the number of minutes the printer takes to print n pages?

A) $\dfrac{60m}{pn}$ **B)** $\dfrac{60mn}{p}$

C) $\dfrac{\frac{pn}{p}}{60m}$ **D)** $\dfrac{mn}{60p}$

979. PS-192 · Hard · Calculator

A bank sells US dollars at 3.77 riyals per dollar. It first deducts a 1% commission from the riyals a customer pays. How many US dollars does a customer receive for $5{,}655$ riyals?

A) $1{,}515$ **B)** $1{,}485$

C) $21{,}319$ **D)** $1{,}500$

Percentages (51)

980. PS-002 · Medium

A jacket is discounted by 30%. A 5% sales tax is then applied to the discounted price. If the final price paid is $\$44.10$, what was the original price of the jacket, before the discount?

A) $\$42.00$ **B)** $\$58.80$

C) $\$60.00$ **D)** $\$63.00$

981. PS-014 · Medium

A laptop's price increased by 20% during a promotional period, then decreased by 15% after the promotion ended. If the final price is SAR $1{,}020$, what was the original price, before the increase?

A) 950 **B)** $1{,}000$

C) $1{,}050$ **D)** $1{,}200$

982. PS-023 · Medium

Which of the following represents the result of decreasing the quantity y by 12%, where $y > 0$?

A) $0.88y$ **B)** $0.12y$

C) $y - 12$ **D)** $y - 0.12$

983. PS-024 · Medium · SPR

The number m is 24% greater than 75. If m is the product of 75 and r, what is the value of r?

Answer:

984. PS-025 · Medium

A household's monthly electricity bill was SAR 340.00. After a rate adjustment, the bill is now SAR 378.20. To the nearest tenth of a percent, by what percent did the amount of the bill increase?

A) 10.1% **B)** 11.2%

C) 11.9% **D)** 12.4%

985. PS-026 · Medium

Fahad purchased a car for SAR $90{,}000$. He estimates that each year the value of the car will depreciate by 15% of the previous year's estimated value. What is the estimated value of the car exactly 2 years after Fahad purchased it?

A) SAR $61{,}200$ **B)** SAR $63{,}750$

C) SAR $65{,}025$ **D)** SAR $76{,}500$

986. PS-064 · Medium

A school's enrollment increased from 640 students to 720 students over one year. What is the percent increase in enrollment?

A) 10% **B)** 11.1%

C) 12.5% **D)** 14.3%

987. PS-065 · Medium

A laptop has a listed price of SAR $3{,}200$. A store offers a 15% discount, and then 15% VAT is added to the discounted price. What is the final price of the laptop, in SAR?

A) $2{,}976$ **B)** $3{,}008$

C) $3{,}128$ **D)** $3{,}200$

988. PS-066 · Medium · SPR

After a 20% discount, the price of a jacket is SAR 256. What was the original price, in SAR, of the jacket before the discount?

Answer:

989. PS-067 · Medium

A survey of 850 Grade 11 students found that 34% plan to take the SAT this year. Which of the following is closest to the number of students who plan to take the SAT?

A) 255 **B)** 272

C) 289 **D)** 306

990. PS-071 · Medium

If 30 percent of 40 percent of x is equal to 50 percent of y minus 20 percent of y, what is the value of x in terms of y?

A) $x = 2y$ **B)** $x = 2.5y$

C) $x = 3y$ **D)** $x = 5y$

991. PS-075 · Medium · SPR

The mass of object A is k grams. The mass of object B is 130% of the mass of object A. If the mass of object B is 66 grams greater than the mass of object A, what is the value of k?

Answer:

992. PS-076 · Medium · SPR

A lab combines 40 grams of a solution that is 0.3% sodium chloride by mass with 60 grams of a solution that is 0.05% sodium chloride by mass. What is the percent of sodium chloride by mass in the combined 100 grams of mixture?

Answer:

993. PS-177 · Medium · Calculator

The monthly rent of an apartment increased by 60%. By what percent must the new rent decrease to return to the original rent?

A) 62.5% **B)** 37.5%

C) 60% **D)** 40%

994. PS-001 · Hard · SPR

The number of Umrah pilgrims visiting Makkah increased by 25% from 2022 to 2023. From 2023 to 2024, the number of Umrah pilgrims decreased by 12%. If the number of Umrah pilgrims in 2024 was $9{,}020{,}000$, what was the number of Umrah pilgrims in 2022?

Answer:

995. PS-016 · Hard · SPR

The number of families visiting a Riyadh Season family zone increased by 40% from Week 1 to Week 2, then decreased by 25% from Week 2 to Week 3. If Week 3 attendance was $8{,}400$ families, what was the Week 1 attendance?

Answer:

996. PS-027 · Hard · SPR

The table below summarizes the number of students in each of four school clubs.

Club	Number of students
A	500
B	72
C	1,100
D	128
Total	1,800

The number of students in club C is $p\%$ of the number of students in club A. What is the value of p?

Answer:

997. PS-028 · Hard

In a factory quality check, 40% of the items inspected in a batch are defective. Of those defective items, 45% are also mislabeled. Which of the following is closest to the percentage of items in the batch that are NOT mislabeled defective items?

A) 18% **B)** 55%

C) 82% **D)** 96.4%

998. PS-068 · Hard

The population of a town increased by 10% in the first year and then decreased by 10% in the second year. If the population at the start of the first year was $5{,}000$, what was the population at the end of the second year?

A) $4{,}900$ **B)** $4{,}950$

C) $5{,}000$ **D)** $5{,}050$

999. PS-069 · Hard

A Vision 2030 renewable-energy report estimated that a new solar farm would generate 480 gigawatt-hours (GWh) of electricity in its first year. The farm actually generated 432 GWh. What is the percent error of the estimate, relative to the actual amount generated?

A) 9.1% **B)** 10.0%

C) 11.1% **D)** 12.5%

1000. PS-082 · Hard

The price of an item is increased by 30%, and then the new price is decreased by 30%. The final price is what percent of the original price?

A) 9% **B)** 91%

C) 100% **D)** 109%

1001. PS-083 · Hard · SPR

A store sells an item at 40% above its cost, then offers a 25% discount. The discounted selling price is $\$210$. What was the store's cost, in dollars?

Answer:

1002. PS-084 · Hard · SPR

A town's population increased by 10% in 2023 and then decreased by 20% in 2024, ending at $8{,}800$. What was the population before the 2023 increase?

Answer:

1003. PS-085 · Hard · SPR

The number x is 50% greater than y, and y is 20% less than z, where $z > 0$. The number x is what percent of z?

Answer:

1004. PS-086 · Hard · SPR

A quantity increases by 10%, then increases by 10% again, and then decreases by 20%. By what percent has the quantity decreased from its original value? (Enter as a decimal.)

Answer:

1005. PS-095 · Hard · SPR · No calculator needed

A rectangular banner has an area of 540 square inches. The length of the banner is 150% of its width, and the width is $c\sqrt{10}$ inches, where c is a positive constant. What is the value of c?

Answer:

1006. PS-097 · Hard · Calculator

A metalworker melts together 150 g of alloy X, which is 20% nickel by mass, and m g of alloy Y, which is 50% nickel by mass. Then 40 g of pure nickel is stirred into the melt. The final alloy is 40% nickel by mass. What is the value of m?

A) 250 **B)** 460

C) 60 **D)** 100

1007. PS-100 · Hard · SPR · Calculator

The area covered by a coral colony increases by 33.1% every 3 years. If the colony grows by the same percent each year, by what percent does its area increase each year?

Answer:

1008. PS-103 · Hard · Calculator

Last year a shop sold only mugs and plates. This year the number of mugs sold rose by 30%, the number of plates sold fell by 10%, and the total number of items sold rose by 6%. Of the items sold this year, what percent were mugs, to the nearest tenth of a percent?

A) 49.1% **B)** 50.9%

C) 52.0% **D)** 40.0%

1009. PS-106 · Hard · SPR · No calculator needed

In 2020, the population of city M was 20% greater than the population of city N. From 2020 to 2024, the population of city M increased by 15% and the population of city N decreased by 8%. The population of city M in 2024 was $p\%$ greater than the population of city N in 2024. What is the value of p?

Answer:

1010. PS-107 · Hard · SPR · Calculator

The positive quantities a, b, c and d satisfy the following: b is 0.6% of a, c is 5% of b, and d is 150% of c. If $b + c + d = 27$, what is the value of a?

Answer:

1011. PS-114 · Hard · SPR · Calculator

A retailer's total sales were the same in March and April. In March, online sales were p percent of the total sales and came to $\$18{,}000$. In April, online sales were $(p + 8)$ percent of the total sales and came to $\$25{,}200$. What were the retailer's total sales, in dollars, for one of these months?

Answer:

1012. PS-115 · Hard · Calculator

A metalworker melts x kilograms of an alloy that is 30% tin and mixes in 15 kilograms of pure tin. She then adds 25 kilograms of a second alloy that is 10% tin. The final alloy is 35% tin. What is the total mass, in kilograms, of the final alloy?

A) 125 **B)** 70

C) 85 **D)** 110

1013. PS-121 · Hard · No calculator needed

A library's collection consists of fiction, nonfiction and reference books. Next year the number of fiction books will increase by 20%, the number of nonfiction books will decrease by 10%, and the number of reference books will not change. After these changes, the collection will contain 1% more books than it does now, and the number of fiction books will equal the number of nonfiction books. What percent of the current collection is reference books?

A) 35% **B)** 65%

C) 15% **D)** 20%

1014. PS-122 · Hard · SPR · Calculator

A store raises the price of a jacket by r percent. Later, it reduces the new price by $(r - 13)$ percent. After both changes, the final price is 10% higher than the original price. What is the value of r?

Answer:

1015. PS-125 · Hard · SPR · No calculator needed

Quantity Q is 60% greater than quantity P. Quantity R is 25% less than Q. Quantity R is 50% greater than quantity S. P is what percent greater than S?

Answer:

1016. PS-126 · Hard · SPR · No calculator needed

A 84-gram sample of an alloy contains only copper, zinc and tin. The mass of the copper is 5 grams more than twice the mass of the zinc. The mass of the tin is half the combined mass of the copper and the zinc. What is the mass, in grams, of the tin in the sample?

Answer:

1017. PS-132 · Hard · SPR

A bag of rice costs 40% more than a bag of beans. Three bags of rice and two bags of beans cost a total of $\$111.60$. What is the cost, in dollars, of one bag of rice?

Answer:

1018. PS-164 · Hard · Calculator

Electric vehicles made up 4% of a country's new-car sales in one year and 6% the next year. Total new-car sales increased by 10% from the first year to the second. By what percent did the number of electric vehicles sold increase?

A) 2% **B)** 60%

C) 65% **D)** 50%

1019. PS-165 · Hard · SPR · No calculator needed

The number a is 150% more than the number b, and b is 60% less than the positive number c. The number a is what percent of c?

Answer:

1020. PS-167 · Hard · SPR · Calculator

Bottle B holds 20% more juice than bottle A and costs 15% more. To the nearest tenth of a percent, by what percent is the price per liter of bottle B less than the price per liter of bottle A?

Answer:

1021. PS-176 · Hard · No calculator needed

If 18% of a is 24, what is $a\%$ of 18?

A) 4.32 **B)** 24

C) $2{,}400$ **D)** $133\frac{1}{3}$

1022. PS-178 · Hard · Calculator

A store's profit on each jacket is 37.5% of the selling price. The profit is what percent of the cost of the jacket to the store?

A) 37.5% **B)** 27.3%

C) 62.5% **D)** 60%

1023. PS-179 · Hard · Calculator

A restaurant adds a 15% tax and a 10% service charge to a bill. Both are calculated on the original subtotal. The total paid is 250 dollars. What was the subtotal, in dollars?

A) 200.00 **B)** 217.39

C) 197.63 **D)** 187.50

1024. PS-180 · Hard · No calculator needed

A club has 40 members, and 30% of the members are girls. How many girls must join the club, with no one leaving, so that girls make up 60% of the members?

A) 20 **B)** 12

C) 30 **D)** 28

1025. PS-181 · Hard · Calculator

A worker spends 30% of her monthly salary on rent, then 25% of the remaining amount on food, then 40% of what is left after that on savings. She has $3{,}780$ dollars left. What is her monthly salary, in dollars?

A) $75{,}600$ **B)** $7{,}200$

C) $12{,}000$ **D)** $6{,}300$

1026. PS-182 · Hard · Calculator

Two positive numbers a and b are in the ratio $a : b = 9 : 5$. The number b is what percent less than a, to the nearest tenth of a percent?

A) 44.4% **B)** 80.0%

C) 55.6% **D)** 28.6%

1027. PS-183 · Hard · No calculator needed

The number y is $p\%$ less than the positive number x, where $0 < p < 100$. Which expression is equal to x?

A) $\dfrac{(100-p)y}{100}$ **B)** $\dfrac{100y}{100+p}$

C) $y\left(1+\dfrac{p}{100}\right)$ **D)** $\dfrac{100y}{100-p}$

1028. PS-184 · Hard · No calculator needed

Fresh dates are 80% water by mass. After drying, the dates are 50% water by mass. How many kilograms of dried dates are produced from 60 kilograms of fresh dates?

A) 30 **B)** 24

C) 42 **D)** 12

1029. PS-185 · Hard · No calculator needed

In a survey, exactly 37.5% of the people answered "yes," exactly 20% answered "no," and the rest were undecided. What is the least possible number of people surveyed?

A) 5 **B)** 40

C) 8 **D)** $1{,}000$

1030. PS-186 · Hard · No calculator needed

At a school, 85% of the boys and 60% of the girls passed an exam, and 70% of all the students passed. What percent of the students at the school are boys?

A) 25% **B)** 50%

C) 60% **D)** 40%

One-variable data: Distributions and measures of center and spread (37)

1031. PS-008 · Medium

The table shows the number of siblings reported by 20 students.

Number of siblings	0	1	2	3	4
Number of students	5	8	4	2	1

What is the mean number of siblings for these 20 students?

A) 1.1 **B)** 1.2

C) 1.3 **D)** 1.5

1032. PS-029 · Medium

The table below shows the distribution of ages of 20 tutors enrolled in a training program.

Age	Frequency
20	7
21	5
22	3
23	2
24	1
30	2

Which of the following gives the correct order of the mean, median, and mode of the ages?

A) mode $<$ median $<$ mean

B) mode $<$ mean $<$ median

C) median $<$ mode $<$ mean

D) mean $<$ mode $<$ median

1033. PS-030 · Medium

Data set A and data set B each consist of 20 values. The table shows the frequencies of the values for each data set. Which of the following statements best compares the means of the two data sets?

Value	Data set A frequency	Data set B frequency
30	2	7
35	5	6
40	6	4
45	7	3

A) The mean of data set A is greater than the mean of data set B.
B) The mean of data set A is less than the mean of data set B.
C) The mean of data set A is equal to the mean of data set B.
D) There is not enough information to compare the means of the data sets.

1034. PS-070 · Medium · SPR

If the average (arithmetic mean) of w and x is 40, the average of p and q is 40, and the average of m and n is 40, what is the average of w, x, p, q, m, and n?

Answer:

1035. PS-079 · Medium · SPR

The mean of 5 numbers is 14. One of the numbers, 20, is replaced by 8. What is the mean of the new set of 5 numbers? (Enter as a decimal.)

Answer:

1036. PS-158 · Medium · SPR · No calculator needed

The daily high temperatures in a city for one week have a mean of $34°$C and a standard deviation of $4°$C. Each temperature is converted to degrees Fahrenheit using $F = 1.8C + 32$. What is the standard deviation of the converted temperatures, in degrees Fahrenheit?

Answer:

1037. PS-200 · Medium · SPR · No calculator needed

In a data set, a value is called a low outlier if it is less than $Q_1 - 1.5 \cdot \text{IQR}$, where Q_1 is the first quartile and IQR is the interquartile range. For a data set, $Q_1 = 28$ and the third quartile is 40. What is the greatest integer value that would be a low outlier?

Answer:

1038. PS-007 · Hard

Box plots summarize final exam scores for two Grade 11 sections.

Class A: minimum 60, $Q1 = 70$, median $= 78$, $Q3 = 85$, maximum 95. Class B: minimum 55, $Q1 = 65$, median $= 78$, $Q3 = 90$, maximum 100.

Which statement is best supported by these box plots?

A) The two classes have the same median and the same interquartile range (IQR).
B) The two classes have the same median, but Class B's scores are more spread out, as shown by its greater IQR.
C) Class A has a greater median than Class B.
D) Class B's scores are less spread out than Class A's, as shown by its smaller IQR.

1039. PS-031 · Hard

A data set of 24 different numbers has a mean of 50 and a median of 50. A new data set is created by adding 5 to each number in the original data set that is greater than the median and subtracting 5 from each number that is less than the median. Which of the following measures does NOT have the same value in both the original and new data sets?

A) Median

B) Mean

C) Sum of the numbers

D) Standard deviation

1040. PS-032 · Hard · SPR

The mean score of 12 students on a quiz was 16.5 points. If the highest individual score is removed, the mean score of the remaining 11 students becomes 15 points. What was the highest score?

Answer:

1041. PS-074 · Hard

Let a, b, and c be positive integers. If the average (arithmetic mean) of a, b, and c is 100, which of the following is NOT a possible value of any of the three integers?

A) 150 **B)** 250

C) 298 **D)** 299

1042. PS-077 · Hard · SPR

The mean of 10 numbers is 38. After one number is removed, the mean of the remaining 9 numbers is 36. What is the value of the number that was removed?

Answer:

1043. PS-078 · Hard

A data set is $10,\ 14,\ 18,\ 22,\ 36$. A new value of 18 is added to the set. Which of the following statements about the new data set is (are) true?

I. The mean is less than the mean of the original set. II. The median is the same as the median of the original set. III. The range is the same as the range of the original set.

A) I only **B)** I and II only

C) II and III only **D)** I, II and III

1044. PS-080 · Hard · SPR

A group of 30 people has a mean age of 70 years. When a second group of 20 people is added, the mean age of all 50 people is 76 years. What is the mean age, in years, of the second group?

Answer:

1045. PS-081 · Hard

The frequency table shows the scores of 30 students on a 5-point quiz.

Score	1	2	3	4	5
Frequency	3	5	8	9	5

What is the median score?

A) 2.5 **B)** 3

C) 3.5 **D)** 4

1046. PS-091 · Hard · No calculator needed

A data set has eight values: $12,\ 15,\ 18,\ 20,\ 20,\ 22,\ 25,\ 28$. The values 12 and 28 are removed and the values 19 and 21 are added in their place. Which of the following describes the mean and the standard deviation (SD) of the new data set compared with the original data set?

A) Mean smaller, SD smaller **B)** Mean same, SD larger

C) Mean same, SD smaller **D)** Mean same, SD same

1047. PS-094 · Hard · Calculator

A data set consists of the five values $5,\ 8,\ 17,\ x,\ x$, where x is a real number. The variance of a data set is the mean of the squared deviations of its values from their mean. What is the least possible value of the variance of this data set?

A) 26 **B)** 78

C) 15.6 **D)** 10

1048. PS-102 · Hard · Calculator

A data set has 10 values: $31,\ 38,\ 42,\ 45,\ 47,\ 52,\ 53,\ 58,\ 65,\ 69$. The mean of the data set is 50. The least value and the greatest value are removed, leaving 8 values. Which of the following is true of the new data set compared with the original? (SD means standard deviation.)

A) Mean greater; SD smaller **B)** Mean unchanged; SD smaller

C) Mean unchanged; SD unchanged **D)** Mean smaller; SD smaller

1049. PS-109 · Hard · SPR · No calculator needed

A list of nine positive integers has a mean of 12 and a median of 9. What is the greatest possible value of the sum of the three largest integers in the list?

Answer:

1050. PS-111 · Hard · No calculator needed

Nine positive integers, not necessarily distinct, have a sum of 100 and a median of 9. What is the greatest possible value of the mean of the four largest of these integers?

A) 25 **B)** 20.25

C) 21.75 **D)** 23.75

1051. PS-118 · Hard · No calculator needed

A data set has six values: $2,\ 3,\ 5,\ 12,\ 18,\ 20$. Every value is increased by 4, and then a seventh value equal to the mean of the six increased values is added to the data set. Consider the following statements about the new data set of seven values compared with the original data set.

I. The mean is 4 greater.
II. The standard deviation is smaller.
III. The median is 4 greater.

Which of the statements must be true?

A) I, II and III **B)** I only

C) I and II only **D)** II and III only

1052. PS-124 · Hard · No calculator needed

A data set has a median of 52 and an interquartile range (IQR) of 14. Each value x in the data set is replaced by $15 - \frac{x}{2}$. What is the sum of the median and the IQR of the new data set?

A) 3 **B)** 11

C) -18 **D)** -4

1053. PS-157 · Hard · No calculator needed

Data sets A and B each contain 20 values, all from the list $10, 20, 30, 40, 50$. The frequencies in A are $2, 3, 10, 3, 2$ and the frequencies in B are $5, 3, 4, 3, 5$, in that order. Which statement is true?

A) The standard deviations are equal because the means and ranges are equal.
B) The standard deviation of B is greater.
C) The standard deviation of A is greater.
D) The standard deviations cannot be compared without calculating them.

1054. PS-159 · Hard · SPR · No calculator needed

The data set $12, 15, 19, 24, 30$ has a range of 18. A sixth value, x, is added, and the range of the new data set is 1.5 times the original range. What is the sum of all possible values of x?

Answer:

1055. PS-160 · Hard · SPR · No calculator needed

The data set $2, 6, 9, 15, x$ has a mean equal to its median, where x is a real number. What is the sum of all possible values of x?

Answer:

1056. PS-161 · Hard · SPR · No calculator needed

A data set contains the values $1, 2, 3, 4, 5$ with frequencies $4, k, 6, 3, 5$, respectively, where k is a positive integer. The median of the data set is 3. What is the greatest possible value of k?

Answer:

1057. PS-162 · Hard · No calculator needed

The commute times, in whole minutes, of 20 students are grouped: 10 students took 0 to 9 minutes, 7 took 10 to 19 minutes, and 3 took 20 to 29 minutes. Which of the following could NOT be the median commute time?

A) 10 **B)** 7.5
C) 14 **D)** 4

1058. PS-163 · Hard · No calculator needed

Data set A has 11 values with median 40. Data set B has 11 values with median 52. The two data sets are combined. Which of the following could be the median of the combined 22 values?

A) 46 **B)** 38
C) 60 **D)** 54

1059. PS-175 · Hard · SPR · No calculator needed

Data set A has 10 integers: 4 are from 10 to 19 and 6 are from 20 to 29. Data set B has 10 integers: 3 are from 30 to 39 and 7 are from 40 to 49. What is the smallest possible value of (mean of B) $-$ (mean of A)?

Answer:

1060. PS-193 · Hard · No calculator needed

The data set $3, 5, 6, 8, 9, 11, 12, 46$ has one outlier, 46, which is removed. Which statement about the effect of removing 46 is true?

A) Only the mean changes.
B) Both the mean and the median decrease, and the mean decreases more.
C) The mean decreases and the median increases.
D) The mean decreases and the median stays the same.

1061. PS-194 · Hard · No calculator needed

A data set consists of $12, 15, 18, 21, 24, 27, 30, 33, 36, 40$. The value 40 is mistakenly recorded as 400. How do the median, the interquartile range (IQR) and the standard deviation of the recorded data compare with those of the correct data?

A) Only the median is unchanged.
B) The median, IQR and standard deviation are all unchanged.
C) The median is unchanged, but the IQR and standard deviation are greater.
D) The median and IQR are unchanged, but the standard deviation is greater.

1062. PS-195 · Hard · No calculator needed

Section P has 60 students and section Q has 24 students. Test-score summaries: Section P: minimum 55, first quartile 68, median 76, third quartile 84, maximum 97. Section Q: minimum 62, first quartile 77, median 83, third quartile 88, maximum 95. Which statement must be true?

A) At least 75% of the students in Q scored above the median of P.
B) The IQR of Q is greater than the IQR of P.
C) More students in Q than in P scored above 84.
D) The range of Q is greater than the range of P.

1063. PS-196 · Hard · No calculator needed

At a company, the median monthly salary is $9{,}000$ dollars and the mean monthly salary is $14{,}500$ dollars. Which statement must be true?

A) The most common salary is $9{,}000$ dollars.
B) The standard deviation of the salaries is $5{,}500$ dollars.
C) No more than half of the employees earn more than $14{,}500$ dollars.
D) More than half of the employees earn more than $14{,}500$ dollars.

1064. PS-197 · Hard · SPR · No calculator needed

In a data set, the values 1, 2, 3, 4 and 5 occur with relative frequencies 10%, 15%, 20%, 30% and 25%, respectively. What is the median of the data set minus the mean of the data set?

Answer:

1065. PS-198 · Hard · No calculator needed

A data set has six values. Their deviations from 40 (value minus 40) are -7, 3, 12, -5, x and 2. The mean of the data set is 42. What is the value whose deviation from 40 is x?

A) 35 **B)** 7

C) 47 **D)** 52

1066. PS-199 · Hard · No calculator needed

A data set of 30 values has a mean of 50 and a standard deviation of 10. The values 38 and 62 are added to the data set. Which statement is true?

A) The mean is unchanged and the standard deviation increases.
B) The mean is unchanged and the standard deviation decreases.
C) The mean and the standard deviation are both unchanged.
D) The mean and the standard deviation both increase.

1067. PS-201 · Hard · No calculator needed

The recorded values of a data set are $18, 22, 27, 30, 33, 36, 41, 47, 52$. The value 52 should have been 25. After the correction, how do the mean and the median change?

A) The mean decreases by 3 and the median is unchanged.
B) The median decreases and the mean is unchanged.
C) The mean decreases by 27 and the median decreases by 3.
D) The mean and the median both decrease by 3.

Two-variable data: Models and scatterplots (25)

1068. PS-005 · Medium

A teacher records hours studied (x) and exam score (y) for 6 students.

Hours studied (x)	1	2	3	4	5	6
Exam score (y)	55	60	68	75	80	86

The line of best fit for this data is approximately $y = 6.2x + 48.7$. Which of the following is the best interpretation of the slope of this line?

A) The predicted exam score for a student who studies 0 hours is 6.2.
B) Each additional hour studied is associated with a predicted increase of about 6.2 points on the exam.
C) Each additional point on the exam is associated with a predicted increase of about 6.2 hours studied.
D) The exam score increases by exactly 6.2 points for every student, regardless of hours studied.

1069. PS-038 · Medium · SPR

A household water-conservation study in Riyadh recorded, for several summer days, the daily high temperature x (in degrees Celsius above $25\,^\circ$C) and the household's water use y (in liters). A line of best fit for the data is given by $y = 12x + 20$. On one recorded day, $x = 10$ and the household's actual water use was 155 liters. What is the positive difference between the actual water use on that day and the water use predicted by the line of best fit?

Answer:

1070. PS-039 · Medium

A scatterplot shows the relationship between x, the distance in kilometers from the Jeddah Corniche, and y, the number of visitors counted per hour at a beach access point. The line of best fit for the data passes through the points $(2, 340)$ and $(10, 60)$. What is the slope of this line of best fit?

A) -35 **B)** -28

C) 28 **D)** 35

1071. PS-040 · Medium

A line of best fit models y, the installed solar capacity (in megawatts) at a NEOM pilot site, in terms of x, the number of years since 2015: $y = 45.2x + 12.5$. Which of the following is closest to the capacity predicted by this model at $x = 8.4$?

A) 380.5 **B)** 388.7

C) 392.2 **D)** 401.0

1072. PS-041 · Medium

A desalination plant's storage tank begins a shift completely full. The tank is drained at a constant rate throughout the shift to supply a neighborhood. What type of model best describes the volume of water remaining in the tank at any time during the shift?

A) A linear model with a positive slope
B) A linear model with a negative slope
C) An exponential growth model
D) An exponential decay model

1073. PS-042 · Medium

A scatterplot shows GAT preparation hours x and practice-exam score y for 10 students, with a line of best fit shown. For the student with the highest practice-exam score, the actual score was 96, and the score predicted by the line of best fit was 89. Which of the following is closest to the difference between the actual score and the predicted score for this student?

A) 10 **B)** 7

C) 4 **D)** 1

1074. PS-043 · Medium

The table shows the age x, in years, and the efficiency loss y, in percent, for a sample of rooftop solar panels.

x	1	2	3	4	5
y	3.5	6.0	8.6	11.0	13.4

Which of the following equations could be the line of best fit for the data?

A) $y = 2.5x + 1$ **B)** $y = x + 2.5$

C) $y = -2.5x + 1$ **D)** $y = 2.5x - 1$

1075. PS-150 · Medium

A teacher modeled the final score y of students in a course with the line of best fit $\hat{y} = 82 - 1.6x$, where x is the number of days the student was absent. Which statement is the best interpretation of the number 82 in this model?

A) A student absent for 82 days is predicted to score 0.
B) The predicted score decreases by 82 for each day absent.
C) The mean score of all students in the class is 82.
D) The predicted score of a student with no absences is 82.

1076. PS-154 · Medium

A model $y = 0.85x + 12$ was fitted to data in which x ranged from 20 to 80. A student uses the model to predict y when $x = 150$. Which statement about this prediction is best?

A) It is reliable, because the slope is positive.
B) It is less reliable, because 150 is outside the range of the data.
C) It is impossible, because y cannot be computed for $x > 80$.
D) It is reliable, because the model is linear.

1077. PS-208 · Medium · No calculator needed

For a data set, the correlation coefficient between height in centimeters and mass in kilograms is $r = 0.72$. The heights are converted to inches, the masses to pounds, and the two variables are swapped between the axes. What is the correlation coefficient for the converted data?

A) 0.72 **B)** -0.72

C) $0.72 \cdot \dfrac{2.2}{2.54}$ **D)** $\dfrac{1}{0.72}$

1078. PS-006 · Hard

As part of tracking Vision 2030 progress, annual tourism revenue (in billion SAR) is recorded for years since 2016 (x):

Approximate data points: $(0, 30)$, $(2, 45)$, $(4, 68)$, $(6, 95)$, $(8, 138)$, with described upward curvature in the later years.

A line of best fit for this data is $y = 13.5x + 20$. What is the residual for the data point at $x = 8$?

A) -10 **B)** 0

C) 10 **D)** 128

1079. PS-015 · Hard

A scatterplot showing the relationship between daily study time (x, in hours) and nightly sleep time (y, in hours) for a group of students has a correlation coefficient of $r = -0.82$. Which of the following best describes this relationship?

A) A strong, positive linear relationship between study time and sleep time.
B) A strong, negative linear relationship between study time and sleep time.
C) A weak, negative linear relationship between study time and sleep time.
D) No linear relationship between study time and sleep time.

1080. PS-044 · Hard · SPR

In which of the following tables is the relationship between the values of x and their corresponding y-values nonlinear?

A)

x	1	2	3	4
y	5	9	13	17

B)

x	1	2	3	4
y	2	4	6	8

C)

x	1	2	3	4
y	5	10	15	20

D)

x	1	2	3	4
y	3	6	12	24

Answer:

1081. PS-045 · Hard

A line of best fit models the relationship between x, the number of years since 2018, and y, the average monthly electricity subsidy, in SAR, provided to households in a Vision 2030 efficiency pilot program: $y = -18.6x + 340$. Which of the following is the best interpretation of the slope of this line of best fit in this context?

A) The estimated subsidy increases by SAR 18.6 each year.
B) The estimated subsidy decreases by SAR 18.6 each year.
C) The estimated subsidy decreases by SAR 340 each year.
D) The estimated initial subsidy was SAR 18.6.

1082. PS-046 · Hard

Ahmad deposits SAR 800 into Account A, which earns 5% interest compounded annually. He deposits SAR $1{,}500$ into Account B, whose balance increases by a fixed SAR 40 each year. Based on these growth patterns, which of the following best describes the long-term behavior of the balances of Account A and Account B?

A) The balance of Account A is always greater than the balance of Account B.
B) The balance of Account A is always less than the balance of Account B.
C) The balance of Account A starts lower than the balance of Account B but eventually exceeds it.
D) The balance of Account A starts higher than the balance of Account B but eventually falls below it.

1083. PS-151 · Hard

The table shows values of x and y for a data set.

x	0	1	2	3	4
y	8	12	18	27	40.5

Which equation fits all five pairs?

A) $y = 8(1.5)^x$ **B)** $y = 4x + 8$
C) $y = 2x^2 + 2x + 8$ **D)** $y = 8(1.2)^x$

1084. PS-152 · Hard · SPR

A line of best fit for a data set is $\hat{y} = 3.2x + 14$. For the data point with $x = 10$, the actual value is $y = 51$. What is the residual for this point (actual value minus predicted value)?

Answer:

1085. PS-153 · Hard

Ten data points lie exactly on the line $y = 2x + 1$, at $x = 1, 2, \ldots, 10$. A new data point $(11, 40)$, far above that line, is added. How does the least-squares line for all eleven points compare with the line $y = 2x + 1$?

A) Smaller slope and greater y-intercept
B) The same slope and the same y-intercept
C) Greater slope and smaller y-intercept
D) Greater slope and greater y-intercept

1086. PS-155 · Hard · SPR

A least-squares line for a data set is $\hat{y} = 1.5x + 4$. The mean of the x-values is 10. What is the mean of the y-values?

Answer:

1087. PS-156 · Hard · SPR

The residuals for the first four of five data points about a least-squares line are -2, 3, 1 and -4. What is the residual for the fifth point?

Answer:

1088. PS-171 · Hard · SPR · No calculator needed

A line of best fit for a data set is $\hat{y} = 2.5x + 40$. For the data point with $x = 24$, the predicted value of y is 20% less than the actual value. What is the residual for this data point?

Answer:

1089. PS-172 · Hard · SPR · No calculator needed

The model $\hat{y} = 4.8x + 62$ estimates a town's population y, in thousands, where x is the number of decades since 1980. According to the model, what is the estimated increase in the population, in people, per year?

Answer:

1090. PS-174 · Hard · SPR · No calculator needed

A line of best fit for a data set is $\hat{y} = 2.4x + 15$. A new data set is formed by adding 5 to every x-value and keeping each y-value. What is the y-intercept of the line of best fit for the new data set?

Answer:

1091. PS-209 · Hard · No calculator needed

For data set P, the line of best fit is $\hat{y} = 4.2x + 3$ and the correlation coefficient is 0.41. For data set Q, the line of best fit is $\hat{y} = 0.6x + 10$ and the correlation coefficient is 0.93. Which statement is true?

A) The associations cannot be compared because the slopes are different.
B) P has the stronger linear association because its slope is greater.
C) Q has a weaker association because its slope is less than 1.
D) Q has the stronger linear association.

1092. PS-210 · Hard · No calculator needed

A model for a data set is $\hat{y} = 2.5x + 10$. The data points are $(2, 17)$, $(4, 19.5)$, $(6, 24)$, $(8, 31)$ and $(10, 34.5)$. For how many of the data points does the model underestimate the actual value of y?

A) 5 **B)** 2

C) 3 **D)** 1

Probability and conditional probability (20)

1093. PS-009 · Medium

As part of a Vision 2030 STEM-participation study, 200 Grade 11 students were surveyed on their intended career track.

	STEM track	Non-STEM track	Total
Male	60	40	100
Female	50	50	100
Total	110	90	200

If a student is selected at random from those surveyed and the student is female, what is the probability that the student intends a STEM track?

A) 0.45 **B)** 0.50

C) 0.55 **D)** 0.60

1094. PS-018 · Medium

The table below shows the number of students in two grade levels at a school who do and do not own a graphing calculator.

	Owns a graphing calculator	Does not own a graphing calculator	Total
Grade 10	18	7	25
Grade 11	22	3	25
Total	40	10	50

If one of these 50 students is selected at random, what is the probability that the student is in Grade 11 and does not own a graphing calculator?

A) $\frac{3}{50}$ **B)** $\frac{7}{50}$

C) $\frac{10}{50}$ **D)** $\frac{22}{50}$

1095. PS-019 · Medium · SPR

The table below shows the distribution of 50 students by club membership.

	Coding club	Not coding club	Total
Robotics club	15	8	23
Not robotics club	10	17	27
Total	25	25	50

If one of these 50 students is selected at random, what is the probability that the student is in the coding club but not the robotics club? (Express your answer as a decimal or fraction, not as a percent.)

Answer:

1096. PS-147 · Medium · SPR

Of 500 residents surveyed, 240 own a car, 180 own a bicycle, and 70 own both. If one of the surveyed residents is chosen at random, what is the probability that the resident owns neither a car nor a bicycle?

Answer:

1097. PS-149 · Medium · SPR

In a trial, 90 of 120 patients given Drug X improved, and 60 of 100 patients given Drug Y improved. How many times as likely is a patient given Drug X to improve as a patient given Drug Y?

Answer:

1098. PS-010 · Hard · SPR

Using the survey from the 90 students on the Non-STEM track (40 male, 50 female), two students are selected at random, one after another, without replacement. What is the probability that both selected students are female? (Round to the nearest thousandth.)

Answer:

1099. PS-020 · Hard · SPR

On January 1, 2024, there were 34.1 million mobile broadband subscriptions in Saudi Arabia. By January 1, 2025, the number of subscriptions had increased to 36.8 million. If a subscriber on January 1, 2025, is selected at random, the probability that the subscriber uses STC is 0.45. There were p million STC subscribers on January 1, 2025. To the nearest integer, what is the value of p?

Answer:

1100. PS-021 · Hard · SPR

The table below shows the distribution of employment sector and nationality for a group of employees at a company.

	Public sector	Private sector
Saudi	45	12
Non-Saudi	9	x

If one of these employees who works in the private sector is chosen at random, the probability that the employee is Non-Saudi is $\frac{1}{5}$. What is the value of x?

Answer:

1101. PS-022 · Hard · SPR

	WhatsApp	Paper form
Robotics	60%	80%
Sports	25%	10%
Art	10%	5%
Music	5%	5%

A school activity survey asked each student to select the one activity they preferred for a school event. Some students responded via WhatsApp, and the others responded on a paper form. The table above shows the distribution of preferred activity, in percent, for each response type. For the survey, the number of paper-form responses was twice the number of WhatsApp responses. If a student who preferred Art is selected at random, what is the probability that the student responded via the paper form? (Express your answer as a decimal or fraction, not as a percent.)

Answer:

1102. PS-092 · Hard · No calculator needed

A conference has 300 participants. Each participant attended either the Technical track or the Business track, and either in person or online. Some of the data are shown.

&Technical&Business&Total
In person&70& &
Online&50& &
Total&120& &300

Of the participants in the Business track, 35% attended online. How many of the 300 participants attended in person?

A) 187 **B)** 63

C) 113 **D)** 145

1103. PS-113 · Hard · No calculator needed

A survey of 240 adults asked whether they commute by bicycle and whether they own a helmet. Of the adults who commute by bicycle, $\frac{3}{5}$ own a helmet. Of the adults who do not commute by bicycle, $\frac{1}{4}$ own a helmet. If one adult is selected at random from those who own a helmet, the probability that the selected adult commutes by bicycle is $\frac{12}{19}$. How many of the 240 adults own a helmet?

A) 60 **B)** 95

C) 144 **D)** 100

1104. PS-128 · Hard · SPR

A school has 150 students in two sessions: 90 in the morning and 60 in the afternoon. In the morning session 40% of the students own a laptop, and in the afternoon session 70% do. A student is chosen at random from all students who own a laptop. What is the probability that the student is in the morning session? (Give your answer as a fraction or a decimal.)

Answer:

1105. PS-143 · Hard · SPR

A company's customers were asked whether they renewed their contract. The results are shown in the table.

	Renewed	Did not renew
Plan A	84	36
Plan B	45	35

If one of the customers who did not renew is chosen at random, what is the probability that the customer was on Plan B?

Answer:

1106. PS-144 · Hard · SPR

Events A and B are independent. The probability of A is 0.4, and the probability that at least one of A and B occurs is 0.7. What is the probability of B?

Answer:

1107. PS-145 · Hard

A bag holds 5 red marbles and 3 blue marbles. Two marbles are drawn at random, one after the other, without replacement. What is the probability that the two marbles are the same color?

A) $\frac{15}{28}$ **B)** $\frac{13}{28}$

C) $\frac{5}{14}$ **D)** $\frac{17}{32}$

1108. PS-146 · Hard · SPR

Two fair six-sided dice are rolled. Given that at least one of the dice shows an even number, what is the probability that the sum of the two dice is 8?

Answer:

1109. PS-148 · Hard · SPR

In a town, 30% of the residents are over 60 years old. Of the residents over 60, 80% vote in elections; of the residents who are not over 60, 50% vote. A voter is chosen at random from all residents who vote. What is the probability that the voter is over 60?

Answer:

1110. PS-168 · Hard · SPR · No calculator needed

In a school, a student is chosen at random. The probability that the student is in grade 10, given that the student is in the robotics club, is $\frac{3}{8}$. The probability that the student is in the robotics club, given that the student is in grade 10, is $\frac{1}{4}$. There are 96 grade 10 students. How many robotics club members are NOT in grade 10?

Answer:

1111. PS-169 · Hard · SPR · No calculator needed

In grade 10, 12 students are in a club and 28 are not. In grade 11, k students are in the club and 42 are not. Being in the club is independent of grade. What is the value of k?

Answer:

1112. PS-170 · Hard · No calculator needed

In grade 10, 18 students are in the robotics club and 42 are not. In grade 11, 30 students are in the robotics club and 60 are not. A student is chosen at random from the students who are in grade 10 OR in the robotics club. What is the probability that the student is in grade 11?

A) $\frac{1}{3}$ **B)** $\frac{1}{5}$

C) $\frac{1}{2}$ **D)** $\frac{5}{8}$

Inference from sample statistics and margin of error (18)

1113. PS-012 · Medium

A random sample of 500 residents in the NEOM region found that 62% support a proposed residential zoning initiative, with a margin of error of 4 percentage points at the 95% confidence level. Which of the following is the most appropriate interpretation of this result?

A) Exactly 62% of all residents in the NEOM region support the initiative.
B) It is plausible that between 58% and 66% of all residents in the NEOM region support the initiative.
C) Exactly 95% of residents in the NEOM region support the initiative.
D) The survey proves beyond doubt that a majority of residents support the initiative.

1114. PS-047 · Medium

Researchers studying a reforestation program near AlUla selected a random sample of 250 acacia trees planted through the program and measured each tree's growth rate. Based on this sample, which of the following is the most appropriate conclusion?

A) The average growth rate of the sampled trees can be used to estimate the average growth rate of all acacia trees planted through the reforestation program.
B) The average growth rate of the sampled trees can be used to estimate the average growth rate of all trees, of any species, in Saudi Arabia.
C) The average growth rate of the sampled trees is exactly equal to the average growth rate of all acacia trees planted through the reforestation program.
D) The average growth rate of the sampled trees cannot be used to estimate the average growth rate of any other trees.

1115. PS-048 · Medium

A survey of a random sample of employees at a company estimates that the mean daily commute time is 34 minutes, with an associated margin of error of 4 minutes. Which of the following is the most plausible value for the actual mean commute time, in minutes, of all employees at the company?

A) 25 **B)** 31

C) 39 **D)** 44

1116. PS-049 · Medium

A poll of a random sample of residents in a Jeddah district estimates that 42% of all residents support a new public transit expansion, with a margin of error of 3.5 percentage points. Which of the following is the most plausible value for the actual percentage of all residents in the district who support the expansion?

A) 36% **B)** 41%

C) 47% **D)** 50%

1117. PS-050 · Medium

A survey of a random sample of students at a university estimates that $22\% \pm 3\%$ of the university's $8{,}000$ students use the campus shuttle daily. Based on this estimate, which of the following is closest to a plausible range for the number of students at the university who use the shuttle daily?

A) $1{,}400 < n < 1{,}800$ **B)** $1{,}520 < n < 2{,}000$

C) $1{,}760 < n < 1{,}840$ **D)** $1{,}600 < n < 2{,}200$

1118. PS-051 · Medium

In a random sample of 60 laptops tested at a distribution center, 15 were found to have a manufacturing defect. Based on this sample, which of the following is the best conclusion about the number of defective laptops in a shipment of $2{,}400$ laptops from the same production run?

A) Exactly 600 laptops in the shipment are defective.
B) At least 600 laptops in the shipment are defective.
C) A best estimate is that 600 laptops in the shipment are defective.
D) At most 600 laptops in the shipment are defective.

1119. PS-052 · Medium · SPR

In a random sample of 500 registered voters surveyed about a proposed municipal budget, 310 said they support it. There are $45{,}000$ registered voters in total. Based on the sample, what is the expected number of registered voters who support the proposed budget?

Answer:

1120. PS-139 · Medium

Two independent random samples were taken from the same population to estimate a proportion at the same confidence level. The first sample had 400 members and a margin of error of 6%. The second sample had $1{,}600$ members. Which is most likely the margin of error for the second sample?

A) 1.5% **B)** 6%
C) 3% **D)** 12%

1121. PS-013 · Hard · SPR

A survey of $n = 400$ randomly selected residents produces a margin of error of 4.5% at a given confidence level. Assuming the margin of error is inversely proportional to $\sqrt{n}$, what would the margin of error be, in percent, for a survey of $n = 1600$ residents at the same confidence level?

Answer:

1122. PS-053 · Hard

Based on a random sample survey, a researcher estimates that $24\% \pm 4\%$ of high school students in a district spend more than 2 hours a day on Arabic-language reading apps. A school administrator claims that 32% of the district's high school students spend more than 2 hours a day on such apps. Is the administrator's claim consistent with the survey's estimate?

A) Yes, because 32% is within the range of 20% to 28% suggested by the estimate.
B) Yes, because 32% is close to 24%.
C) No, because the sample size is too small to draw any conclusion.
D) No, because 32% is outside the range of 20% to 28% suggested by the estimate.

1123. PS-096 · Hard · SPR · Calculator

Two polls use the same confidence level. For a poll of n people with sample proportion $\hat{p}$, the margin of error is proportional to $\sqrt{\dfrac{\hat{p}(1-\hat{p})}{n}}$. Poll A surveyed 625 people and had $\hat{p} = 0.50$. Poll B had $\hat{p} = 0.10$ and a margin of error half as large as the margin of error of Poll A. How many people did Poll B survey?

Answer:

1124. PS-099 · Hard · Calculator

A researcher surveys a random sample of 625 workers in a city and calculates a 95% confidence interval of 24.2 to 27.8 minutes for the mean one-way commute time of all workers in the city. Assume a second random sample of $2{,}500$ workers has similar variability and the same method is used. Which of the following statements must be true?

A) About 95% of sampled commutes lie in the interval.
B) The new margin is about 0.45 min.
C) The population mean cannot be 28.1 min.
D) The new margin is about 0.9 min.

1125. PS-105 · Hard · Calculator

County P has $2{,}500$ adult residents and county Q has 900. A random sample of county P residents found that 52% support a proposed bike lane, with a margin of error of 3 percentage points. A separate random sample of county Q residents found that 38% support it, with a margin of error of 4 percentage points. Based on the margins of error, what is the greatest possible value of (number of adult supporters in county P) $-$ (number of adult supporters in county Q)?

A) $1{,}033$ **B)** $1{,}069$
C) 525 **D)** 958

1126. PS-137 · Hard

In a poll of a random sample of voters, 51% said they support a proposed tax. The margin of error was 2.5% at the 95% confidence level. Based on the poll, which statement is best supported?

A) A majority of all voters support the tax.
B) It is plausible that a majority support the tax, but less than half supporting it is also plausible.
C) Fewer than half of all voters support the tax.
D) Exactly 51% of all voters support the tax.

1127. PS-138 · Hard · SPR

A random sample of students at a university reported a mean of 3.2 hours of study per day, with a margin of error of 0.4 hours. The university has $18{,}000$ students. Based on the sample, what is the greatest total number of study hours per day that is plausible for the whole student body?

Answer:

1128. PS-140 · Hard

A random sample of 500 residents of Town P gave an estimate of 30% for the percent who favor a new park, with a margin of error of 4%. A random sample of 500 residents of Town Q gave an estimate of 38%, also with a margin of error of 4%. Which statement is best supported?

A) It is plausible that the percent in favor is the same in both towns.
B) The percent in favor in Town P is exactly 4 points less than that in Town Q.
C) Town Q definitely has a greater percent in favor than Town P.
D) It is not possible for the percents to be equal.

1129. PS-141 · Hard

A school has $1{,}200$ seniors and $2{,}800$ students in other grades. A random sample of 300 seniors was surveyed, and an estimated 62% plan to attend college, with a margin of error of 5%. Which statement is best supported by the survey?

A) Exactly 62% of the seniors at the school plan to attend college.
B) Between 57% and 67% of the 300 surveyed seniors plan to attend college.
C) Between 57% and 67% of all students at the school plan to attend college.
D) It is plausible that between 57% and 67% of the seniors at the school plan to attend college.

1130. PS-142 · Hard · SPR

To estimate the number of fish in a lake, a biologist caught and tagged 150 fish and released them. A week later she caught a random sample of 60 fish from the lake, and 9 of them were tagged. Assuming the sample is representative and the tagged fish mixed evenly, what is the best estimate of the number of fish in the lake?

Answer:

Evaluating statistical claims: Observational studies and experiments (16)

1131. PS-033 · Medium

A researcher surveyed a random sample of chemistry teachers selected from all chemistry teachers in the Makkah region. The teachers were asked to name the textbook publisher they currently use. What is the largest population to which the results of the survey can be generalized?

A) All teachers in Saudi Arabia
B) All chemistry teachers in Saudi Arabia
C) All chemistry teachers in the Makkah region
D) All teachers in the Makkah region

1132. PS-034 · Medium

A random sample of 150 residents of a city was surveyed about whether they support a new recycling program. All 150 responded, and 64% said they support it. Based on this information, which of the following statements must be true?

I. Of all city residents, 64% support the recycling program.

II. If another random sample of 150 residents were surveyed, 64% would say they support it.

A) Neither
B) I only
C) II only
D) I and II

1133. PS-135 · Medium

A website for a cycling club posted a poll asking, “Should the city spend more on bike lanes?” Of the $1{,}850$ people who chose to respond, 91% voted yes. Which is the main reason the poll result should not be used to estimate the opinion of all city residents?

A) Respondents selected themselves and are likely to hold similar views.
B) The sample size is too large.
C) The question has too few possible answers.
D) The poll was conducted online, and no online poll can be accurate.

1134. PS-136 · Medium

A random sample of 500 adults reported their coffee consumption. The adults who drink coffee daily had a lower rate of a certain heart condition than those who do not. Which change to the study would make it possible to conclude that drinking coffee reduces the rate of the condition?

A) Increase the sample to $5{,}000$ adults.
B) Include only adults older than 40.
C) Randomly assign the participants to drink coffee daily or not, then compare the rates.
D) Ask each adult about more lifestyle habits.

1135. PS-205 · Medium · No calculator needed

An observational study found that children from homes with more books tend to have higher reading scores. Which of the following is the most plausible confounding variable?

A) Parents' education level
B) The scale used for the reading test
C) The number of pages per book
D) The number of children in the study

1136. PS-206 · Medium · No calculator needed

To estimate the average weekly exercise time of adults in a city, a researcher randomly selected 400 people from the membership lists of the city's gyms. Why is the estimate likely to be biased?

A) The sample size is too small.
B) The selection was random, so the estimate is not biased.
C) The list leaves out adults who are not gym members.
D) There was no random assignment.

1137. PS-207 · Medium · No calculator needed

A teacher randomly selected 50 students and asked each one, face to face, "Have you ever copied homework?" Of the students, 6% answered yes. Which is the most likely problem with this result?

A) The sample was not selected at random.
B) There was no control group.
C) Students may give untrue answers because the question is sensitive and asked in person.
D) A sample of 50 students is too large.

1138. PS-011 · Hard

A researcher wants to determine whether a flipped-classroom method (Method A) causes higher unit-test scores than traditional lecture (Method B) among Grade 11 Algebra II students in a large school district. From all Grade 11 Algebra II sections in the district, the researcher randomly assigns 10 sections to use Method A and 10 sections to use Method B for one unit, then compares the sections' mean test scores. Which statement is best supported by this study design?

A) Because sections were randomly assigned to a method, a causal conclusion about the effect of the method on test scores can be drawn for the Grade 11 Algebra II sections in this district, but the results should not be generalized beyond this district.
B) Because sections, rather than individual students, were randomly assigned, no causal conclusion can be drawn at all.
C) Because an entire school district was involved, the results can be generalized to all high school students nationally.
D) Because the researcher chose which sections received which method, this is an observational study, so no causal conclusions are valid.

1139. PS-035 · Hard

For a date-palm research station near Al-Ahsa, a scientist randomly selected 40 date palm saplings that were 3 years old and randomly assigned them to two groups. Each group received a different irrigation schedule for 2 consecutive years to observe whether the schedule affects fruit yield. Based on the design of the study, what is the largest group to which these results can be applied?

A) All 40 saplings that were selected at this station
B) All date palm saplings that were 3 years old at this station
C) All date palm saplings that were 3 years old in Saudi Arabia
D) All date palm saplings that were 5 years old at this station

1140. PS-036 · Hard

At the end of a Saudi radio program, the host invited listeners to text in and respond to a poll asking, "Do you support extending school hours by 30 minutes?" By the end of the show, the host reported that 62% responded "Yes" and 35% responded "No." Which of the following best explains why the results are unlikely to represent the sentiments of all students' families in the country?

A) The percentages do not add up to 100%, so any possible conclusions from the poll are invalid.
B) Those who responded to the poll were not a random sample of all students' families in the country.
C) There were not 50% "Yes" responses and 50% "No" responses.
D) The show did not allow listeners enough time to respond to the poll.

1141. PS-037 · Hard

A polling agency surveyed 800 adults selected at random from a large Saudi city and asked, "Are you satisfied with the quality of public transportation in the city?" Of those surveyed, 71% responded that they were satisfied. Based on the results of the survey, which of the following statements must be true?

1. Of all adults in the city, 71% are satisfied with the quality of public transportation.

2. If another 800 adults selected at random from the city were surveyed, 71% of them would report they are satisfied.

3. If 800 adults selected at random from a different city were surveyed, 71% of them would report they are satisfied.

A) None
B) 2 only
C) 1 and 2 only
D) 1 and 3 only

1142. **PS-133** · Hard

A researcher recruited 120 volunteers who responded to an online advertisement. She randomly assigned 60 of them to study with method X and 60 to study with method Y for a month. The method X group had a higher mean score on a final test, and the difference was larger than would be expected from random assignment alone. Which conclusion is best supported by the study?

A) The study provides no evidence about either method.
B) Method X caused higher scores for volunteers like those in the study.
C) Method X is associated with higher scores, but no causal conclusion is possible.
D) Method X causes higher scores for all students.

1143. **PS-134** · Hard

A random sample of 500 adults was selected from a large city, and each adult reported how many hours per week they exercise and how many hours they sleep each night. The adults who exercise more tend to sleep more. Which conclusion is best supported by this study?

A) Exercising more causes adults in the city to sleep more.
B) Sleeping more causes adults in the city to exercise more.
C) Among the 500 adults only, exercise and sleep are associated; nothing can be said about the city.
D) Among adults in the city, exercise and sleep are positively associated, but exercise has not been shown to cause more sleep.

1144. **PS-202** · Hard · No calculator needed

A company randomly selected 200 of its employees and gave them new office chairs. Their average productivity rose 8% during the next 3 months. Which conclusion is best supported?

A) The new chairs caused the rise for all employees.
B) The rise cannot be attributed to the chairs, because there was no comparison group.
C) The new chairs had no effect on productivity.
D) The new chairs caused the rise for the 200 employees only.

1145. **PS-203** · Hard · No calculator needed

In an experiment, garden plots were randomly assigned to receive 0, 20 or 40 grams of a fertilizer. The plots given 40 grams had the highest mean yield. Which conclusion is best supported?

A) Doubling the fertilizer doubles the yield.
B) For plots like these, increasing the fertilizer from 0 to 40 grams caused higher yields.
C) No causal conclusion is possible.
D) Using 80 grams would give an even higher yield.

1146. **PS-204** · Hard · No calculator needed

A survey was mailed to $1{,}200$ randomly selected households, and 180 surveys were returned. Of these, 72% supported a new fee. What is the main reason the 72% may not represent all households?

A) There was no random assignment to treatments.
B) The households were selected at random.
C) The sample of $1{,}200$ is too small.
D) Many selected households did not respond, and those who responded may differ from those who did not.